Chapter 3 - Section 12

Vision Language Models – II

Dr. Li Yali

Friday, May 22, 2026

Acknowledge: Zeyuan Wu

Vision Language Models

- Vision-Language pretrained Model (MoEs)
  - VLMo, VL-BEIT, BEIT-3, Florence-2
- Instruction tuning with LLMs
  - BLIP-2, InstructBlip, LLaVA
- Multimodal Large Language Model
  - Qwen-VL, Qwen2-VL, Qwen3-VL
  - DeepSeek-VL
  - Kimi-VL

Preliminary – Mixture of Experts （MoE）

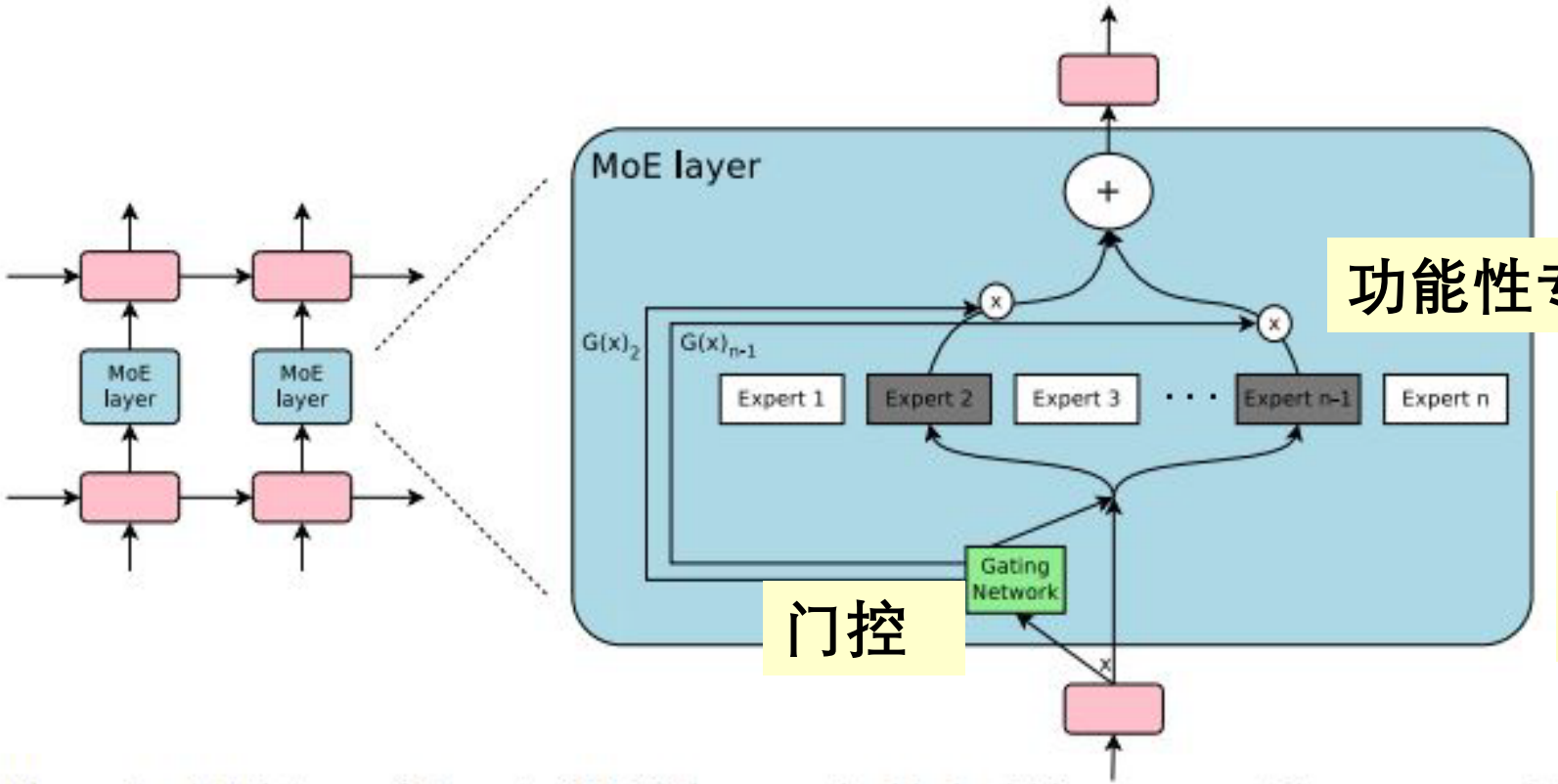


Figure 1: A Mixture of Experts (MoE) layer embedded within a recurrent language model. In this case, the sparse gating function selects two experts to perform computations. Their outputs are modulated by the outputs of the gating network.

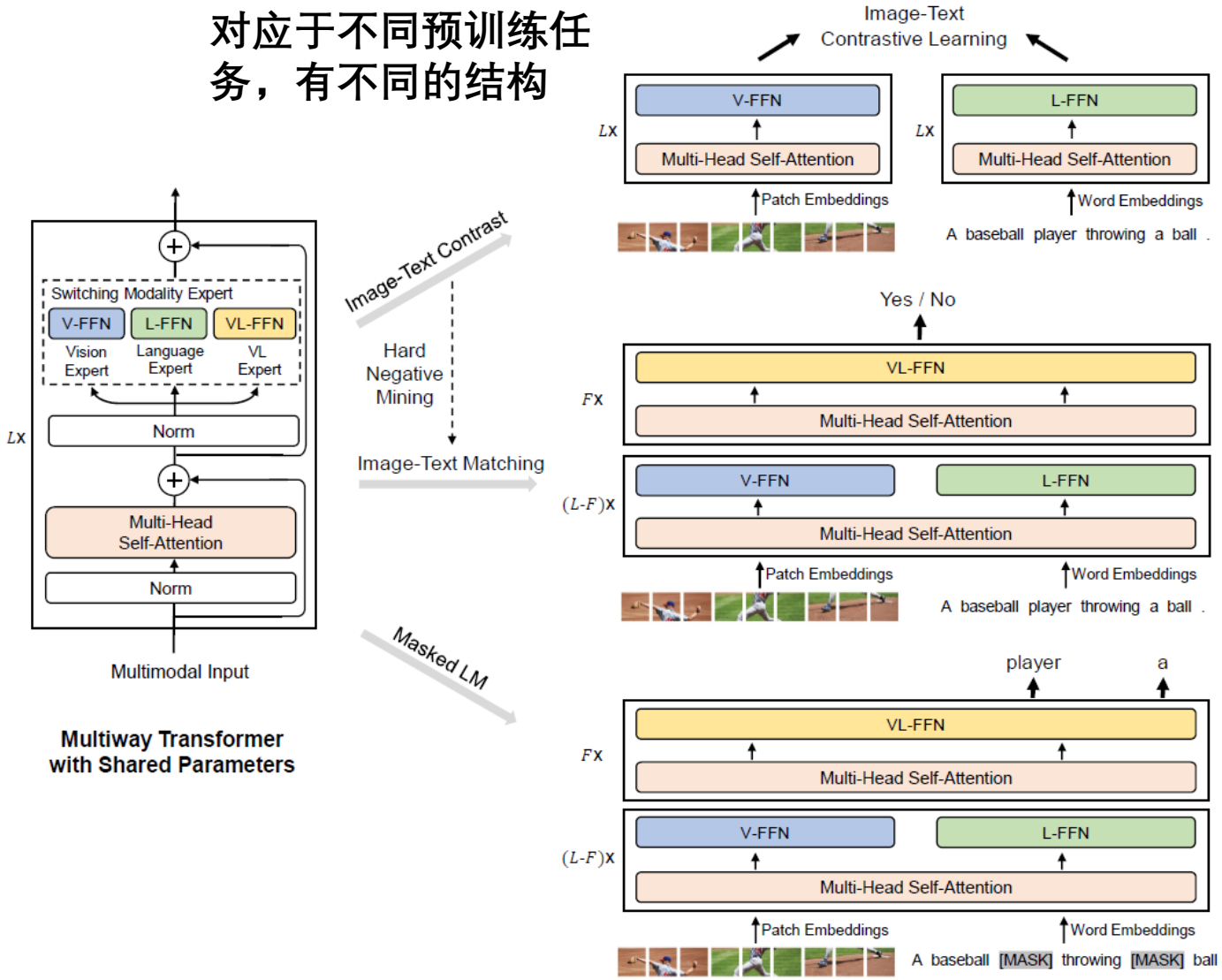
VLMO: Vision-Language pretrained Model

- Multiway transformer: Mixture-of-Modality-Experts (MoME)
  - A pool of modality experts as a substitute of feed forward network in standard Transformer to capture modality-specific information, and uses the modality of input token to conduct hard routing over the pool of feed-forward networks.
  - Using a shared self-attention module across different modalities to learn the alignment of different modalities.
- Pre-training Tasks
  - VLMO is jointly pretrained by image-text contrastive learning on the image and text representations, masked language modeling and image-text matching on the image-text pair representations with shared parameters.

Hangbo Bao et al. VLMO: Unified Vision-Language Pre-Training with Mixture-of-Modality-Experts . NeurIPS 2022.

VLMO: Vision-Language pretrained Model

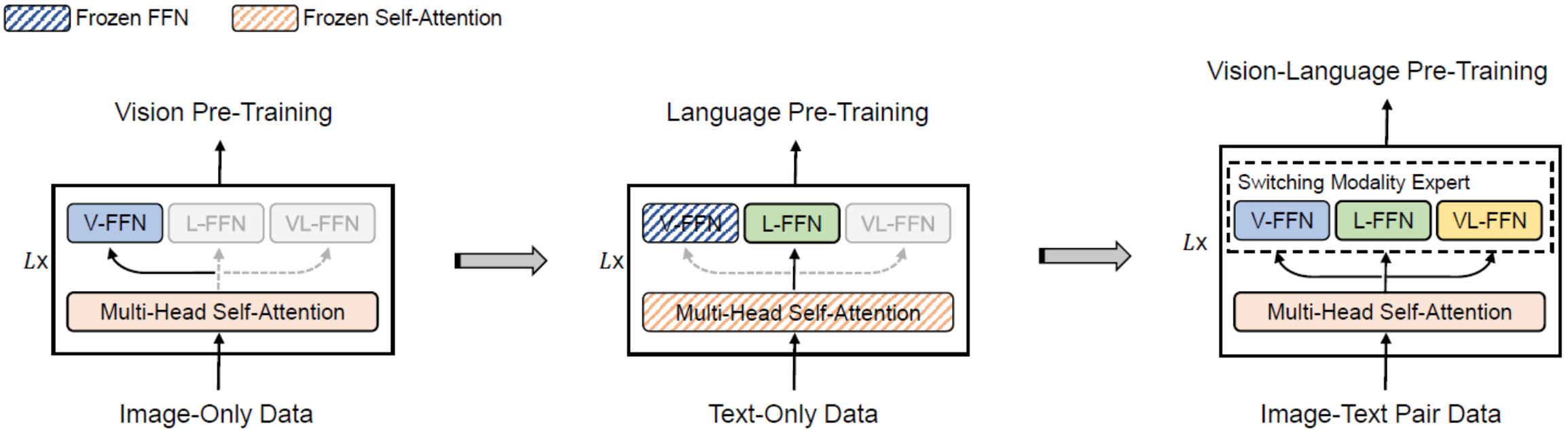
对应于不同预训练任务，有不同的结构



Hangbo Bao et al. VLMO: Unified Vision-Language Pre-Training with Mixture-of-Modality-Experts . NeurIPS 2022.

VLMO: Vision-Language pretrained Model

- Multi-stage Pre-training



1. Pretrain the vision expert (V-FFN) and self-attention module on large-scale image-only data as in BEiT

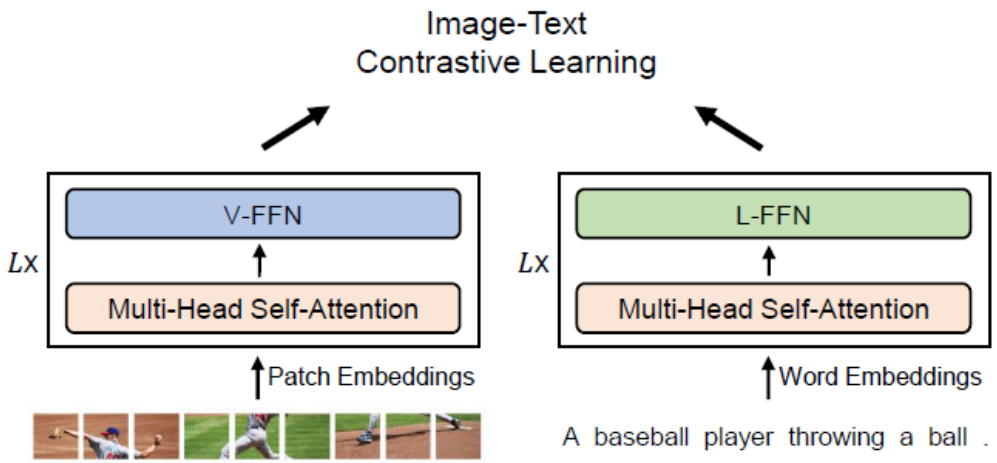
2. Freeze vision expert and self-attention module, train the language expert (L-FFN) by masked language modeling on large amounts of text-only data

3. Train the whole model with vision-language pre-training

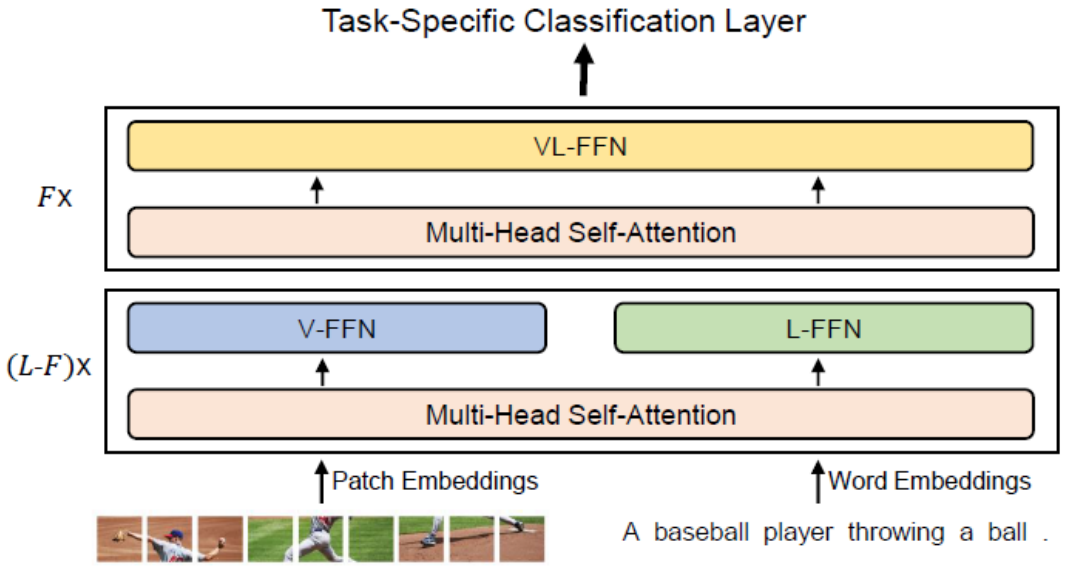
Hangbo Bao et al. VLMO: Unified Vision-Language Pre-Training with Mixture-of-Modality-Experts . NeurIPS 2022.

VLMO: Vision-Language pretrained Model

- Finetuning Cross Tasks



(a) Fine-Tuning on VL Retrieval Tasks



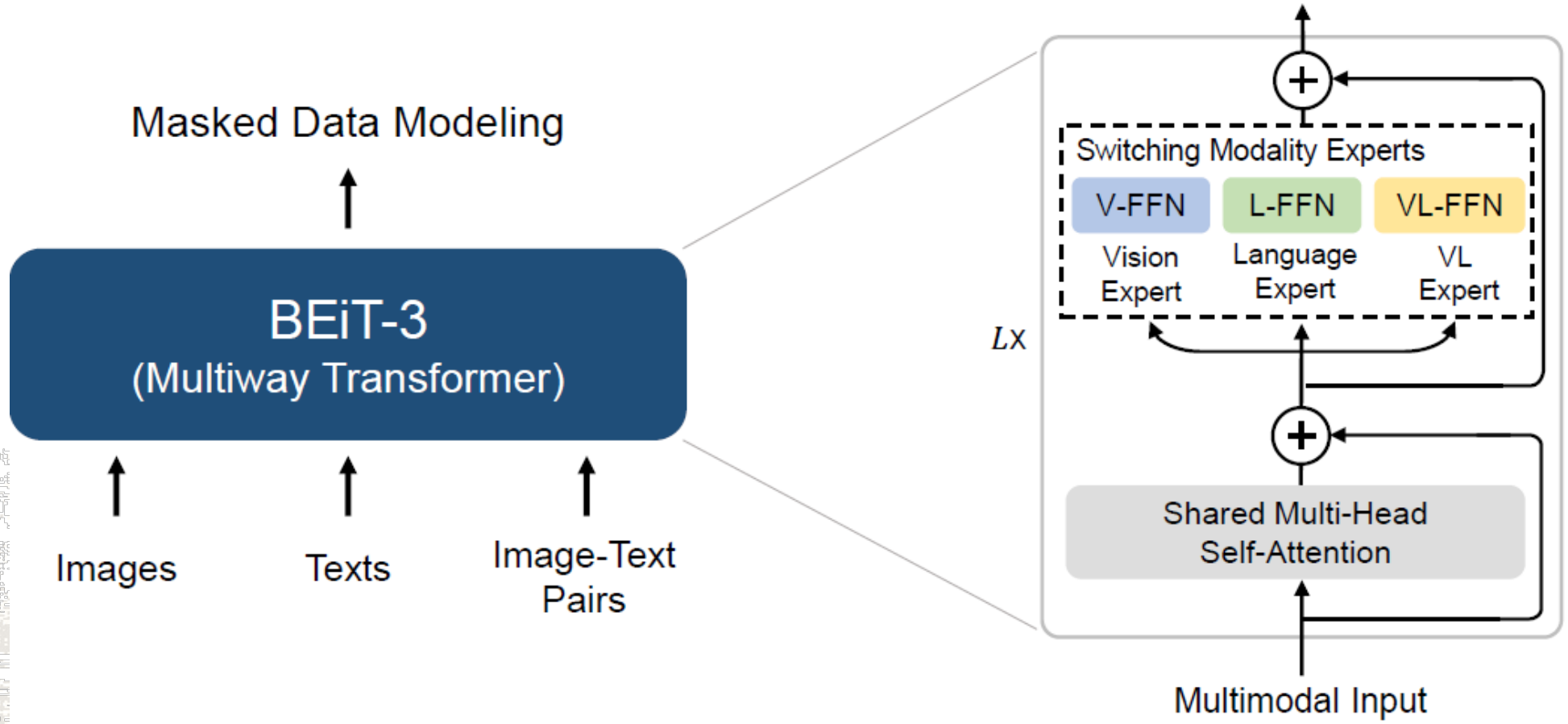
(b) Fine-Tuning on VL Classification Tasks

Multiway transformer as a two-tower vision-language model for efficient image-text retrieval, encoding images and text separately.

Multiway transformer as a single-tower vision-language model to jointly encode image-text pairs and model the deep interaction of image/text

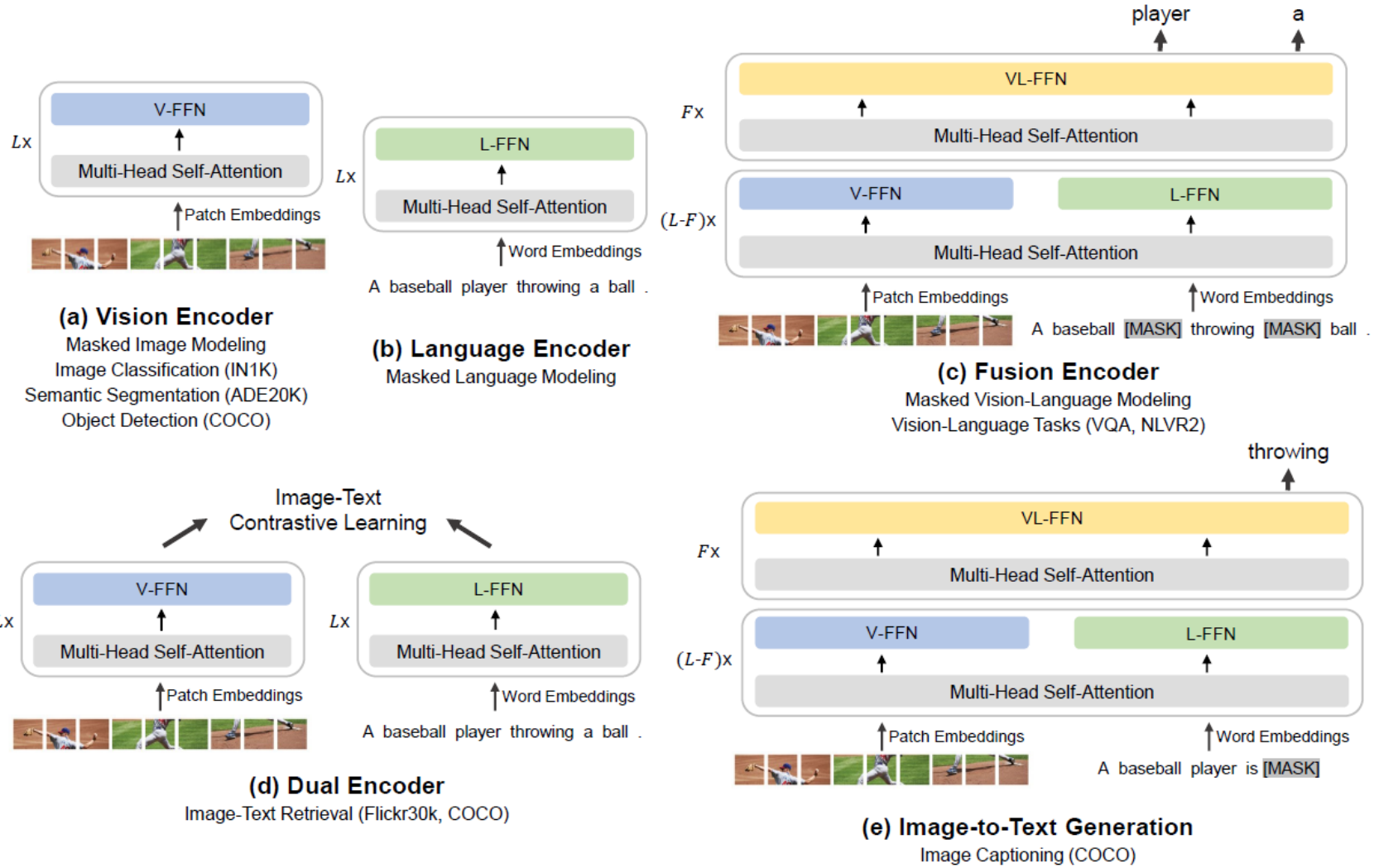
Hangbo Bao et al. VLMO: Unified Vision-Language Pre-Training with Mixture-of-Modality-Experts . NeurIPS 2022.

BEiT-3: foundation model



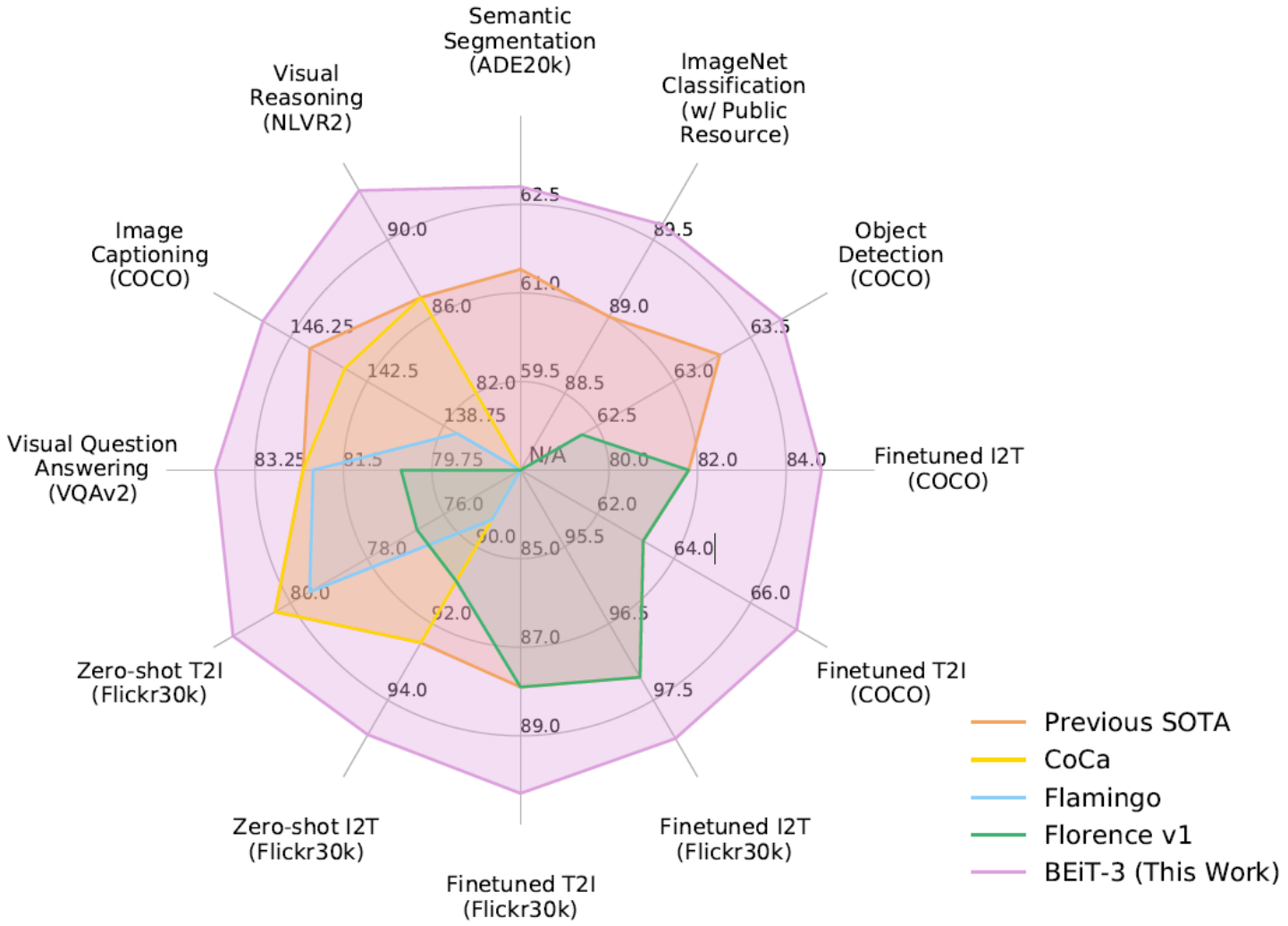
Wenhui Wang et al. Image as a Foreign Language: BEIT Pretraining for All Vision and Vision-Language Tasks. CVPR 2023.

BEIT-3: foundation model



Wenhui Wang et al. Image as a Foreign Language: BEIT Pretraining for All Vision and Vision-Language Tasks. CVPR 2023.

BEiT-3: foundation model



Wenhui Wang et al. Image as a Foreign Language: BEIT Pretraining for All Vision and Vision-Language Tasks. CVPR 2023. (Microsoft)

BEIT-3: foundation model

Category	Task	Dataset	Metric	Previous SOTA	BEIT-3
Vision	Semantic Segmentation	ADE20K	mIoU	61.4 (FD-SwinV2)	**62.8** (+1.4)
	Object Detection	COCO	AP	63.3 (DINO)	**63.7** (+0.4)
	Instance Segmentation	COCO	AP	54.7 (Mask DINO)	**54.8** (+0.1)
	Image Classification	ImageNet†	Top-1 acc.	89.0 (FD-CLIP)	**89.6** (+0.6)
Vision-Language	Visual Reasoning	NLVR2	Acc.	87.0 (CoCa)	**92.6** (+5.6)
	Visual QA	VQAv2	VQA acc.	82.3 (CoCa)	**84.0** (+1.7)
	Image Captioning	COCO‡	CIDEr	145.3 (OFA)	**147.6** (+2.3)
	Finetuned Retrieval	COCO	R@1	72.5 (Florence)	**76.0** (+3.5)
		Flickr30K	R@1	92.6 (Florence)	**94.2** (+1.6)
	Zero-shot Retrieval	Flickr30K	R@1	86.5 (CoCa)	**88.2** (+1.7)

Table 1: Overview of BEIT-3 results on various vision and vision-language benchmarks. We compare with previous state-of-the-art models, including FD-SwinV2 [WHX+22], DINO [ZLL+22], Mask DINO [ZLL+22], FD-CLIP [WHX+22], CoCa [YWV+22], OFA [WYM+22], Florence [YCC+21]. We report the average of top-1 image-to-text and text-to-image results for retrieval tasks. "†" indicates ImageNet results only using publicly accessible resources. "‡" indicates image captioning results without CIDEr optimization.

Wenhui Wang et al. Image as a Foreign Language: BEIT Pretraining for All Vision and Vision-Language Tasks. CVPR 2023.

BEIT-3: foundation model

Model	#Layers	Hidden Size	MLP Size	#Parameters				
				V-FFN	L-FFN	VL-FFN	Shared Attention	Total
BEIT-3	40	1408	6144	692M	692M	52M	317M	1.9B

Table 2: Model configuration of BEIT-3. The architecture layout follows ViT-giant [ZKHB21].

Data	Source	Size
Image-Text Pair	CC12M, CC3M, SBU, COCO, VG	21M pairs
Image	ImageNet-21K	14M images
Text	English Wikipedia, BookCorpus, OpenWebText, CC-News, Stories	160GB documents

Table 3: Pretraining data of BEIT-3. All the data are academically accessible.

Wenhui Wang et al. Image as a Foreign Language: BEIT Pretraining for All Vision and Vision-Language Tasks. CVPR 2023.

BEIT-3: foundation model

Model	VQAv2		NLVR2		COCO Captioning			
	test-dev	test-std	dev	test-P	B@4	M	C	S
Oscar [LYL+20]	73.61	73.82	79.12	80.37	37.4	30.7	127.8	23.5
VinVL [ZLH+21]	76.52	76.60	82.67	83.98	38.5	30.4	130.8	23.4
ALBEF [LSG+21]	75.84	76.04	82.55	83.14	-	-	-	-
BLIP [LLXH22]	78.25	78.32	82.15	82.24	40.4	-	136.7	-
SimVLM [WYY+21]	80.03	80.34	84.53	85.15	40.6	33.7	143.3	**25.4**
Florence [YCC+21]	80.16	80.36	-	-	-	-	-	-
OFA [WYM+22]	82.00	82.00	-	-	43.9	31.8	145.3	24.8
Flamingo [ADL+22]	82.00	82.10	-	-	-	-	138.1	-
CoCa [YWV+22]	82.30	82.30	86.10	87.00	40.9	**33.9**	143.6	24.7
BEIT-3	**84.19**	**84.03**	**91.51**	**92.58**	**44.1**	32.4	**147.6**	**25.4**

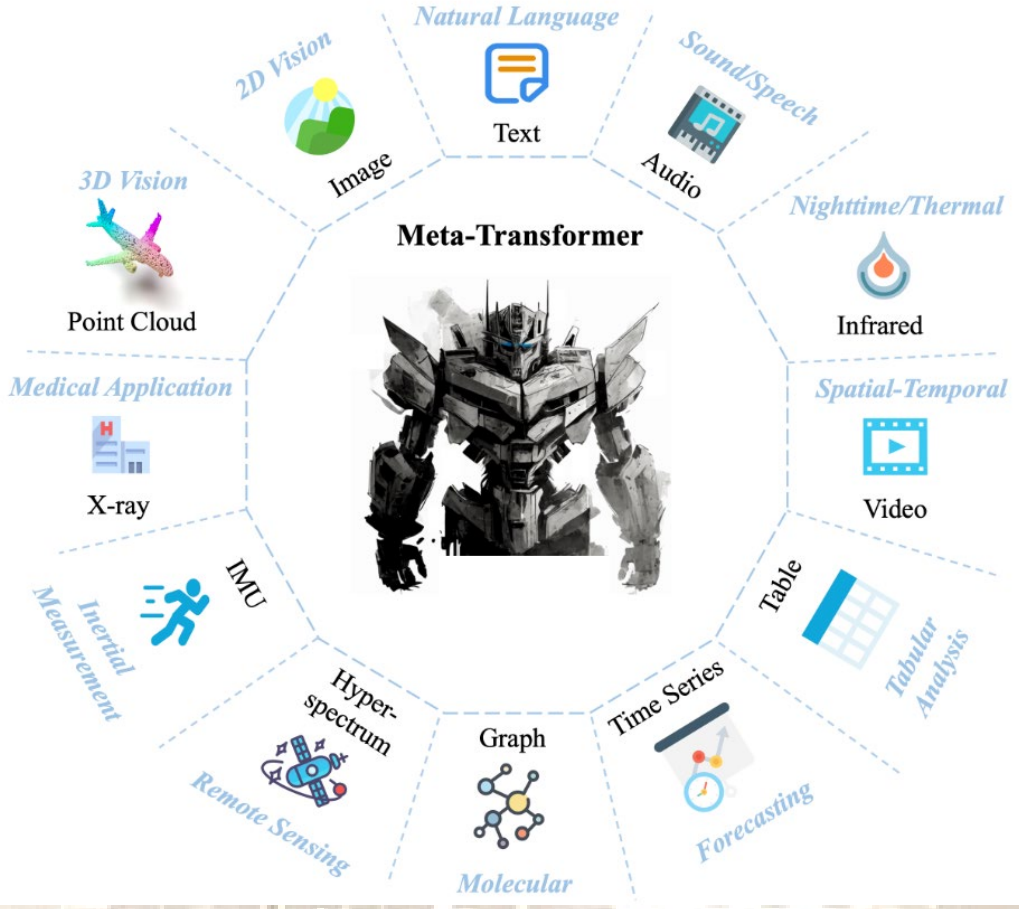
Table 4: Results of visual question answering, visual reasoning, and image captioning tasks. We report *vqa-score* on VQAv2 test-dev and test-standard splits, accuracy for NLVR2 development set and public test set (test-P). For COCO image captioning, we report BLEU@4 (B@4), METEOR (M), CIDEr (C), and SPICE (S) on the Karpathy test split. For simplicity, we report captioning results without using CIDEr optimization.

Wenhui Wang et al. Image as a Foreign Language: BEIT Pretraining for All Vision and Vision-Language Tasks. CVPR 2023.

Meta-transformer

- Unified encoder: simultaneously extract representations from multiple modalities with the same set of parameters

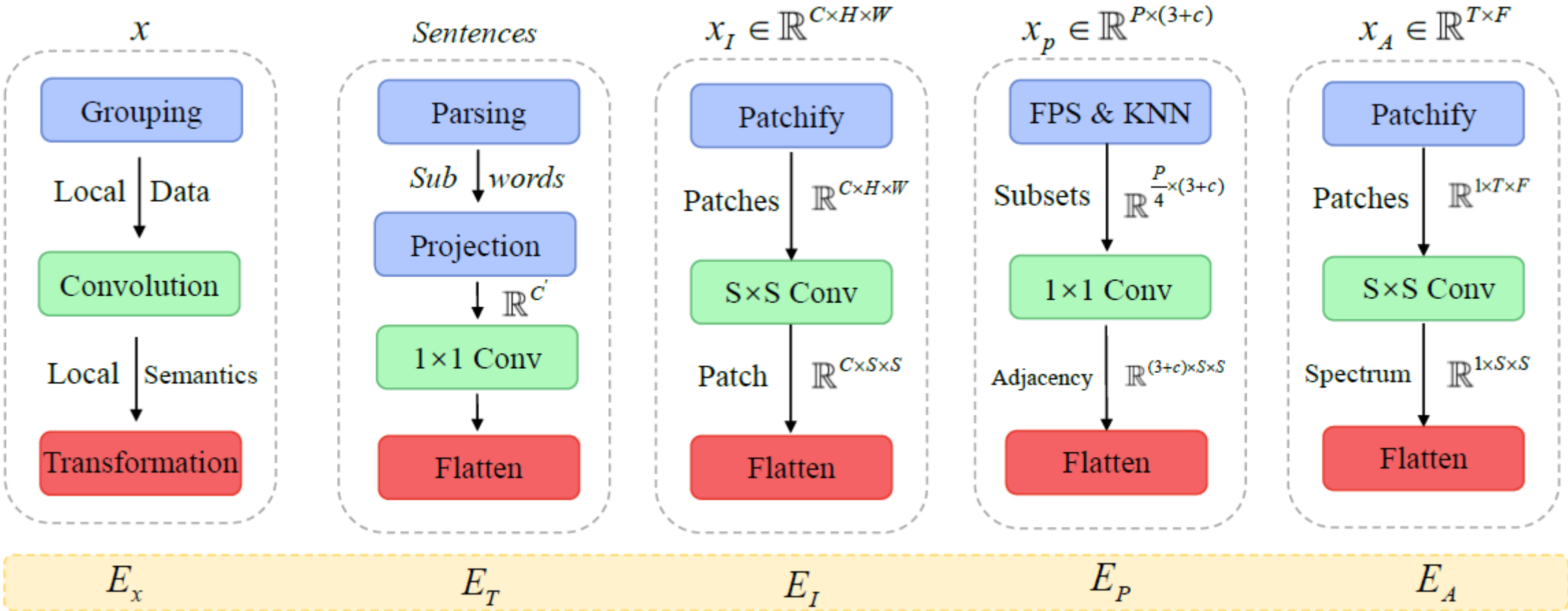
12 modalities: image, text, audio, infrared, video, table, time series, graph, hyper-spectrum, IMU, X-ray, point cloud



Meta-Transformer: A Unified Framework for Multimodal Learning.2023.7
https://kxgong.github.io/meta_transformer/

Meta-transformer

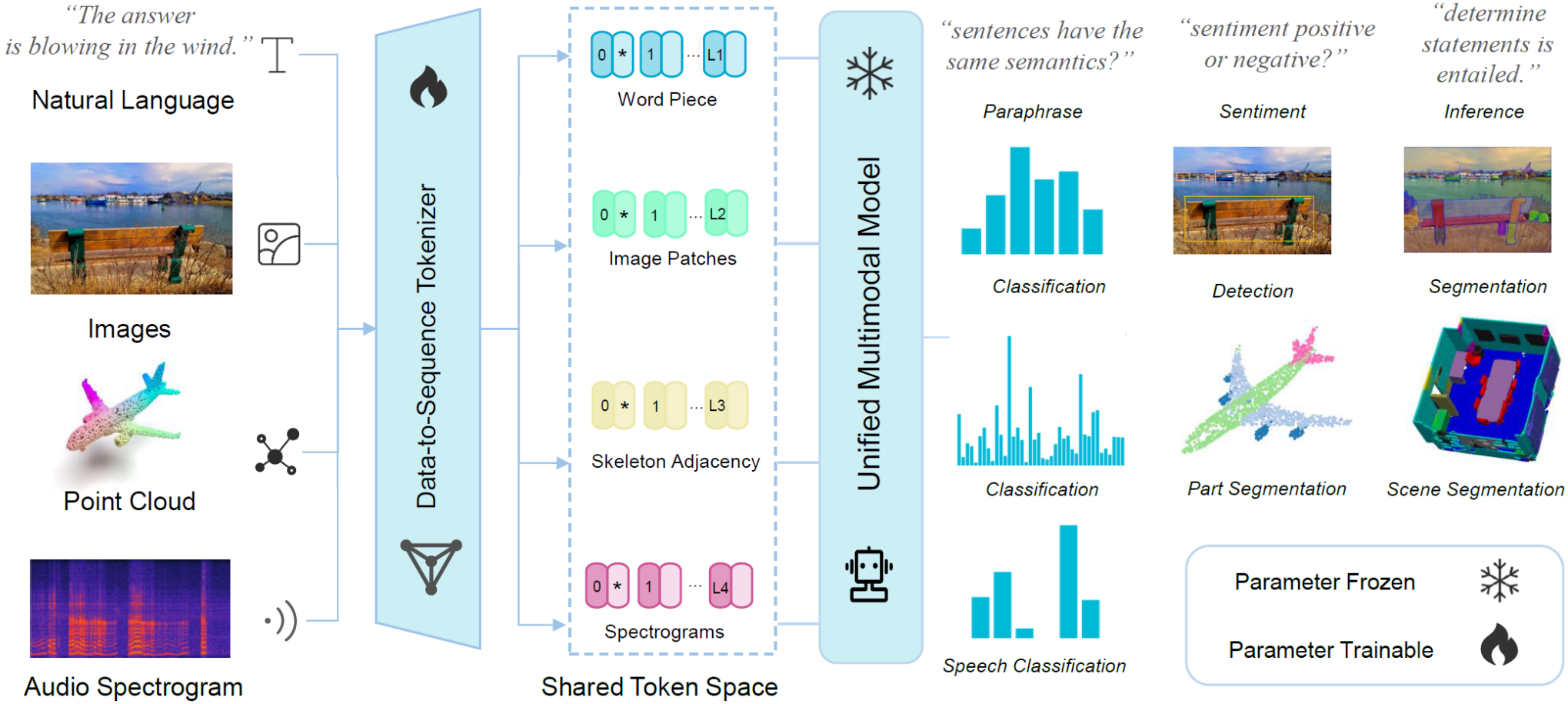
- **Data-to-sequence tokenization**



Meta-Transformer: A Unified Framework for Multimodal Learning.2023.7
https://kxgong.github.io/meta_transformer/

Meta-transformer

- **Unified feature encoding, down-stream task learning**



Meta-Transformer: A Unified Framework for Multimodal Learning.2023.7
https://kxgong.github.io/meta_transformer/

Florence-2

- Built on **large-scale data** from both **visual and textual sources** to understand complex relationships.



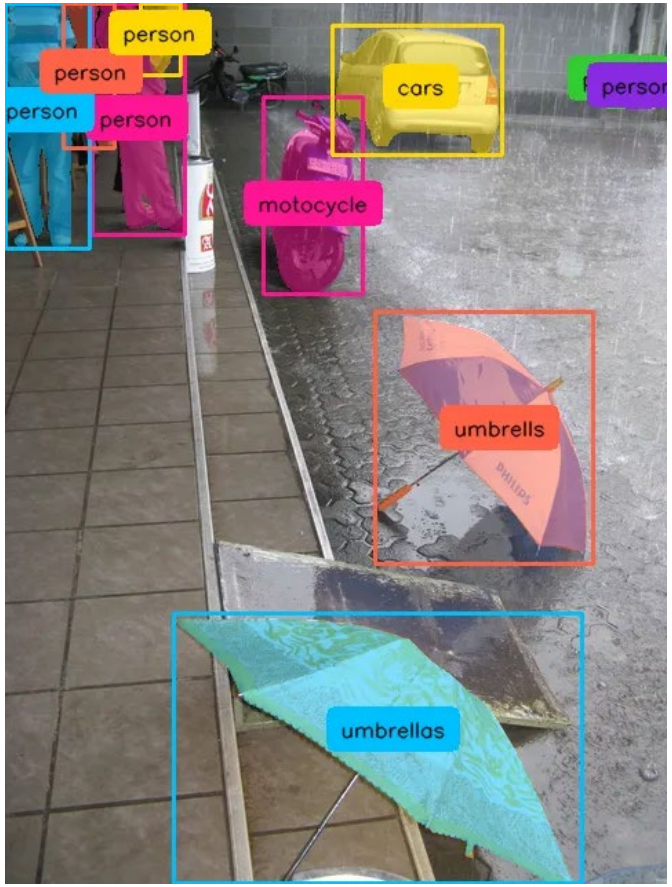
Bin Xiao et al. Florence-2: Advancing a Unified Representation for a Variety of Vision Tasks. CVPR 2024.

Florence-2

Input Image



Object Detection + Segmentation



Phrase Grounding



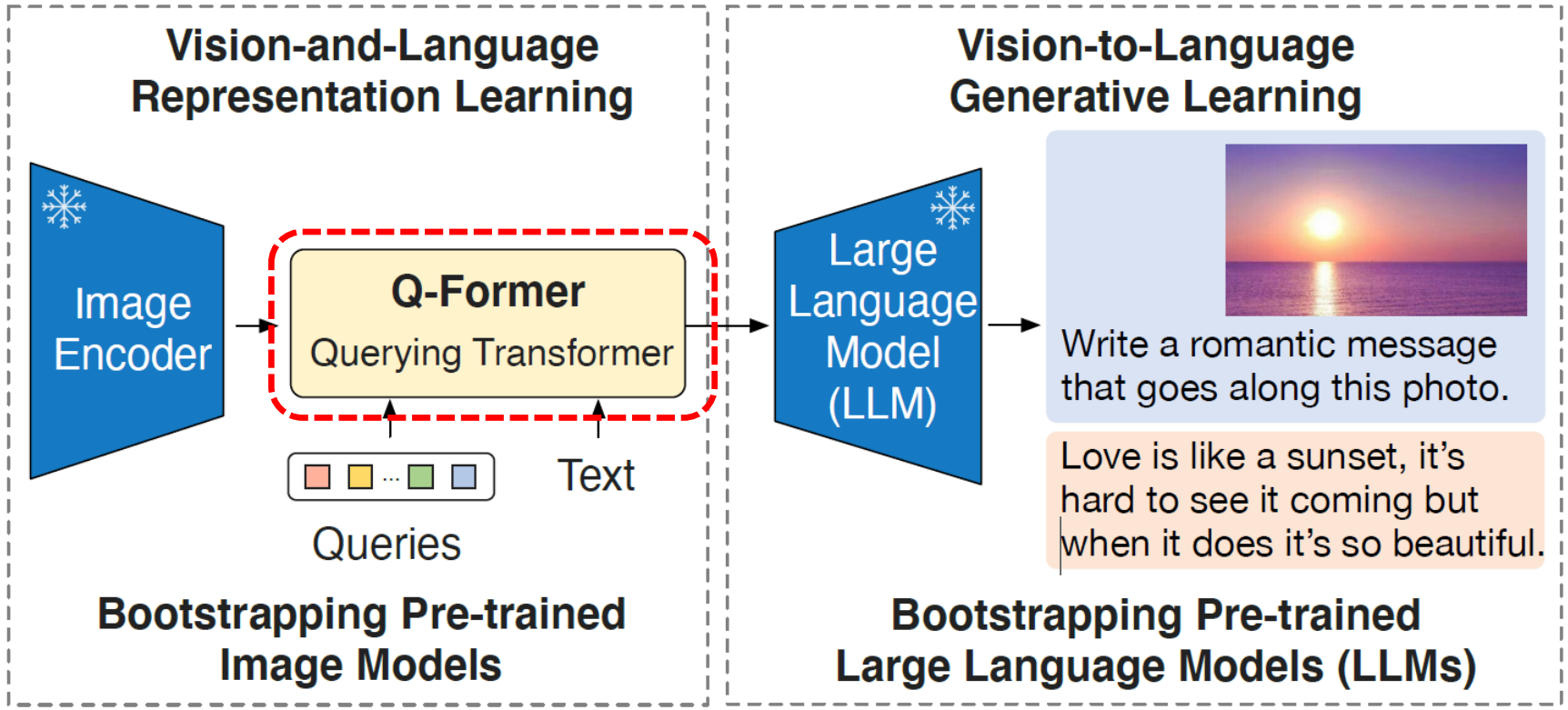
Bin Xiao et al. Florence-2: Advancing a Unified Representation for a Variety of Vision Tasks. CVPR 2024.

Vision Language Models

- Vision-Language pretrained Model
  - VLMo, VL-BEIT, BEIT-3, Florence-2
- Instruction tuning with LLMs (adapter)
  - BLIP-2, InstructBlip, LLaVA
- Multimodal Large Language Model
  - Qwen-VL, Qwen2-VL, Qwen3-VL
  - DeepSeek-VL
  - Kimi-VL

BLIP-2

- **BLIP-2: BLIP with Frozen Image Encoders and Large Language Models**



Junnan Li et al. BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. ICML 2023.

BLIP-2

- BLIP-2: BLIP with Frozen Image Encoders and Large Language Models
  - Challenges: How to bridge the modality gap?
  - Q-Former: pre-train a lightweight Querying Transformer
  - Two-stage strategy:
    - The first stage bootstraps vision-language representation learning from a frozen image encoder.
    - The second stage bootstraps vision-to-language generative learning from a frozen LLM, which enables zero-shot instructed image-to-text generation

Junnan Li et al. BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. ICML 2023.

BLIP-2

- **BLIP-2: BLIP with Frozen Image Encoders and Large Language Models**

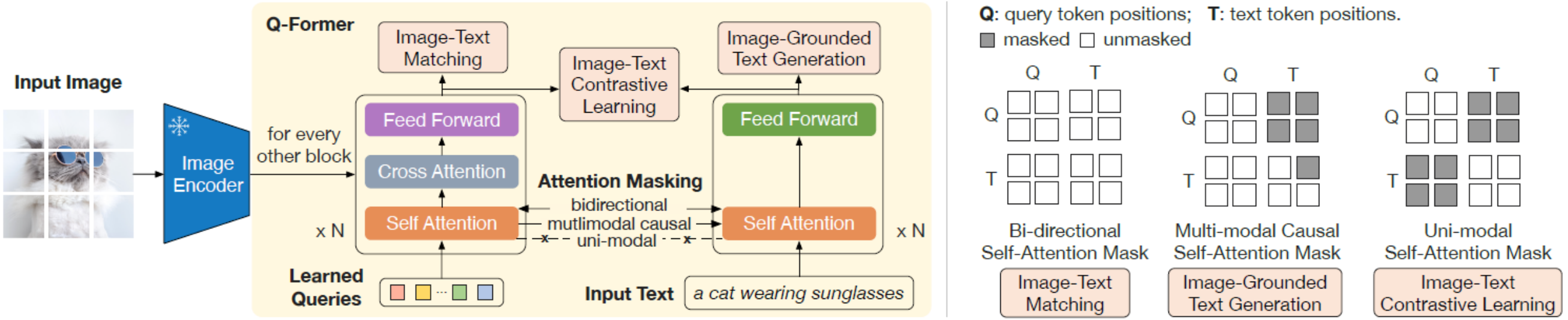


Figure 2. (**Left**) Model architecture of Q-Former and BLIP-2's first-stage vision-language representation learning objectives. We jointly optimize three objectives which enforce the queries (a set of learnable embeddings) to extract visual representation most relevant to the text. (**Right**) The self-attention masking strategy for each objective to control query-text interaction.

Junnan Li et al. BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. ICML 2023.

BLIP-2

- **BLIP-2: BLIP with Frozen Image Encoders and Large Language Models**

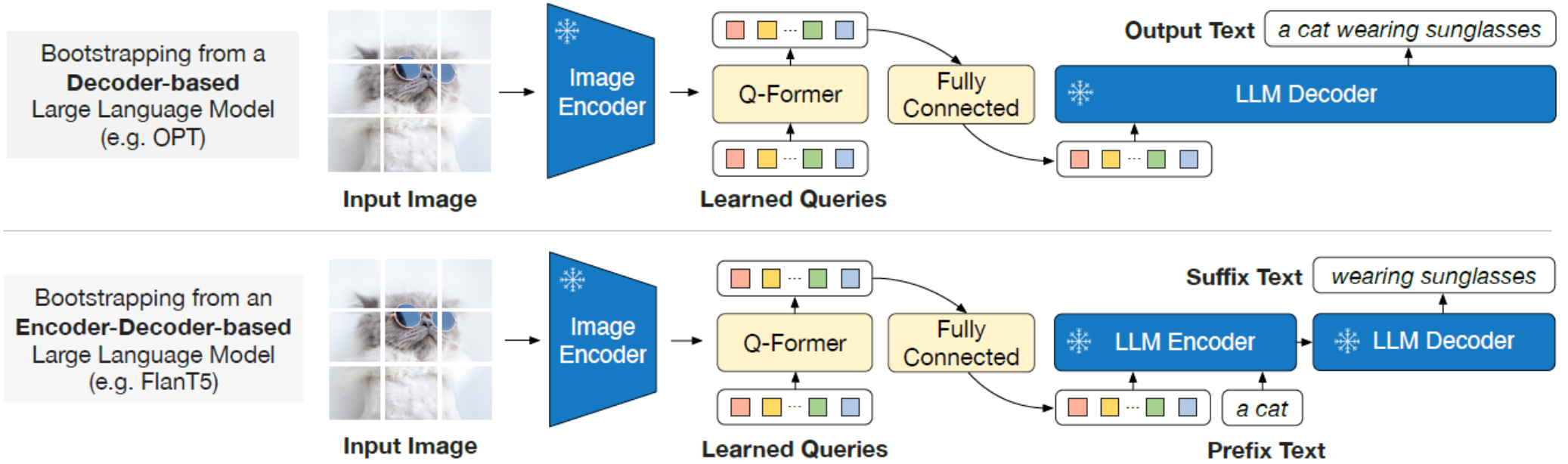


Figure 3. BLIP-2's second-stage vision-to-language generative pre-training, which bootstraps from frozen large language models (LLMs). (**Top**) Bootstrapping a decoder-based LLM (e.g. OPT). (**Bottom**) Bootstrapping an encoder-decoder-based LLM (e.g. FlanT5). The fully-connected layer adapts from the output dimension of the Q-Former to the input dimension of the chosen LLM.

Junnan Li et al. BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. ICML 2023.

InstructBLIP

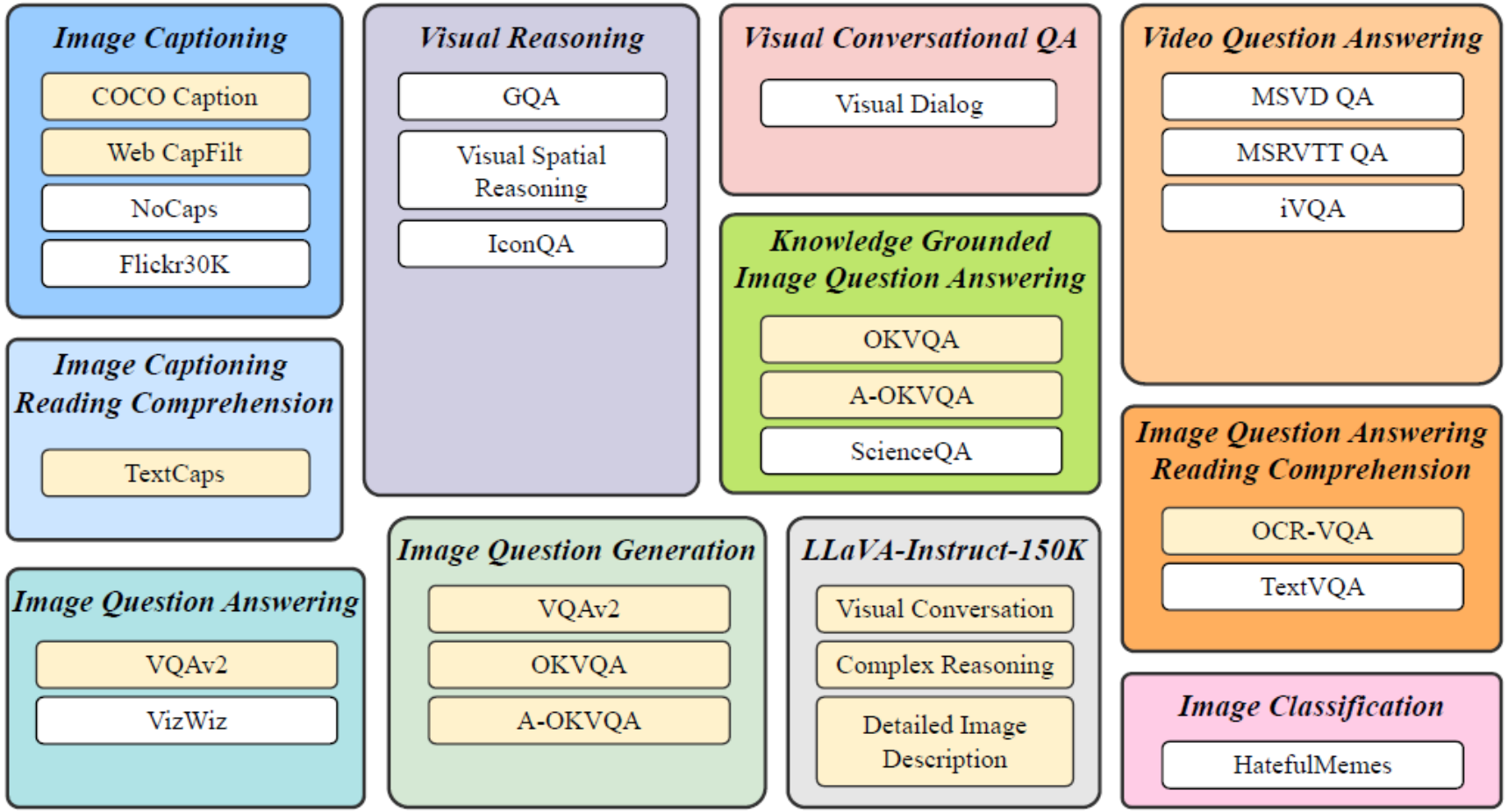


Figure 2: Tasks and their corresponding datasets used for vision-language instruction tuning. The held-in datasets are indicated by yellow and the held-out datasets by white.

Wenliang Dai et al. InstructBLIP: Towards General-purpose Vision-Language Models with Instruction Tuning. NeurIPS 2023.

InstructBLIP

Task	Instruction Template
Image Captioning	<Image>A short image caption: <Image>A short image description: <Image>A photo of <Image>An image that shows <Image>Write a short description for the image. <Image>Write a description for the photo. <Image>Provide a description of what is presented in the photo. <Image>Briefly describe the content of the image. <Image>Can you briefly explain what you see in the image? <Image>Could you use a few words to describe what you perceive in the photo? <Image>Please provide a short depiction of the picture. <Image>Using language, provide a short account of the image. <Image>Use a few words to illustrate what is happening in the picture.
VQA	<Image>{Question} <Image>Question: {Question} <Image>{Question} A short answer to the question is <Image>Q: {Question} A: <Image>Question: {Question} Short answer: <Image>Given the image, answer the following question with no more than three words. {Question} <Image>Based on the image, respond to this question with a short answer: {Question}. Answer: <Image>Use the provided image to answer the question: {Question} Provide your answer as short as possible: <Image>What is the answer to the following question? "{Question}" <Image>The question "{Question}" can be answered using the image. A short answer is
VQG	<Image>Given the image, generate a question whose answer is: {Answer}. Question: <Image>Based on the image, provide a question with the answer: {Answer}. Question: <Image>Given the visual representation, create a question for which the answer is "{Answer}". <Image>From the image provided, craft a question that leads to the reply: {Answer}. Question: <Image>Considering the picture, come up with a question where the answer is: {Answer}. <Image>Taking the image into account, generate an question that has the answer: {Answer}. Question:

Wenliang Dai et al. InstructBLIP: Towards General-purpose Vision-Language Models with Instruction Tuning. NeurIPS 2023.

InstructBLIP

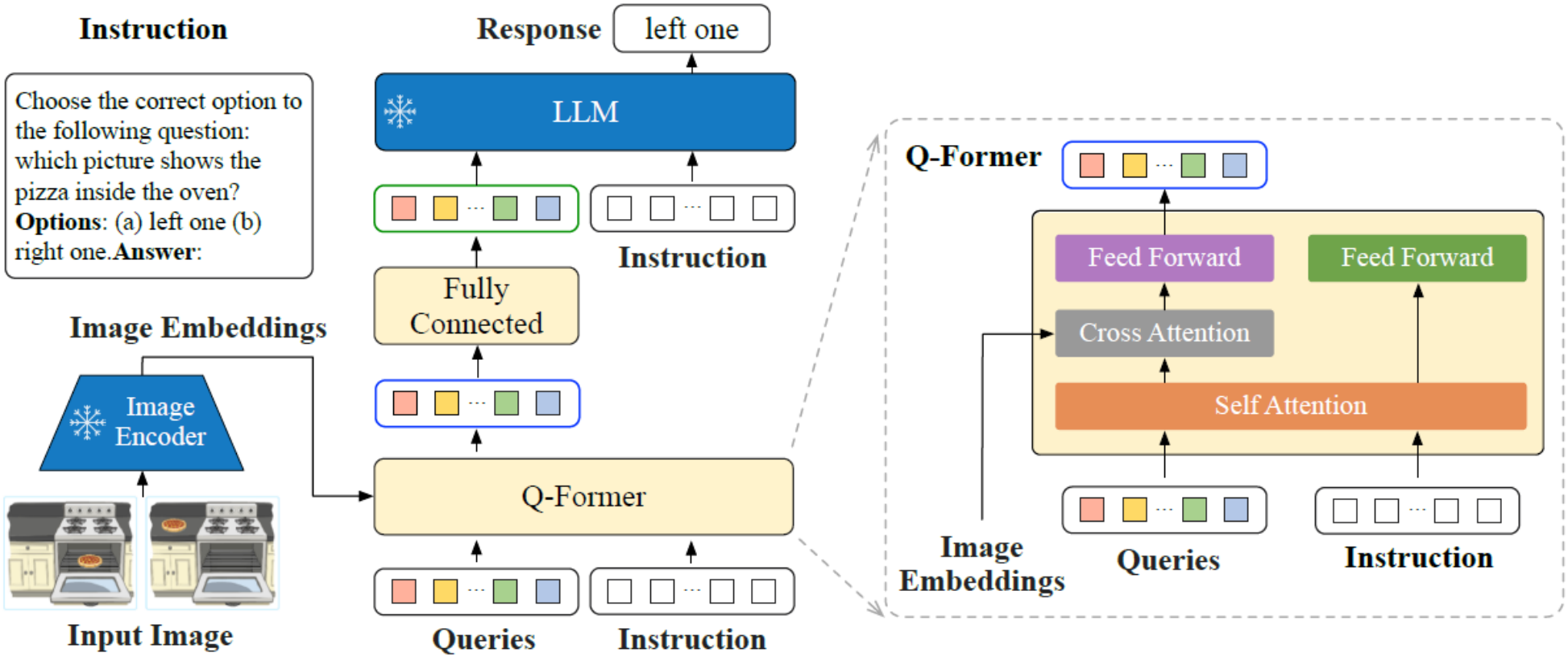


Figure 3: Model architecture of InstructBLIP. The Q-Former extracts instruction-aware visual features from the output embeddings of the frozen image encoder, and feeds the visual features as soft prompt input to the frozen LLM. We instruction-tune the model with the language modeling loss to generate the response.

Wenliang Dai et al. InstructBLIP: Towards General-purpose Vision-Language Models with Instruction Tuning. NeurIPS 2023.

InstructBLIP



[Input Instruction] Do you know who drew this painting?

[Model Response]

InstructBLIP Leonardo da Vinci.

LLaVA The painting depicts a woman, commonly believed to be Mona Lisa, the famous artwork by Leonardo da Vinci. It is a portrait painting that showcases the woman's enigmatic smile and has become one of the most famous and iconic art pieces in the world. The original work is displayed in the Louvre Museum in Paris, and it is known for its intricate details, use of oil paint, and the artist's innovative techniques that contributed to its enduring appeal and mystery.

MiniGPT-4 The painting is the Mona Lisa, also known as the Mona Lisa, by Leonardo da Vinci. It is one of the most famous paintings in the world and is known for its enigmatic smile and the mysterious identity of the sitter. The painting is housed in the Louvre Museum in Paris, France.

Figure 6: In this example, we illustrate that long-form responses are not always preferable. InstructBLIP can directly address the user's intent by adaptively adjusting the response length, while other models tend to generate lengthy paragraphs with less-relevant sentences. The response from LLaVA is taken from the paper, and for MiniGPT-4, we utilize its official demo.

[Input Image]



[Input Instruction] Introduce me this painting in detail.

[Model Response]

InstructBLIP The painting depicts a young girl with a pearl earring, also known as "The Girl with the Pearl Earring" or "Girl with a Pearl Earring". She is wearing a blue dress and has a pearl earring on her left ear. Her hair is pulled back into a bun, and she is looking directly at the viewer. This painting is a famous work of art by Dutch painter Johannes Vermeer, created in 1665. It is considered one of his masterpieces and is widely recognized for its intricate details and realistic portrayal of the young girl's facial expression.

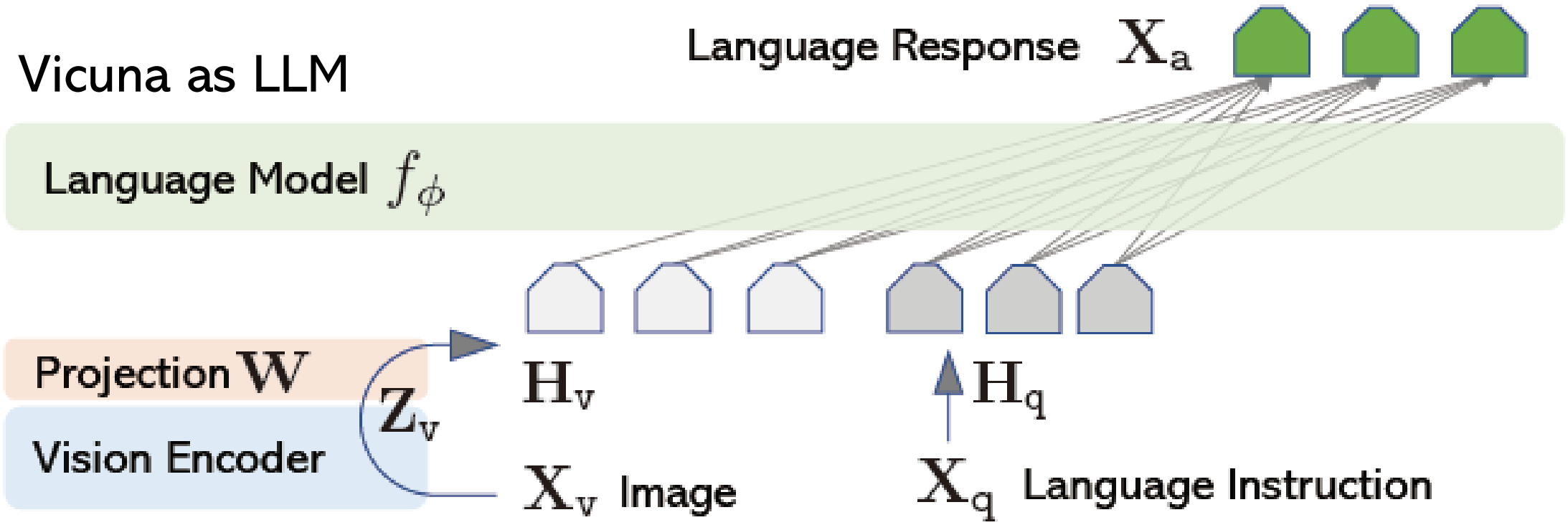
LLaVA The painting features a woman with a headscarf, possibly of Dutch origin, dressed in blue and white clothing. She is wearing a large blue hat and has a red mouth, which adds a vibrant touch to the portrait. The woman is looking directly at the viewer, creating a sense of connection. The background of the painting is predominantly blue, further emphasizing the color theme of the piece. The overall composition and attire of the woman suggest that the painting might be inspired by the style of Rembrandt.

MiniGPT-4 The painting depicts a young woman with a pearl earring, wearing a blue dress and a white apron. She is shown with a serious expression, looking directly at the viewer. The background is a dark, neutral tone, with no visible objects or elements. The overall mood of the painting is somber and contemplative.
The painting is well-executed, with a high level of detail and realism in the depiction of the woman's features and clothing. The use of shading and lighting creates a sense of depth and dimension, making the woman appear three-dimensional. The artist has used a limited color palette, with the woman's dress and apron providing the only sources of color in the painting.
The painting is well-composed, with the woman's head and shoulders positioned in the center of the canvas, creating a sense of balance and symmetry. The use of negative space around the woman's figure adds to the overall somber mood of the painting.
Overall, this is a well-executed and thought-provoking painting that effectively conveys a sense of contemplation and introspection.

Wenliang Dai et al. InstructBLIP: Towards General-purpose Vision-Language Models with Instruction Tuning. NeurIPS 2023.

LLaVA

$$\mathbf{H}_\mathbf{v} = \mathbf{W} \cdot \mathbf{Z}_\mathbf{v}, \text{ with } \mathbf{Z}_\mathbf{v} = g(\mathbf{X}_\mathbf{v})$$



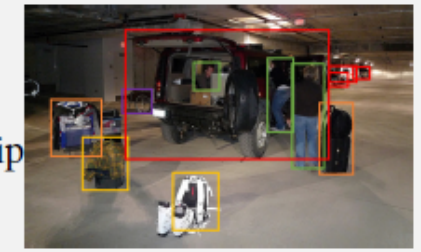
Haotian Liu et al. Visual Instruction Tuning . NeurIPS 2023.

LLaVA

Human : $X_q X_v$<STOP> Assistant : X_c<STOP>.

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation

Question: What type of vehicle is featured in the image?
Answer: The image features a black sport utility vehicle (SUV) ...<omitted>

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>

Response type 3: complex reasoning

Question: What challenges do these people face?
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

Haotian Liu et al. Visual Instruction Tuning . NeurIPS 2023.

LLaVA

- Two-stage instruction-tuning
  - Stage 1: Pre-training for Feature Alignment. Filter CC3M to 595K image-text pairs. These pairs are converted to the instruction-following data
  - Stage 2: Fine-tuning End-to-End. Keep the visual encoder weights frozen, and continue to update both the pre-trained weights of the projection layer and LLM in LLaVA

Haotian Liu et al. Visual Instruction Tuning . NeurIPS 2023.

MoE-LLaVA

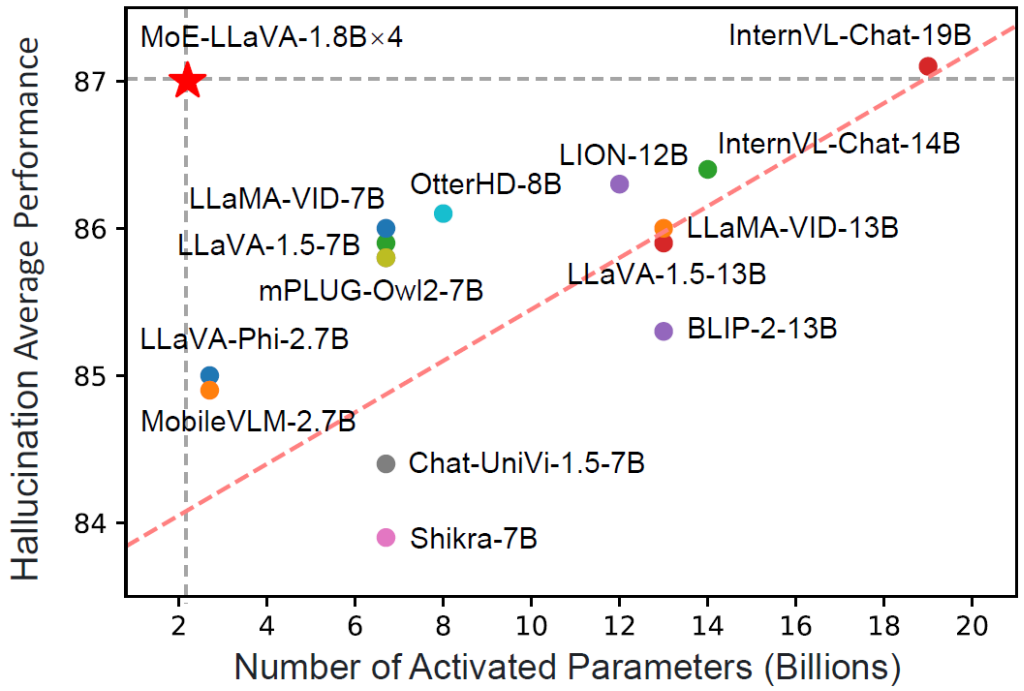


Figure 1. **Comparison between MoE-LLaVA-1.8B×4 and open-source LVLMs on object hallucination benchmark.** We report the average performance on the POPE (Li et al., 2023d) benchmark, which includes three subsets of Adversarial, Random, and Popular. The red dashed line represents the linear fit to the data points of all models except MoE-LLaVA.

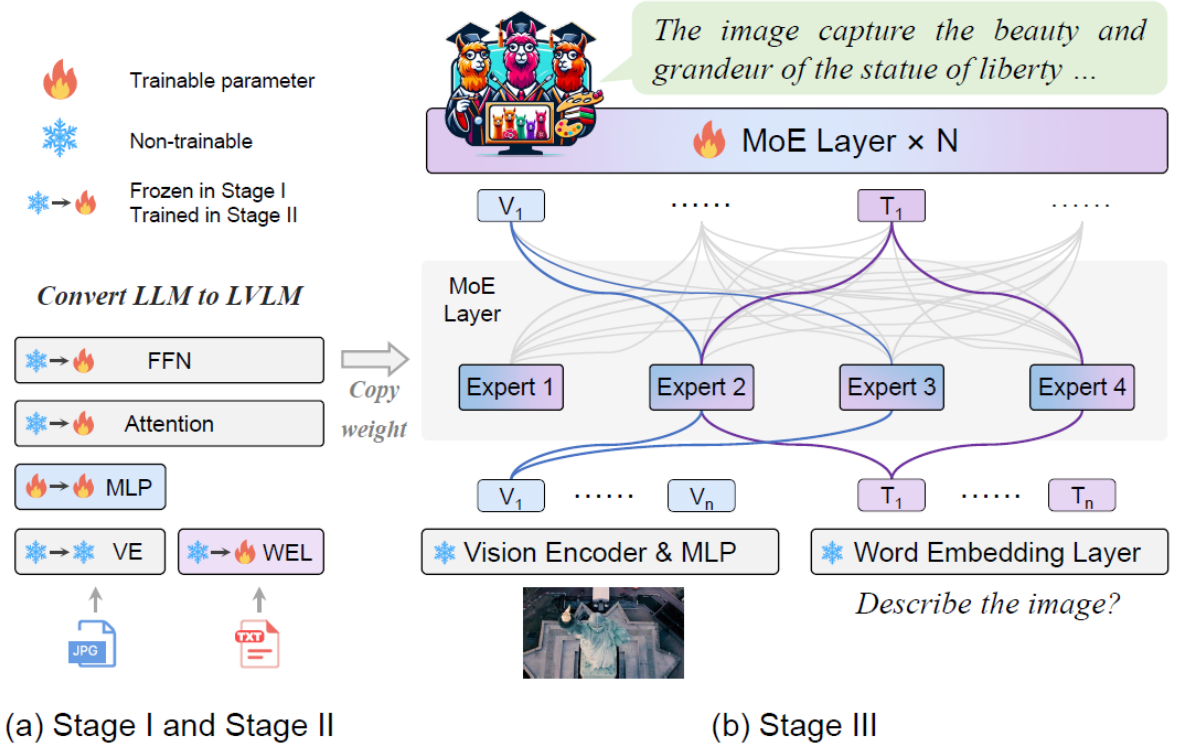


Figure 2. **Illustration of MoE-Tuning.** The MoE-Tuning consists of three stages. In stage I, only the MLP is trained. In stage II, all parameters are trained except for the Vision Encoder (VE). In stage III, FFNs are used to initialize the experts in MoE, and only the MoE layers are trained. For each MoE layer, only two experts are activated for each token, while the other experts remain silent.

Bin Lin et al. MoE-LLaVA: Mixture of Experts for Large Vision-Language Models. ArXiv 2024.

MoE-LLaVA

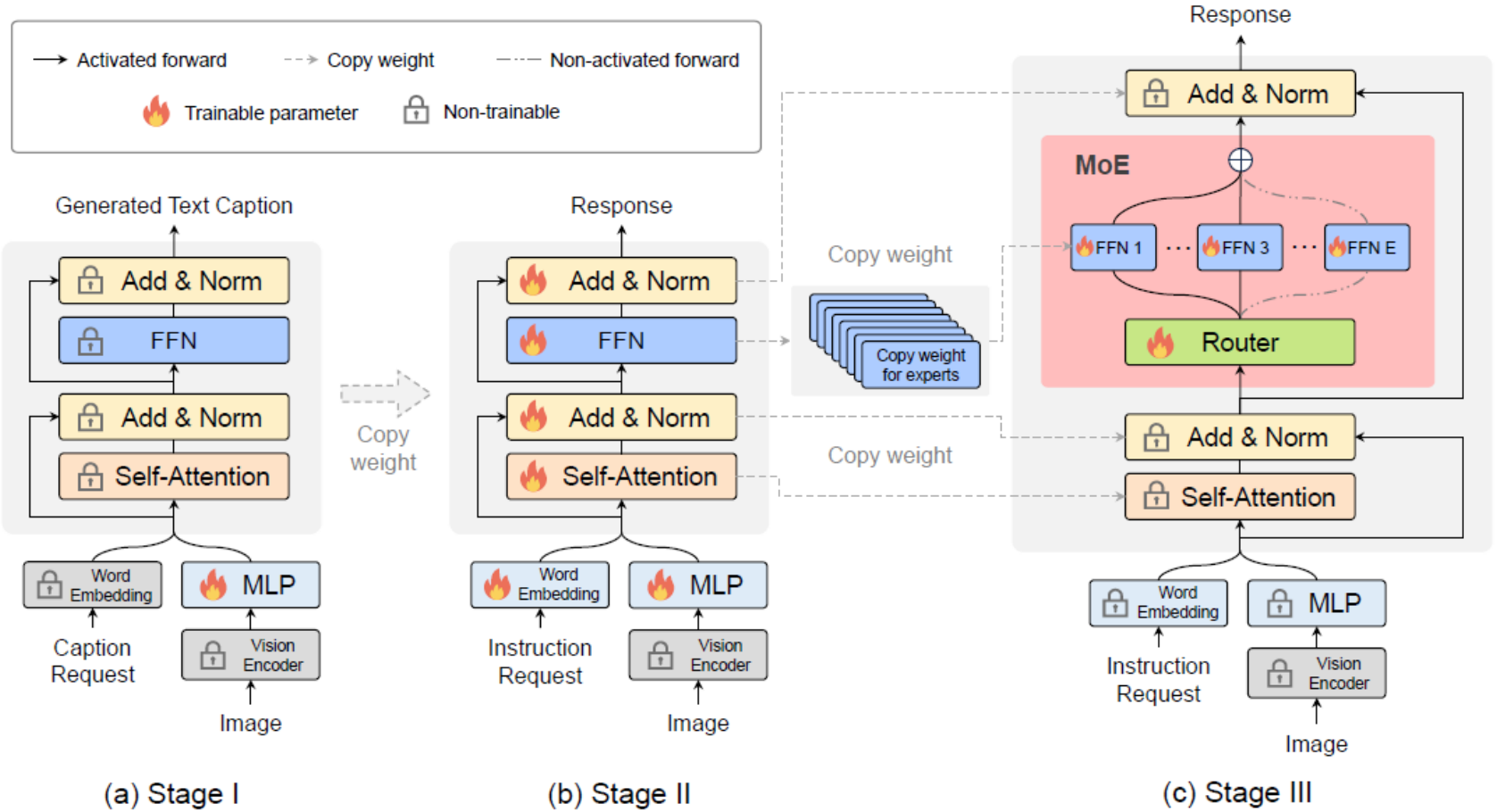


Figure 3. **Training framework and strategy.** MoE-LLaVA adopts a three-stage training strategy. (a) We solely train the MLP to adapt the LLM to visual inputs. (b) Training the LLM backend empowers multi-modal understanding capability and MoE layers are not involved. (c) In this stage, we replicate the weights of the FFN to initialize each expert.

Bin Lin et al. MoE-LLaVA: Mixture of Experts for Large Vision-Language Models. ArXiv 2024.

MiniGPT4

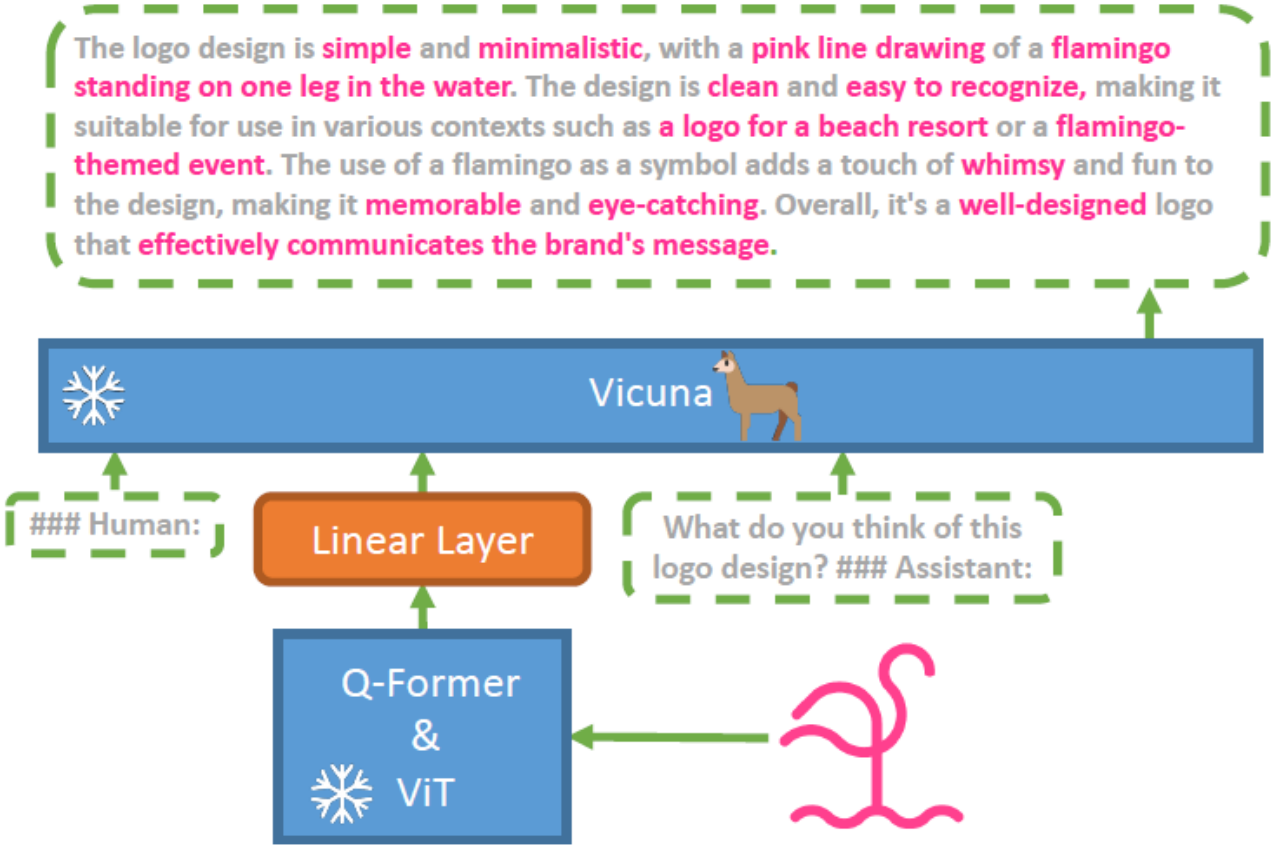


Figure 1: **The architecture of MiniGPT-4.** It consists of a vision encoder with a pretrained ViT and Q-Former, a single linear projection layer, and an advanced Vicuna large language model. MiniGPT-4 only requires training the linear projection layer to align the visual features with the Vicuna.

Deyao Zhu et al. Mini-GPT4: MiniGPT-4 Enhancing Vision-Language Understanding with Advanced Large Language Models. ArXiv 2023.

MiniGPT4

- **Mini-GPT4:**
  - **By aligning visual features with advanced large language models like Vicuna, MiniGPT-4 can achieve advanced vision-language capabilities comparable to those exhibited in the GPT-4 demonstrations.**
  - **Training merely one projection layer can effectively align a pretrained vision encoder with the large language model. MiniGPT-4 only requires training approximately 10 hours on 4 A100 GPUs.**

Deyao Zhu et al. Mini-GPT4: MiniGPT-4 Enhancing Vision-Language Understanding with Advanced Large Language Models. ArXiv 2023.

MiniGPT4

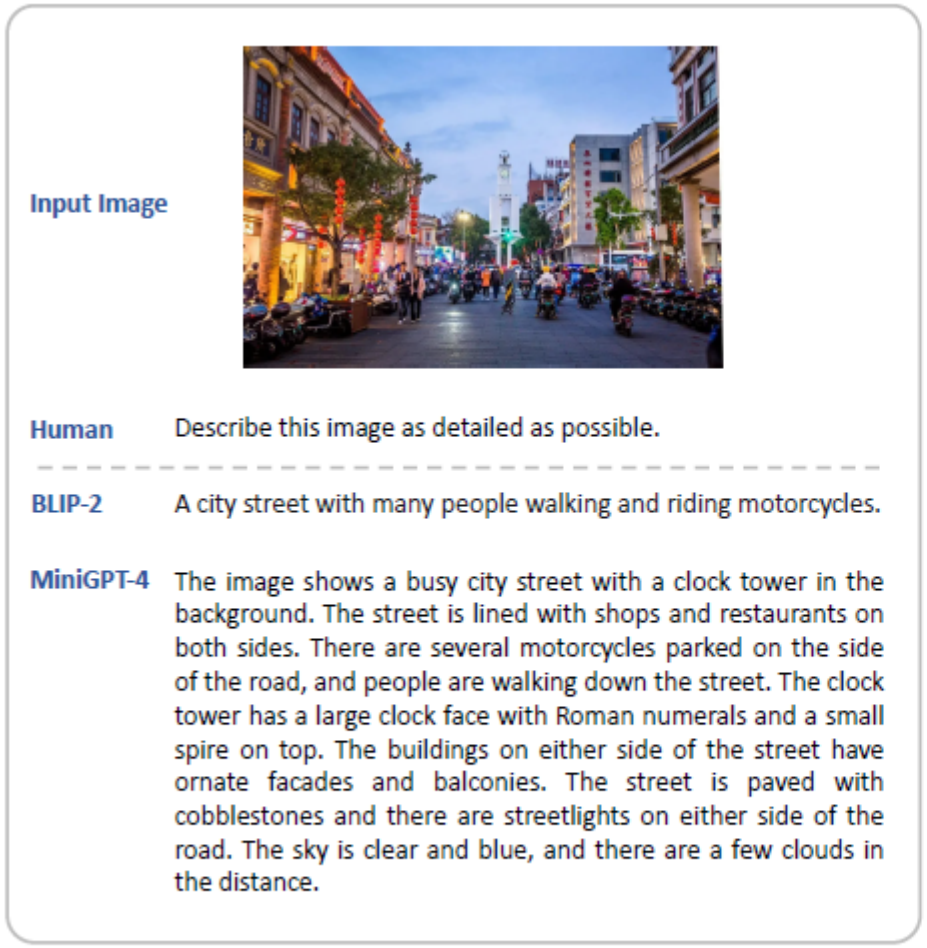
- 

Figure 2: Detailed description

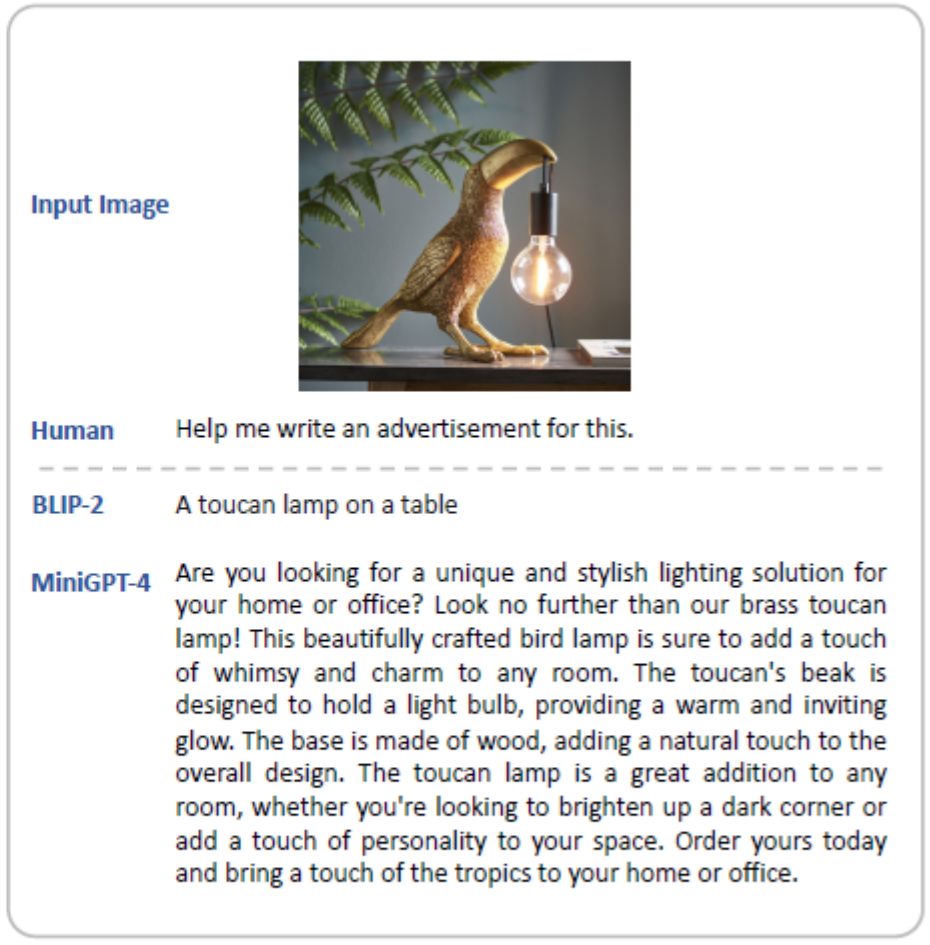
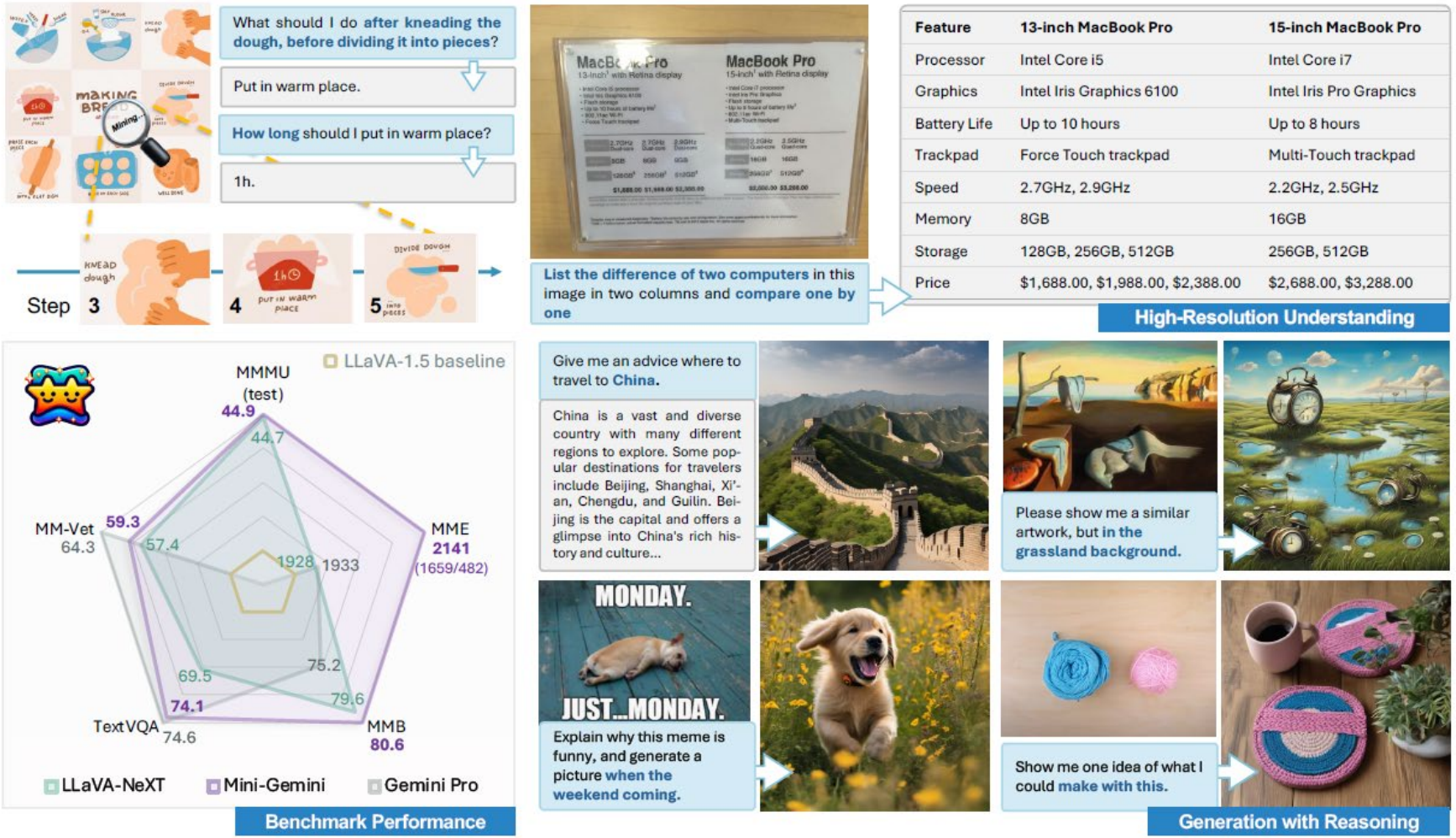


Figure 3: Advertisement promotion

Deyao Zhu et al. Mini-GPT4: MiniGPT-4 Enhancing Vision-Language Understanding with Advanced Large Language Models. ArXiv 2023.

Mini-Gemini



Feature	13-inch MacBook Pro	15-inch MacBook Pro
Processor	Intel Core i5	Intel Core i7
Graphics	Intel Iris Graphics 6100	Intel Iris Pro Graphics
Battery Life	Up to 10 hours	Up to 8 hours
Trackpad	Force Touch trackpad	Multi-Touch trackpad
Speed	2.7GHz, 2.9GHz	2.2GHz, 2.5GHz
Memory	8GB	16GB
Storage	128GB, 256GB, 512GB	256GB, 512GB
Price	$1,688.00, $1,988.00, $2,388.00	$2,688.00, $3,288.00

Figure 1: Mini-Gemini is advanced in various vision-related tasks.

Yanwei Li et al. Mini-Gemini: Mining the Potential of Multi-modality Vision Language Models. ArXiv 2024.

Mini-Gemini

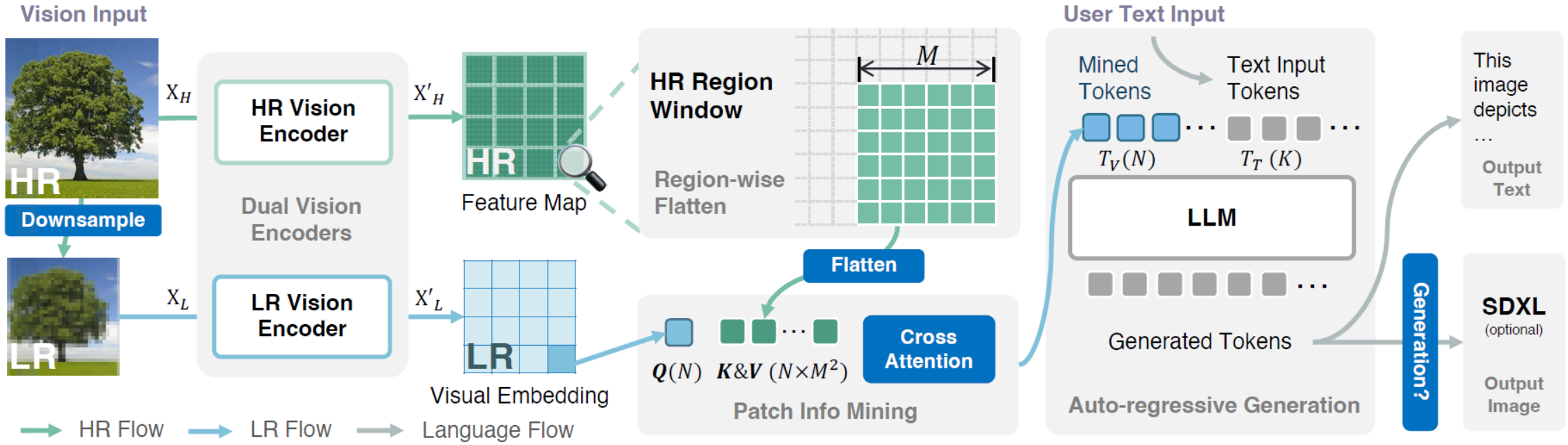


Figure 2: The framework of Mini-Gemini with any-to-any workflow.

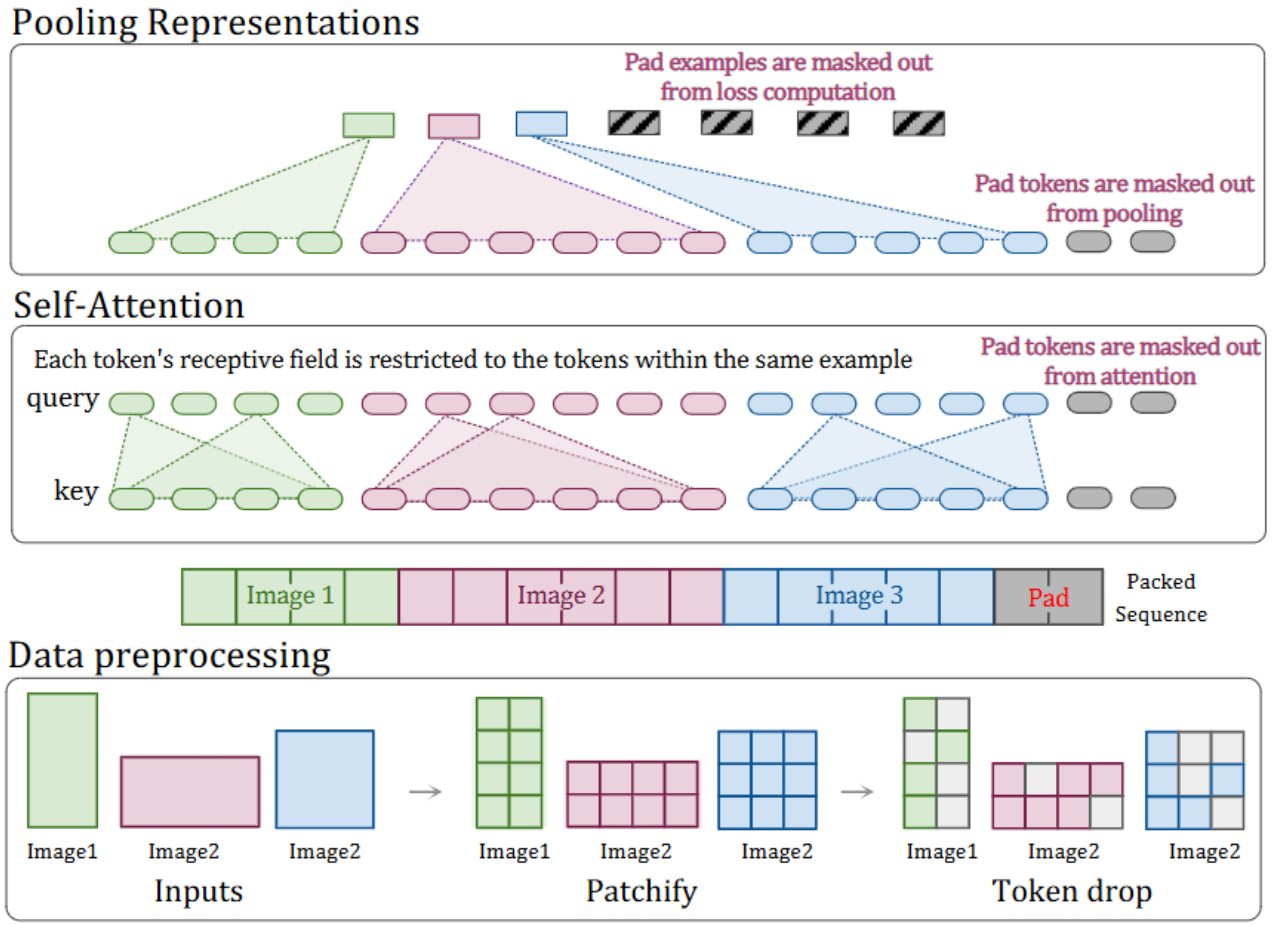
Yanwei Li et al. Mini-Gemini: Mining the Potential of Multi-modality Vision Language Models. ArXiv 2024.

Vision Language Models

- Vision-Language pretrained Model (MoEs)
  - VLMo, VL-BEIT, BEIT-3, Florence-2
- Instruction tuning with LLMs
  - BLIP-2, InstructBlip, LLaVA
- Multimodal Large Language Model
  - Qwen-VL, Qwen2-VL, Qwen3-VL
  - DeepSeek-VL
  - Kimi-VL

Preliminary

- **Native resolution of Vision Transformer: NaViT**



Patch n' Pack: NaViT, a Vision Transformerfor any Aspect Ratio and Resolution. 2023.

Preliminary

- Dynamic resolution （LLaVA-Next, AnyRes）

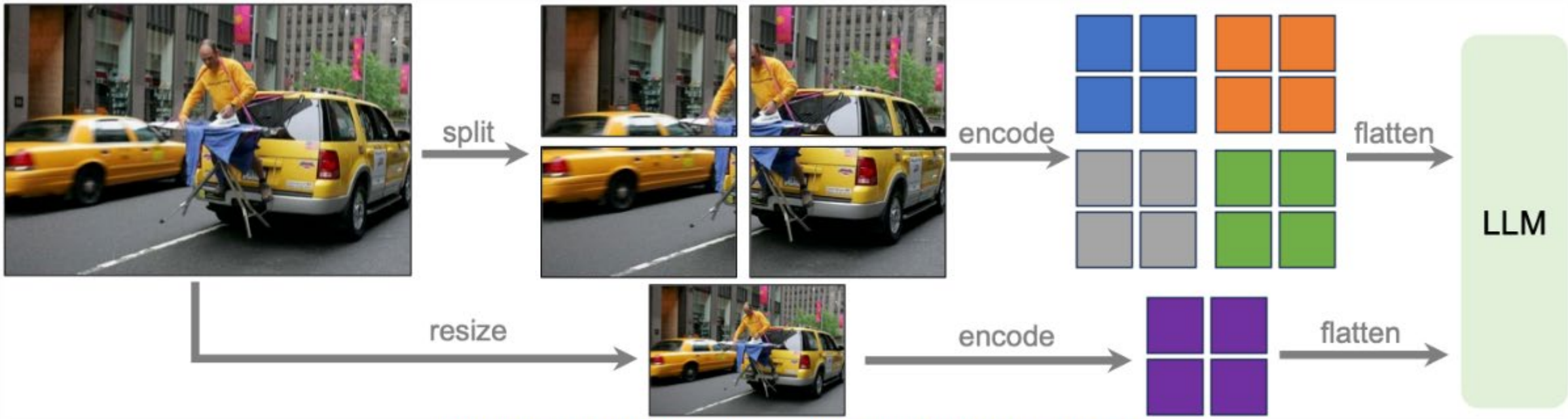


Illustration of dynamic high resolution scheme: a grid configuration of 2×2

grid configuration of {2×2,1×{2,3,4},{2,3,4}×1}

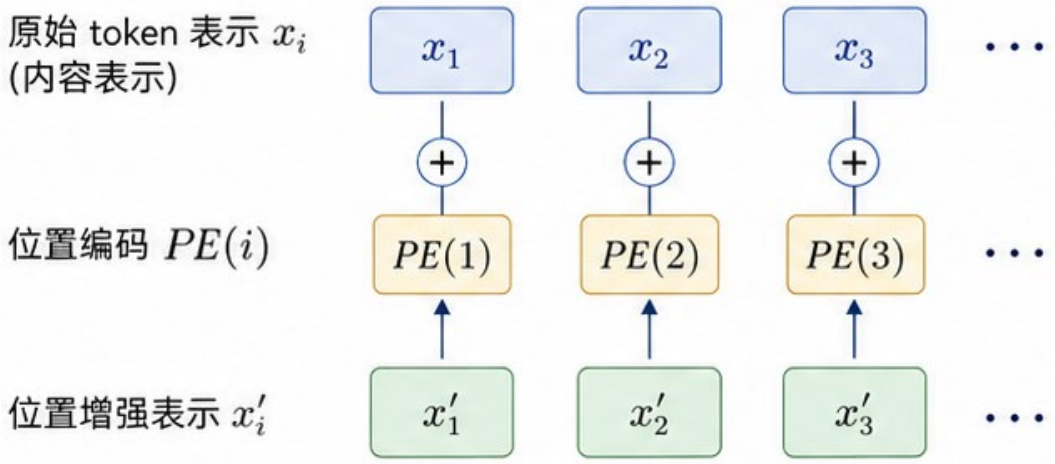
https://llava-vl.github.io/blog/2024-01-30-llava-next/

Preliminary

- **Positional Encoding – Absolute Positional Encoding**
  - **A sequence of N input tokens:** $\mathbb{S}_N = \{w_i\}_{i=1}^N$;
  - **Word embedding:** $\mathbb{E}_N = \{\boldsymbol{x}_i\}_{i=1}^N$
  - $\boldsymbol{q}_m = f_q(\boldsymbol{x}_m, m), \boldsymbol{k}_n = f_k(\boldsymbol{x}_n, n), \boldsymbol{v}_n = f_v(\boldsymbol{x}_n, n)$
  - $f_{t:t\in\{q,k,v\}}(x_i, i) := \boldsymbol{W}_{t:t\in\{q,k,v\}}(\boldsymbol{x}_i + \boldsymbol{p}_i)$

$$\begin{cases} \boldsymbol{p}_{i,2t} = sin\left(k/10000^{\frac{2t}{d}}\right) \\ \boldsymbol{p}_{i,2t+1} = cos\left(k/10000^{\frac{2t}{d}}\right) \end{cases}$$

$\boldsymbol{p}_{i,2t}$ is the $2t^{\text{th}}$ element of the d-dimensional vector $\boldsymbol{p}_i$



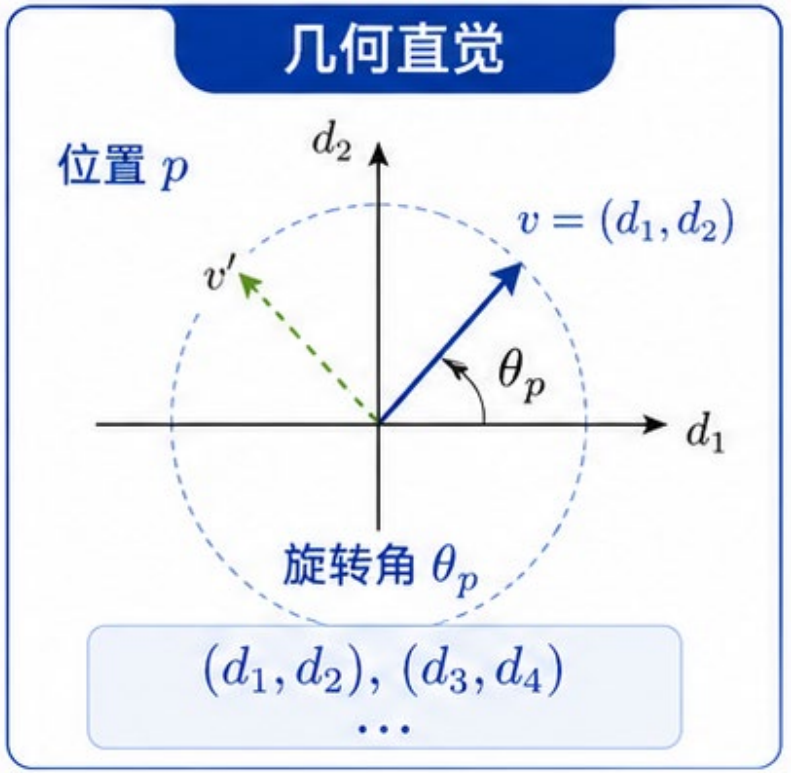
Preliminary

- **Positional Encoding – Rotary position embedding**
  - **A sequence of N input tokens: $\mathbb{S}_N = \{w_i\}_{i=1}^N$;**
  - **Word embedding: $\mathbb{E}_N = \{\boldsymbol{x}_i\}_{i=1}^N$**
  - $\boldsymbol{q}_m = f_q(\boldsymbol{x}_m, m), \boldsymbol{k}_n = f_k(\boldsymbol{x}_n, n), \boldsymbol{v}_n = f_v(\boldsymbol{x}_n, n)$
  - $\langle f_q(\boldsymbol{x}_m, m), f_k(\boldsymbol{x}_n, n)\rangle = g(\boldsymbol{x}_m, \boldsymbol{x}_n, m-n)$

$$f_q(\boldsymbol{x}_m, m) = (\boldsymbol{W}_q \boldsymbol{x}_m) e^{im\theta}$$

$$f_k(\boldsymbol{x}_n, n) = (\boldsymbol{W}_k \boldsymbol{x}_n) e^{in\theta}$$

$$g(\boldsymbol{x}_m, \boldsymbol{x}_n, m-n) = Re\left[(\boldsymbol{W}_q \boldsymbol{x}_m)(\boldsymbol{W}_k \boldsymbol{x}_n)^* e^{i(m-n)\theta}\right]$$



RoFormer: Enhanced Transformer with Rotary Position Embedding. arxiv.org/abs/2104.09864

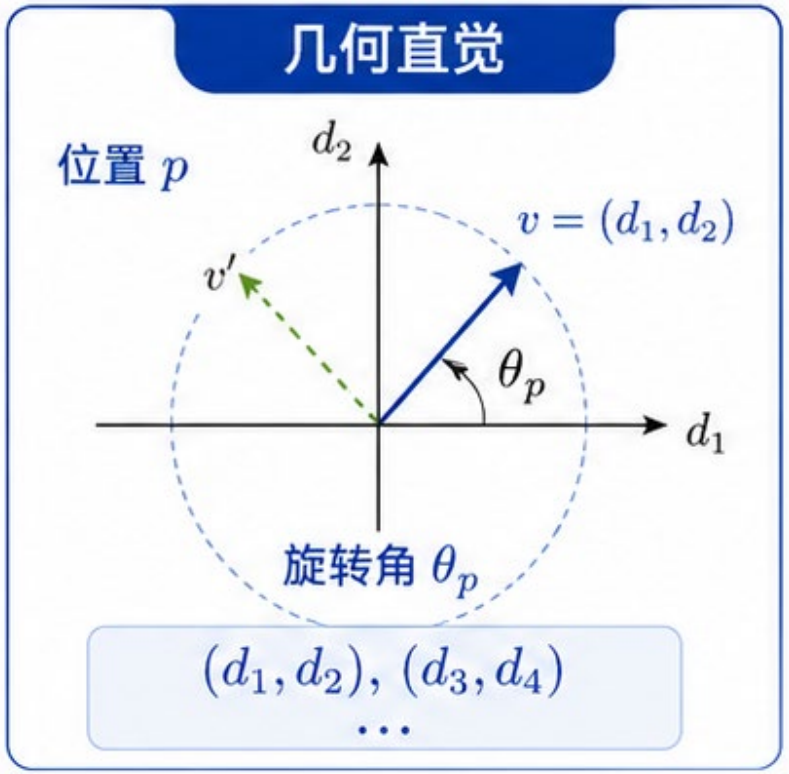
Preliminary

- **Positional Encoding – Rotary position embedding**
  - **A sequence of N input tokens: $\mathbb{S}_N = \{w_i\}_{i=1}^N$;**
  - **Word embedding: $\mathbb{E}_N = \{\boldsymbol{x}_i\}_{i=1}^N$**

1D RoPE

$$f_{\{q,k\}}(\boldsymbol{x}_m, m)$$

$$= \begin{pmatrix} \cos m\theta & -\sin m\theta \\ \sin m\theta & \cos m\theta \end{pmatrix} \begin{pmatrix} W_{\{q,k\}}^{(11)} & W_{\{q,k\}}^{(12)} \\ W_{\{q,k\}}^{(21)} & W_{\{q,k\}}^{(22)} \end{pmatrix} \begin{pmatrix} x_m^{(1)} \\ x_m^{(2)} \end{pmatrix}$$



RoFormer: Enhanced Transformer with Rotary Position Embedding. arxiv.org/abs/2104.09864

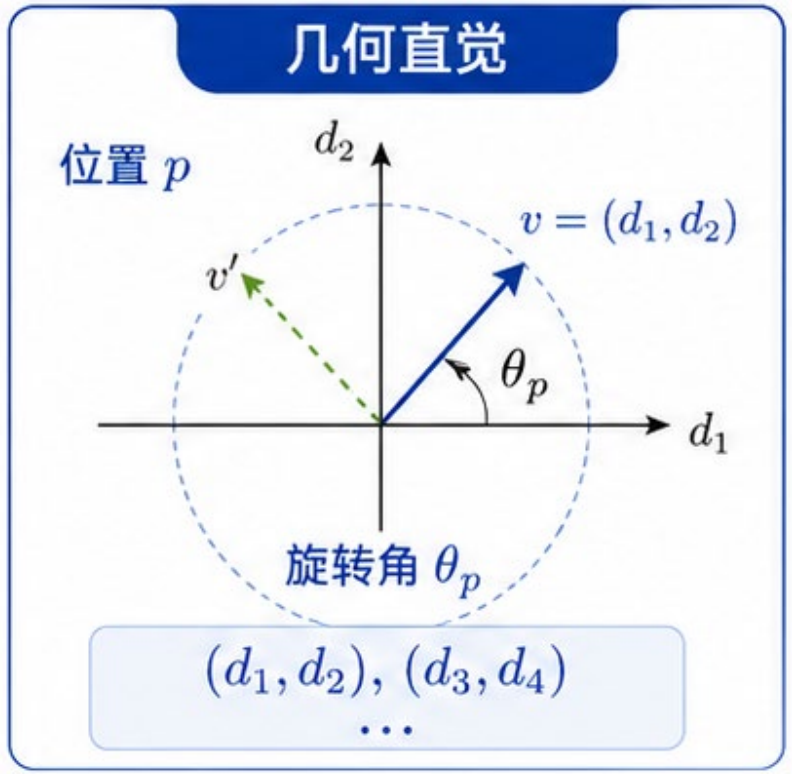
Preliminary

- **Positional Encoding – Rotary position embedding**
  - **A sequence of N input tokens: $\mathbb{S}_N = \{w_i\}_{i=1}^N$;**
  - **Word embedding: $\mathbb{E}_N = \{\boldsymbol{x}_i\}_{i=1}^N$**

2D RoPE

$$f_{\{q,k\}}(\boldsymbol{x}_m, m) = \boldsymbol{R}^d_{\Theta,m}\boldsymbol{W}_{\{q,k\}}\boldsymbol{x}_m$$

$$\boldsymbol{R}^d_{\Theta,m} = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \cdots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix}$$



RoFormer: Enhanced Transformer with Rotary Position Embedding. arxiv.org/abs/2104.09864

Preliminary

- **Positional Encoding – Rotary position embedding**
  - **A sequence of N input tokens: $\mathbb{S}_N = \{w_i\}_{i=1}^N$;**
  - **Word embedding: $\mathbb{E}_N = \{\boldsymbol{x}_i\}_{i=1}^N$**

2D RoPE

$$\boldsymbol{q}_m^T \boldsymbol{k}_n = \left(\boldsymbol{R}_{\Theta,m}^d \boldsymbol{W}_q \boldsymbol{x}_m\right)^T \left(\boldsymbol{R}_{\Theta,n}^d \boldsymbol{W}_k \boldsymbol{x}_n\right) = \boldsymbol{x}_m^T \boldsymbol{W}_q \boldsymbol{R}_{\Theta,n-m}^d \boldsymbol{W}_k \boldsymbol{x}_n$$

$$\Theta = \{\theta_i = 10000^{-\frac{2(i-1)}{d}}, i \in [1,2,\dots,d/2]\}$$

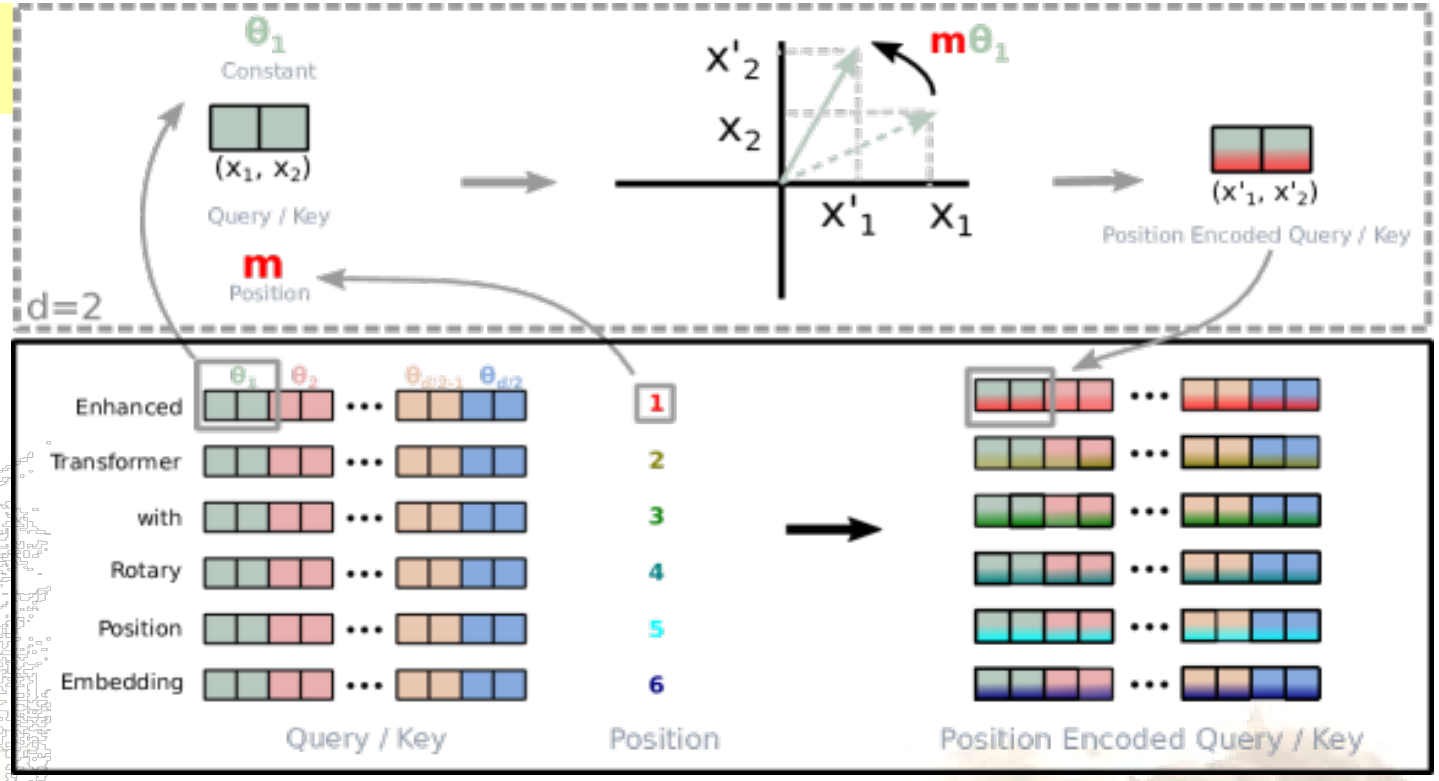
$$\boldsymbol{R}_{\Theta,n-m}^d = \left(\boldsymbol{R}_{\Theta,m}^d\right)^T \boldsymbol{R}_{\Theta,n}^d$$

RoFormer: Enhanced Transformer with Rotary Position Embedding. arxiv.org/abs/2104.09864

Preliminary

- **Positional Encoding – Rotary position embedding**
  - **A sequence of N input tokens: $\mathbb{S}_N = \{w_i\}_{i=1}^N$;**
  - **Word embedding: $\mathbb{E}_N = \{\boldsymbol{x}_i\}_{i=1}^N$**

2D RoPE



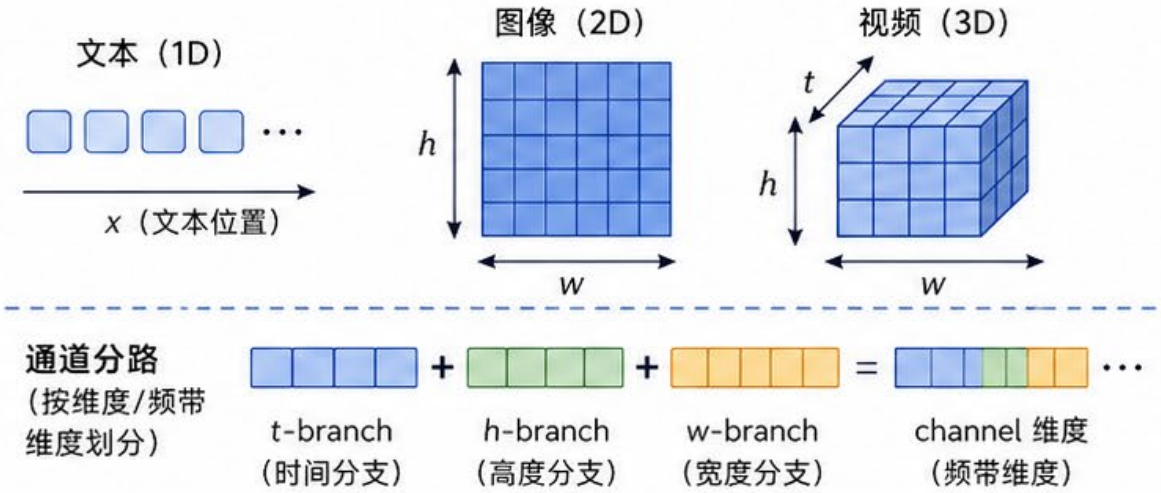
RoFormer: Enhanced Transformer with Rotary Position Embedding. arxiv.org/abs/2104.09864

Preliminary

- **Positional Encoding – M-RoPE, Interleave MRoPE**

M-RoPE

- M-RoPE 可看成 2D-RoPE 的进一步推广
- 文本是 1D，图像是 2D，视频是 3D
- 把通道分为三路，分别对应 t、h、w



Interleave-MRoPE

- 普通 MRoPE 的频带分配

freq dims:	0	1	2	3	4	5	6	7	8	9	10	11
axis:	T	T	T	T	H	H	H	H	W	W	W	W

⚠ **问题：** 每个轴只拿到一部分频带，而不是完整的低频到高频范围。

- Qwen3-VL 的 Interleave-MRoPE

freq dims:	0	1	2	3	4	5	6	7	8	9	10	11
axis:	T	H	W	T	H	W	T	H	W	T	H	W

- T / H / W 交错分配频带
- 每个轴都能更均衡地覆盖低频到高频范围

ℹ 视频时间还可以进一步显式映射为文本时间戳。

RoFormer: Enhanced Transformer with Rotary Position Embedding. arxiv.org/abs/2104.09864

Qwen VL 系列

Qwen-VL	预训练ViT-bigG 绝对位置编码 固定分辨率	单层cross-attention 加入绝对位置编码	Qwen	
Qwen2-VL	重新训练ViT: 2D RoPE 压缩图像token 动态分辨率	两层MLP	Qwen2 位置编码：MRoPE	三阶段训练
Qwen2.5-VL	重新训练ViT: 2D RoPE 交错Window-Attention RMSNorm + SwiGLU 动态分辨率	两层MLP LayerNorm换位RMSNorm	Qwen2.5 位置编码：MRoPE, 视频输入对齐了绝对时间 从视觉理解走向 visual agent，可以在手机和电脑场景里进行 reasoning、tool use、task execution	三阶段训练
Qwen3-VL	Interleaved-MRoPE	DeepStack：跨层视觉注入	原生 256K 的 interleaved text-image-video context 视频时间对齐从“位置编码”转向“显式文本时间戳” 采用 **square-root reweighting** 去平衡 text-only 和 multimodal learning objectives，并在后训练阶段拆分成 **non-thinking** 和 **thinking** 两个分支	
Qwen3.5-VL			从 late fusion/adapter 思路，走向更原生的 early fusion 架构从 Transformer-only，走向 hybrid attention 把 **线性注意力（通过 Gated Delta Networks）** 和 **sparse MoE** 结合起来	Agent 训练被正式提升到核心地位

Qwen VL系列

视觉输入表示：从固定图像到原生动态视觉序列

- Qwen-VL：固定分辨率图像，视觉特征压缩成固定长度序列
- Qwen2-VL：动态分辨率，视觉 token 数可变
- Qwen2.5-VL：native dynamic-resolution ViT + Window Attention，图像与视频都能以更自然方式进入模型
- Qwen3-VL：继续保留动态视觉输入，但把它纳入 256K interleaved 序列
- Qwen3.5：进一步朝 early fusion 原生多模态序列建模推进。

Qwen VL系列

清華大學 Tsinghua University

视觉—语言桥接：从 adapter 到更深层融合

- Qwen-VL：cross-attention adapter，把视觉压成 256 token
- Qwen2-VL / 2.5-VL：更轻量的 MLP merger，配合动态 token 序列
- Qwen3-VL：不只 merger，还用 DeepStack 把多层视觉特征注入 LLM 多层
- Qwen3.5：更强调 early fusion，而不是简单后接 merger

Qwen VL系列

位置与时间建模：从 2D 位置到统一时空编码

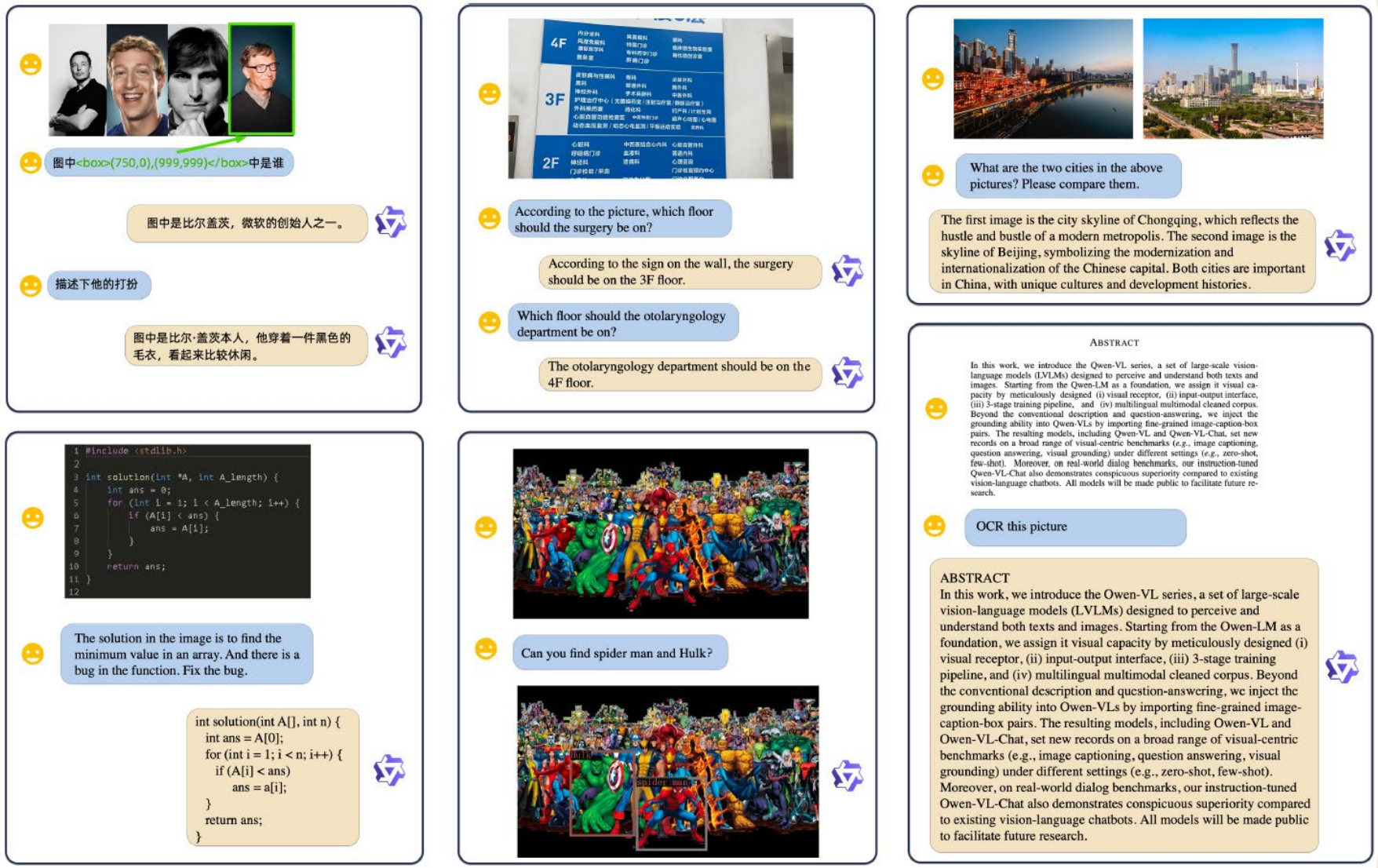
- Qwen-VL：adapter 中显式保留 2D 位置信息
- Qwen2-VL：M-RoPE 统一 text/image/video 的 t-h-w 三轴
- Qwen2.5-VL：时间轴与真实 absolute time 对齐
- Qwen3-VL：Interleaved-MRoPE 修正频谱不均衡，并把视频时间换成显式文本时间戳

Qwen VL系列

训练目标：从“看图对话”到“文档/视频/Agent/Reasoning 一体化”

- Qwen-VL：caption、VQA、OCR、grounding、多图对话
- Qwen2-VL：加入统一图像/视频理解与 scaling
- Qwen2.5-VL：文档解析、长视频、GUI、agent、point grounding
- Qwen3-VL：再叠加 long-context、thinking model、tool-integrated RL
- Qwen3.5：进一步把 agent RL 和 unified multimodal foundation 放在核心位置

Qwen VL



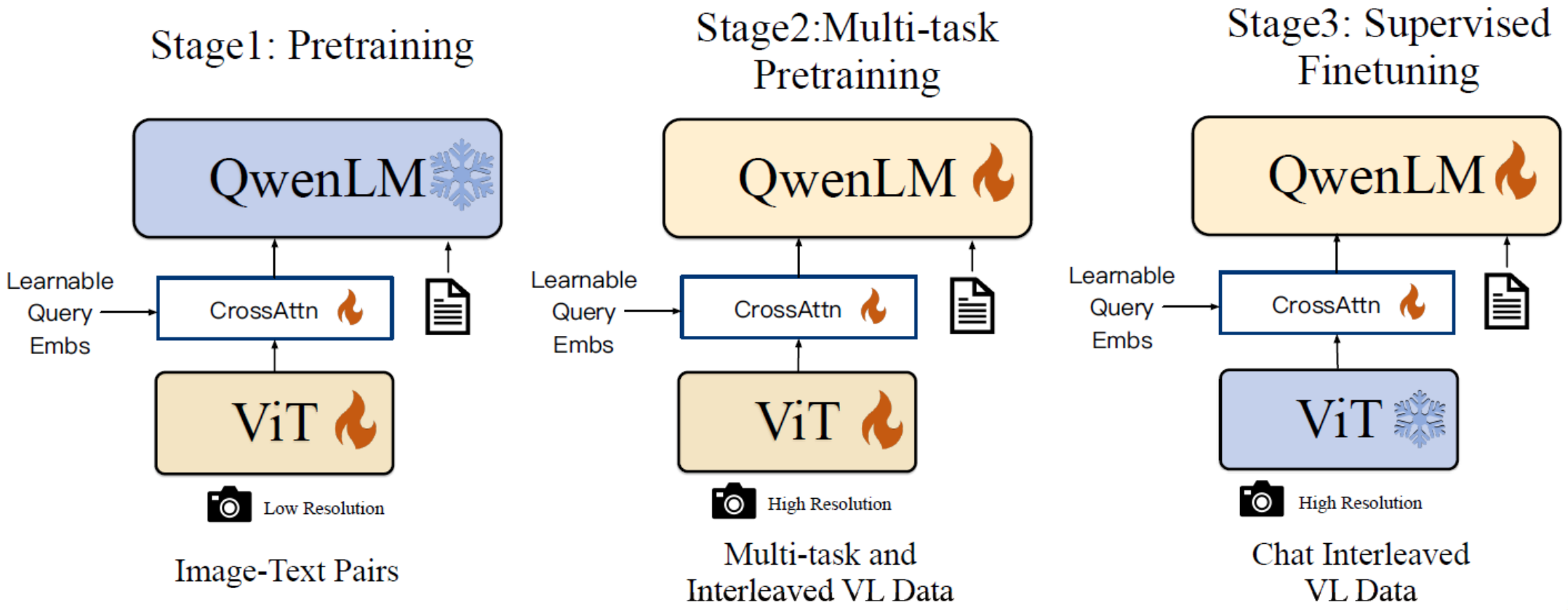
Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond. 2023.08

Qwen VL

- **Large Language Model:** The model is initialized with pre-trained weights from Qwen-7B.
- **Visual Encoder:** The visual encoder of Qwen-VL uses ViT architecture, initialized with pre-trained weights from Openclip's ViT-bigG. During both training and inference, input images are resized to a specific resolution. The visual encoder processes images by splitting them into patches with a stride of 14, generating a set of image features.
- **Position-aware Vision-Language Adapter:** This adapter comprises a single-layer cross-attention module initialized randomly. The module uses a group of trainable vectors (Embeddings) as query vectors and the image features from the visual encoder as keys for cross-attention operations.

Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond. 2023.08

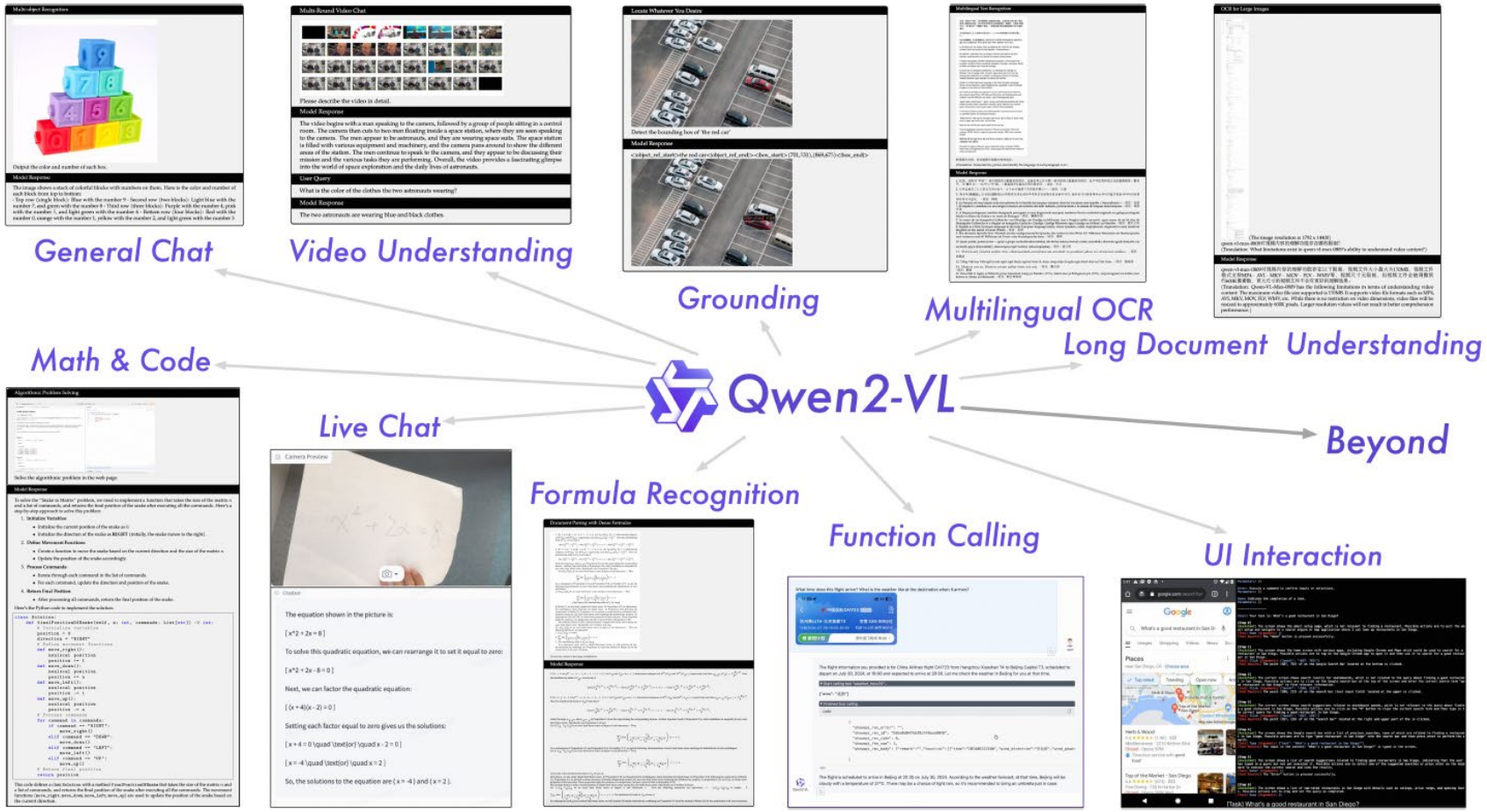
Qwen VL



Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond. 2023.08

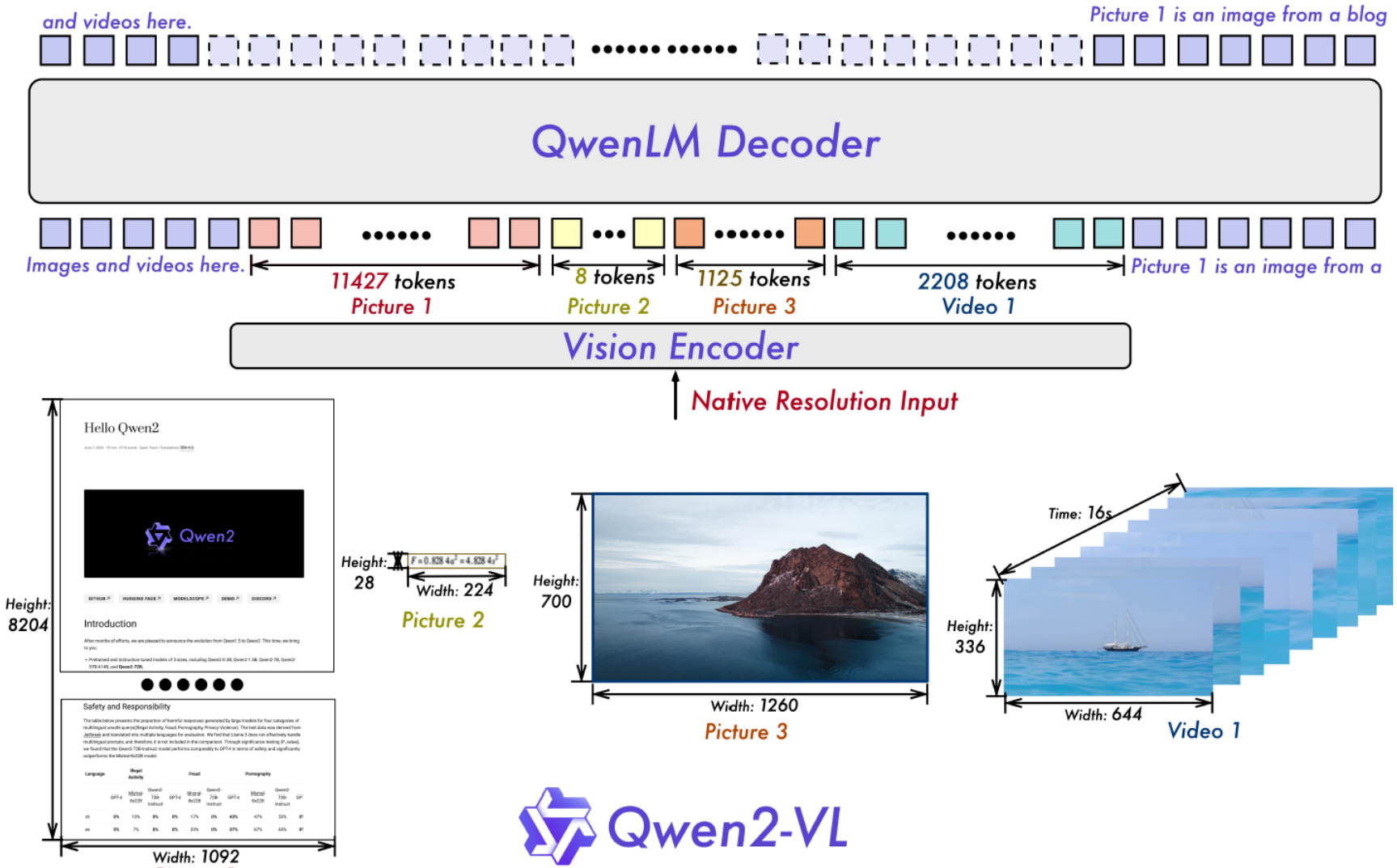
Qwen2 VL

- **Qwen2-VL capabilities: Multilingual image text understanding, code/math reasoning, video analysis, live chat, agent potential, and more**



Qwen2-VL: Enhancing Vision-Language Model's Perception of theWorld at Any Resolution. 2024.10

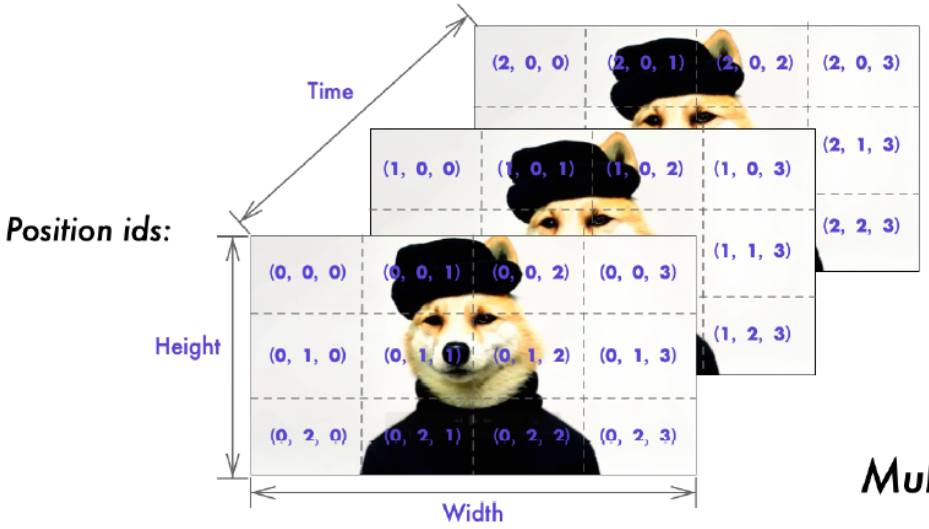
Qwen2 VL

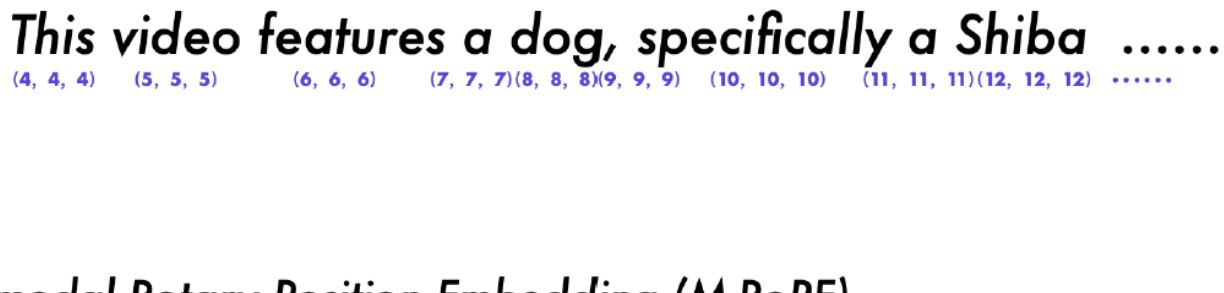


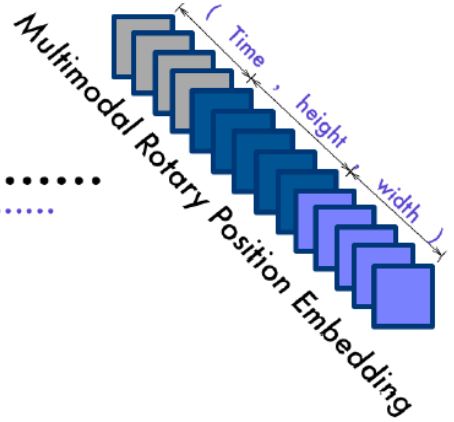
Qwen2-VL: Enhancing Vision-Language Model's Perception of theWorld at Any Resolution. 2024.10

Qwen2 VL

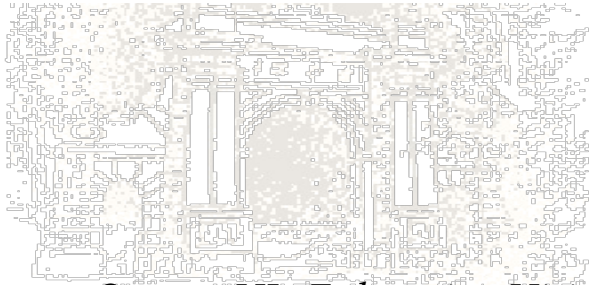
- **Multimodal Rotary Position Embedding (M-RoPE)**





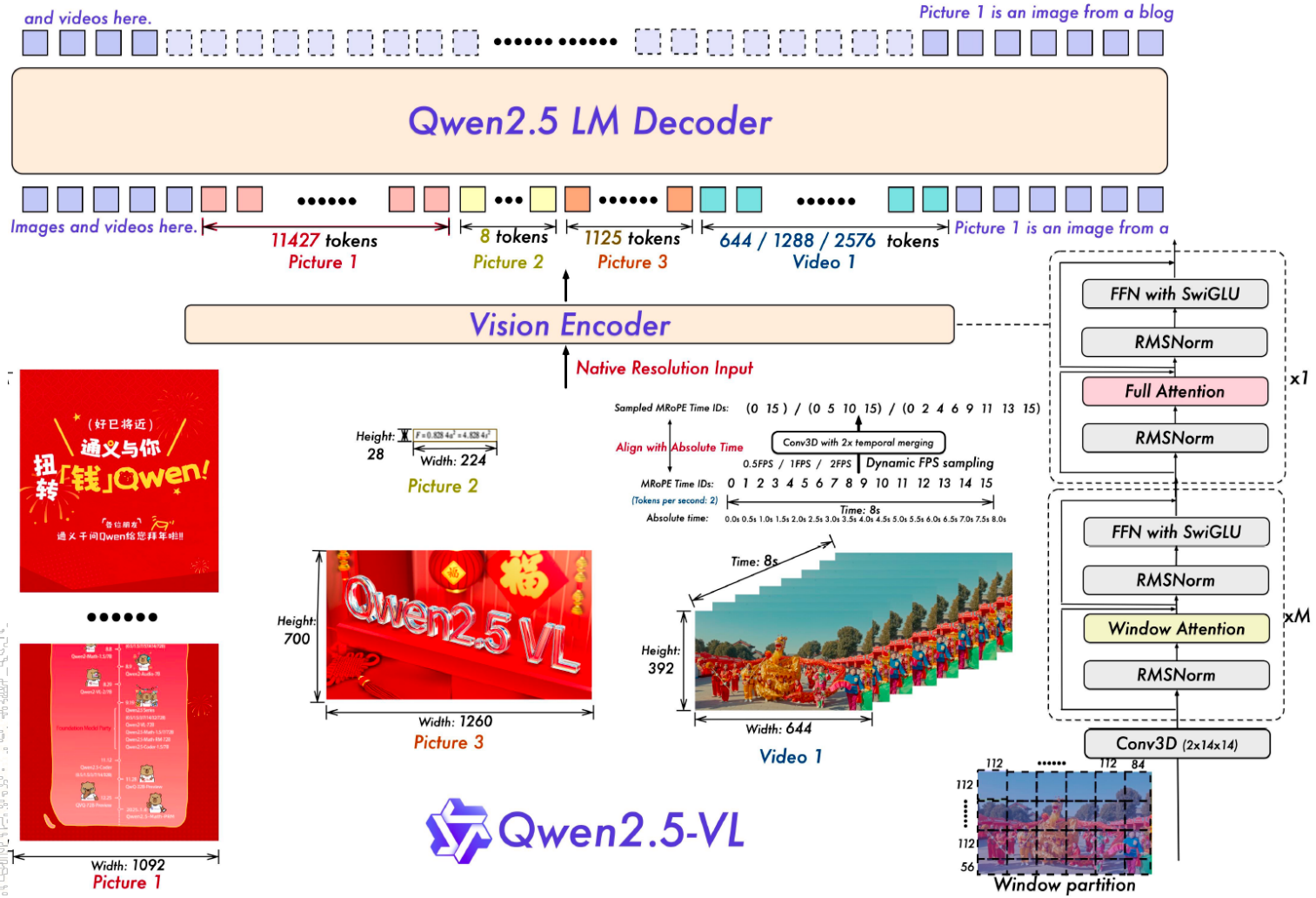


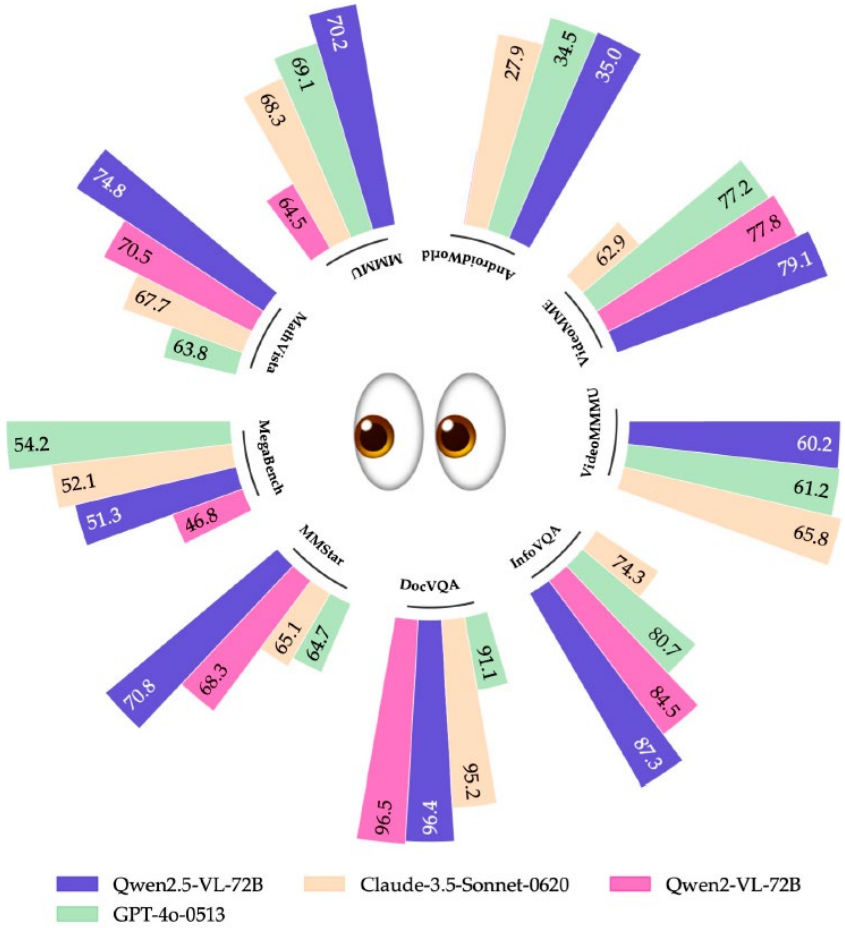
Multimodal Rotary Position Embedding (M-RoPE)



Qwen2-VL: Enhancing Vision-Language Model's Perception of theWorld at Any Resolution. 2024.10

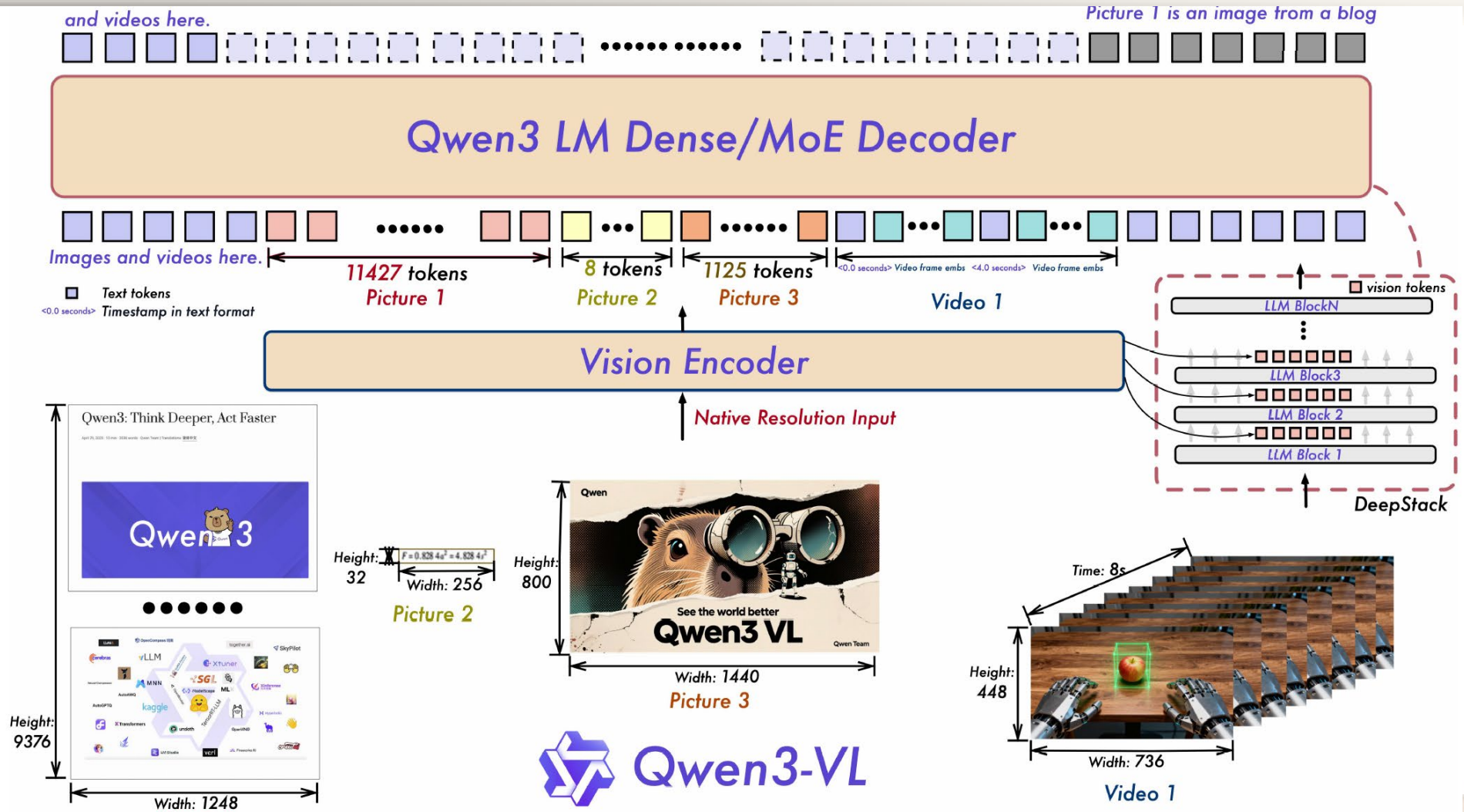
Qwen2.5 VL





Qwen2.5-VL Technical Report 2025.3

Qwen3 VL



Qwen3-VL Technical Report 2025.12

Qwen3 VL

- **Model Architecture**
  - Large Language Model: Qwen3-VL is instantiated in three dense variants (Qwen3-VL-2B/4B/8B/32B) and two MoE variants (Qwen3-VL-30B-A3B, Qwen3-VL-235B-A22B), all built upon Qwen3 backbones. The flagship model, Qwen3-VL-235B-A22B, has 235B total parameters with 22B activated per token.
  - Vision Encoder: We utilize the SigLIP-2 architecture (Tschannen et al., 2025) as our vision encoder and continue training it with dynamic input resolutions, initialized from official pretrained checkpoints. To accommodate dynamic resolutions effectively, we employ 2D-RoPE and interpolate absolute position embeddings based on input size, following the methodology of CoMP (Chen et al., 2025). Specifically, we default to the SigLIP2-SO-400M variant and use SigLIP2-Large (300M) for small-scale LLMs (2B and 4B).
  - MLP-based Vision-Language Merger: As in Qwen2.5-VL, we use a two-layer MLP to compress 2 × 2 visual features from the vision encoder into a single visual token, aligned with the LLM's hidden dimension.

Qwen3 VL

- **Pre-Training (multi-stage)**

Stage	Objective	Training	Token Budget	Sequence Length
S0	Vision-Language Alignment	Merger	67B	8,192
S1	Multimodal Pre-Training	All	~1T	8,192
S2	Long-Context Pre-Training	All	~1T	32,768
S3	Ultra-Long-Context Adaptation	All	100B	262,144

Qwen3-VL Technical Report 2025.12

Qwen3 VL

- Post-Training
  - Supervised Fine-Tuning (SFT).
    - The first stage imparts instruction-following abilities and activates latent reasoning skills. This is conducted in two phases: an initial phase at a 32k context length, followed by an extension to a 256k context window that focuses on long-document and long-video data.
    - Bifurcate the training data into standard formats for non-thinking models and Chain-of-Thought (CoT) formats for thinking models, the latter of which explicitly models the reasoning process.
  - Strong-to-Weak Distillation.
    - perform this distillation using text-only data to fine-tune the LLM backbone.
  - Reinforcement Learning (RL).
    - This phase is divided into Reasoning RL and General RL. We apply large-scale reinforcement learning across a comprehensive set of text and multimodal domains, including but not limited to math, OCR, grounding, and instruction-following, to improve finer-grained capabilities.

Qwen3-VL Technical Report 2025.12

Deepseek VL

Prompt

Is the cyclist on the left or right side of the woman's handbag?



Response

The cyclist is on the left side of the woman's handbag. This can be determined by observing the relative positions of the objects in the image:
1. The woman is holding a handbag that is positioned to her right side, as we view the image.
2. There is a cyclist depicted on the road behind the woman and to her left when looking at the image.
3. The cyclist appears to be moving away from the viewpoint, given their direction and the motion blur effect on their wheels
4. The cyclist is further down the road compared to the woman, which supports the conclusion that they are on the left side of the handbag based on our perspective.
These visual clues confirm that the cyclist is indeed con the left side of the woman's handbag.

Stage 1: Training VL Adaptor

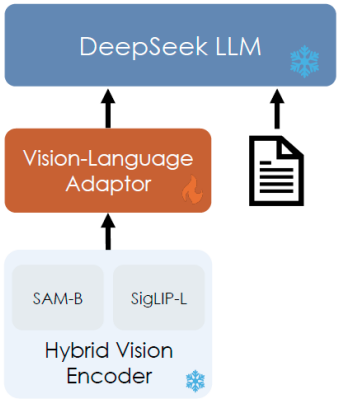
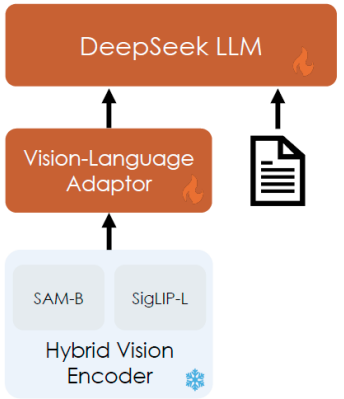


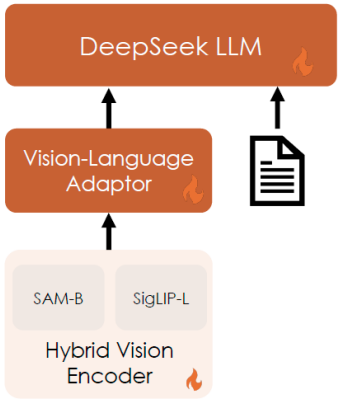
Image-Text Pairs

Stage 2: Joint VL Pre-training



Interleaved VL +
Pure Language Sequences

Stage 3: Supervised Finetuning

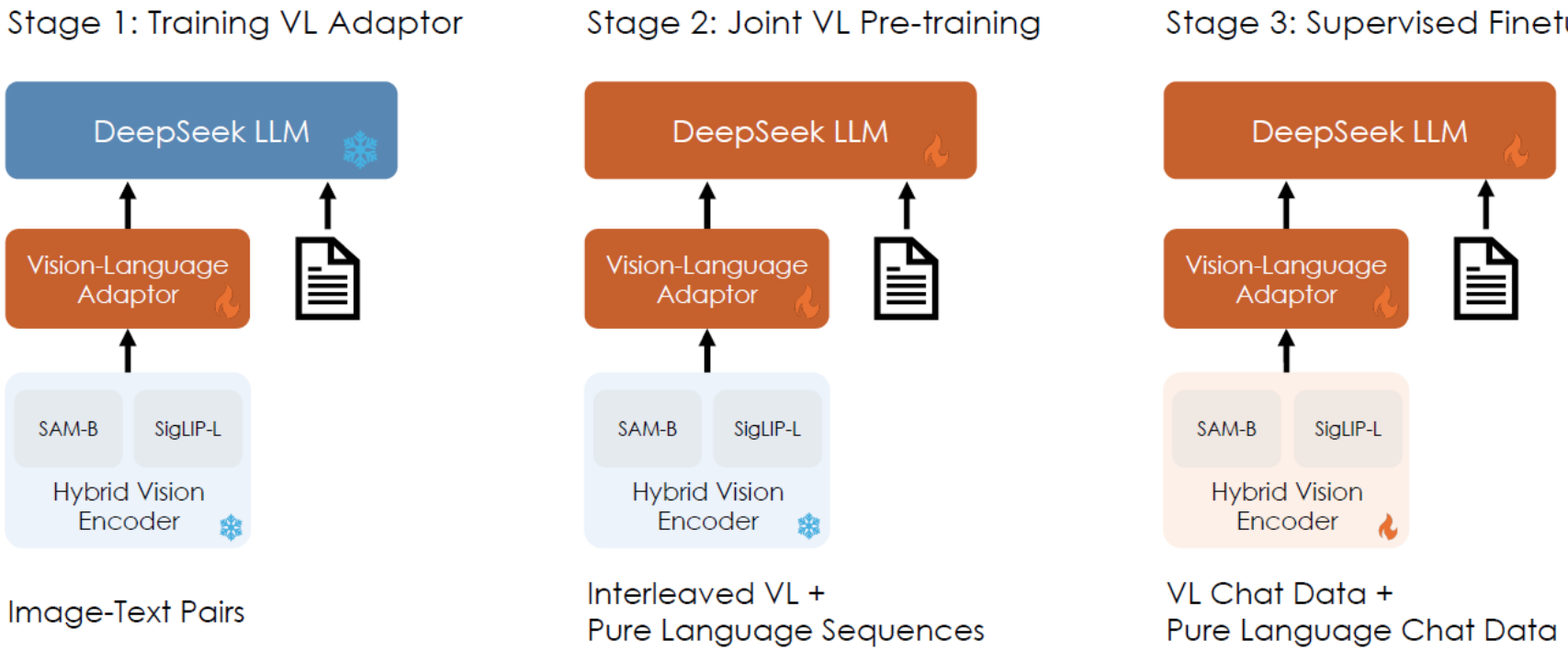


VL Chat Data +
Pure Language Chat Data

DeepSeek-VL: Towards Real-World Vision-Language Understanding. 2024.3

Deepseek VL

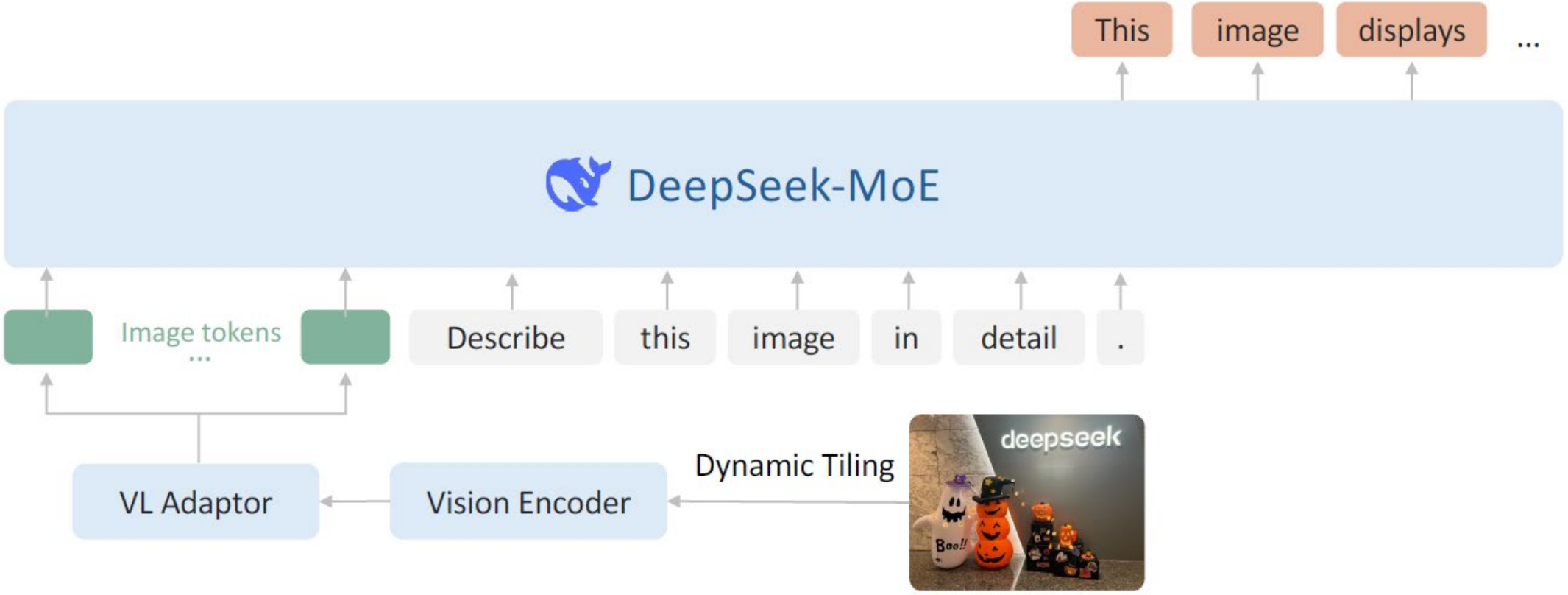
- Stage 1 involves training the Vision-Language (VL) adaptor while keeping the hybrid vision encoder and language model fixed.
- Stage 2 is the crucial part of the joint vision and language pretraining, where both VL adaptor and language model are trainable.
- Stage 3 is the supervised fine-tuning phase, during which the low-resolution vision encoder SigLIP-L, VL adaptor, and language model will be trained.



DeepSeek-VL: Towards Real-World Vision-Language Understanding. 2024.3

Deepseek VL2

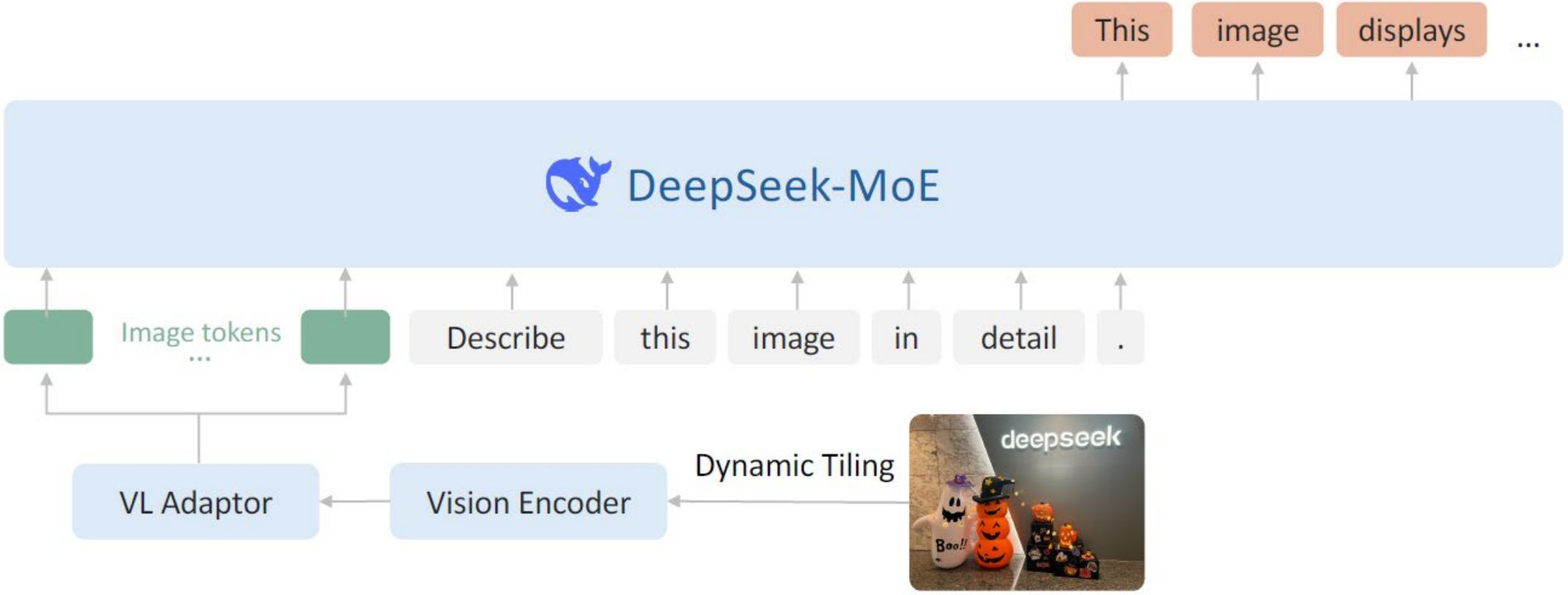
- **Deepseek-VL2: llava-style architecture, which includes a vision encoder, a VL adaptor, and a MoE-based LLM.**



DeepSeek-VL2: Mixture-of-Experts Vision-Language Models for Advanced Multimodal Understanding. 2024.12

Deepseek VL2

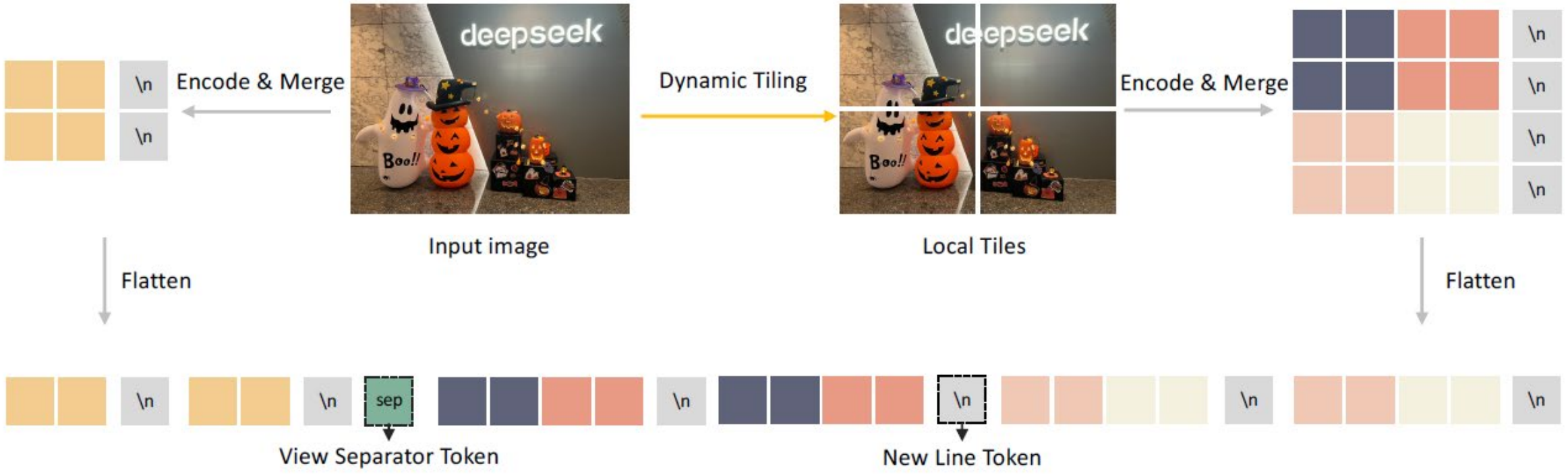
- **Deepseek-VL2: llava-style architecture, which includes a vision encoder, a VL adaptor, and a MoE-based LLM.**



DeepSeek-VL2: Mixture-of-Experts Vision-Language Models for Advanced Multimodal Understanding. 2024.12

Deepseek VL2

- **Dynamic tiling strategy: dividing images into multiple tiles**



DeepSeek-VL2: Mixture-of-Experts Vision-Language Models for Advanced Multimodal Understanding. 2024.12

Deepseek VL2

Tsinghua University

- **Architectural configuration**

	DeepSeek-VL2-Tiny	DeepSeek-VL2-Small	DeepSeek-VL2
Vocabulary size	129,280	102,400	129,280
Embedding size	1,280	2,048	2,560
#Attention heads	10	16	32
#Layers	12	27	30
Attention	Multi-Head Attention	MLA (rank=512)	MLA (rank=512)
#Routed experts	64	64	72
#Shared experts	2	2	2
Top-K for expert selection	6	6	6
Routing function	Softmax	Softmax	Sigmoid
Expert correction bias	×	×	✓

DeepSeek-VL2: Mixture-of-Experts Vision-Language Models for Advanced Multimodal Understanding. 2024.12

Deepseek VL2

- Training pipelines.
  - (1) an initial stage where we train the vision encoder and vision-language adaptor MLP while keeping the language model fixed, using image-text paired data
  - (2) a pretraining stage where we conduct vision-language pre-training using the data
  - (3) a fine-tuning stage where we perform supervised fine-tuning with the data.

DeepSeek-VL2: Mixture-of-Experts Vision-Language Models for Advanced Multimodal Understanding. 2024.12

Deepseek VL2

清華大學 Tsinghua University

- General question answering capability.

Image Description



Describe this scene in detail.

An expansive coastal landscape dominated by towering limestone stacks rising dramatically from turquoise waters of the ocean. These natural formations are part of The Twelve Apostles geological formation located on the Great Ocean Road in Victoria, Australia. In the foreground stands a prominent stack closest to the viewer's perspective; it is flanked by smaller stacks extending into the distance towards the horizon where they gradually diminish due to atmospheric haze or mist shrouding them partially. To the right side of the frame lies a sandy beach that curves gently around as if embracing the rugged cliffs which form the backdrop for this scene. The sky above appears overcast yet diffused sunlight illuminates parts of the cliff faces casting subtle shadows across some areas while highlighting others, adding depth and texture to the overall composition. Small waves break against both the base of the stacks and nearer shoreline creating white foam patterns contrasting sharply with the deep blue-green sea water below. Vegetation can also be seen at various points including atop certain stacks providing hints of greenery amidst predominantly earth-toned rock structures. This serene but powerful seascape captures nature's grandeur through its interplay between landforms, water bodies, light conditions, and vegetation all harmoniously coexisting within the same visual space.

Landmark Recognition

这幅图描绘了一个湖边的日落景象,周围有山峦和一些停泊在水面的船只。特别显眼的是中间的一艘大船,船上有着类似中国传统建筑风格的屋檐。这种建筑风格非常像中国古代建筑,尤其是皇家或寺庙建筑。

这种场景可能位于中国的某个著名湖泊,例如杭州的西湖或南京的玄武湖等。特别是如果这艘船是仿古式的游船,那么它很可能位于西湖景区。西湖是中国最著名的湖泊之一,周围有着秀丽的山景和古建筑,常常成为游客观光和体验传统文化的地方。

因此,图片中展示的可能是中国的西湖,那个有着丰富的历史文化和自然美景的地方。



猜猜这是哪里?

General Knowledge



Can you introduce the background story of this cake?

The cake in the image is themed around the iconic cartoon series "Tom and Jerry." This series, created by William Hanna and Joseph Barbera, first aired in 1940 and became one of the most famous animated shows of all time. The story revolves around the never-ending chase between the cunning Tom the Cat and Jerry the Mouse. Each episode typically features Tom making various plans to catch Jerry, only to be outsmarted or thwarted by Jerry's cleverness and cunning. The dynamic duo is a symbol of constant rivalry and humor, and their interactions have been a part of popular culture for decades. This cake captures the essence of their playful and chaotic relationship through its design and decoration.

Poet Writing

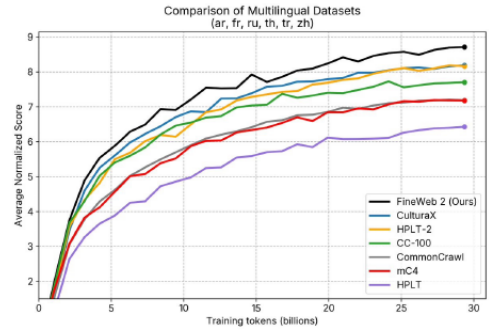


请给这幅景象写一首诗

《望公山》

浮云出岫晓轻寒,
郁郁苍山展画屏。
几树孤影来绿野,
一泓深水抱金城。
鸿鸣远岫千山起,
鹤伴闲居万物兴。
尽望烟霞山色秀,
却寻春色过江亭。

Chart Understanding



Can you explain this chart to me?

The chart shows a comparison of multilingual datasets based on the average normalized score across six languages: Arabic (ar), French (fr), Russian (ru), Thai (th), Turkish (tr), and Chinese (zh). The x-axis represents the number of training tokens in billions, while the y-axis shows the average normalized score, which is a measure of model performance.

Key Points:

1. **Datasets Compared**:
   - FineWeb 2 (Ours)
   - CulturaX
   - HPLT-2
   - CC-100
   - CommonCrawl
   - mC4
   - HPLT
2. **Trend Analysis**:
   - FineWeb 2 consistently achieves the highest average normalized score, indicating superior performance compared to other datasets across the given range of training tokens.
   - The other datasets show varying levels of performance but tend to converge towards a similar score as the number of training tokens increases.
3. **Performance Insights**:
   - Initially, FineWeb 2 has a significant advantage over other datasets, which highlights the effectiveness of the data collected in this dataset.
   - As training tokens increase, the performance gap between datasets narrows, suggesting diminishing returns in terms of dataset size.
   - FineWeb 2 demonstrates robustness, maintaining a lead over the other datasets throughout the training process.

Interpretation:

The chart underscores the importance of dataset quality and size in achieving high model performance in multilingual tasks. The superior results of FineWeb 2 indicate its potential to enhance language model training for various applications across multiple languages.

Text Recognition



What's written on the large black board.

Serving Soul Since Twenty Twelve



图中在宣传什么东西?

图中展示的是与西湖醋鱼相关的食品和饮品。左边的广告牌上展示的是西湖醋鱼蛋糕（West Lake Fish Cake），它是一种以西湖醋鱼为主题的食品。右边的广告牌则是在宣传西湖醋鱼咖啡，这种咖啡被称为"铁狗咖啡"，其特色是将西湖醋鱼的特色和杭州当地的经验结合在一起，把西湖醋鱼的味道融入到咖啡中。

西湖醋鱼是杭州的一道著名美食，主要使用草鱼制作，经过料酒、醋、糖等调料烹饪，酸甜可口。现在通过将西湖醋鱼加入到其他食品和饮品中，这一传统美食得到了全新的诠释。

DeepSeek-VL2: Mixture-of-Experts Vision-Language Models for Advanced Multimodal Understanding. 2024.12

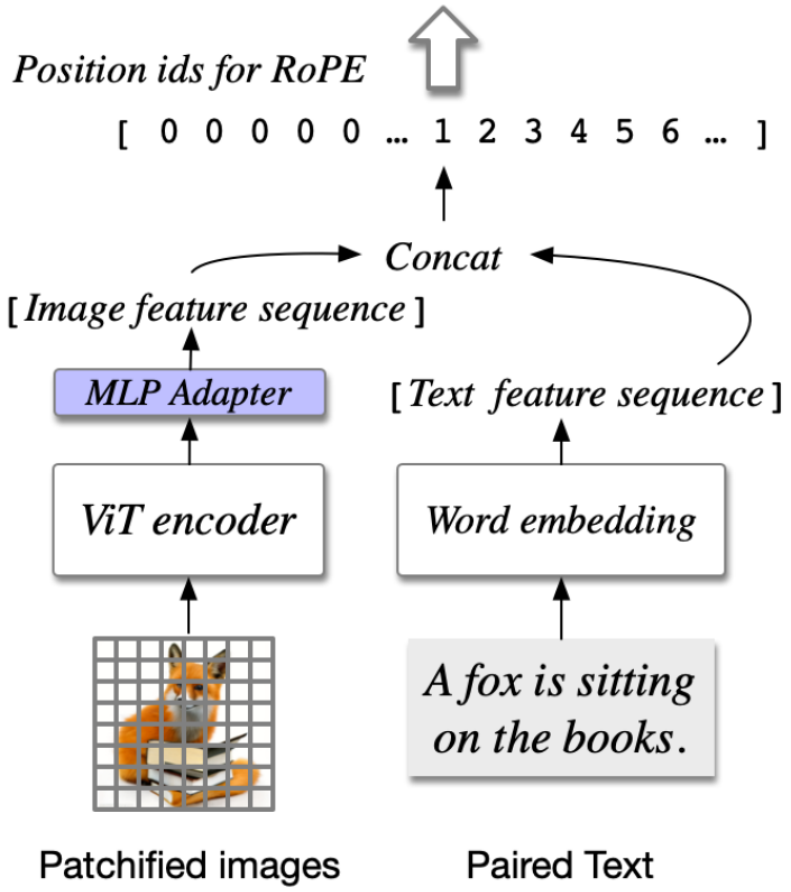
CogVLM





Wang et al. CogVLM: Visual Expert for Pretrained Language Models. NeurIPS 2024.

CogVLM



(a) The input of visual language model

- ViT encoder
- MLP adapter
- Pretrained large language model
- Visual expert module

Wang et al. CogVLM: Visual Expert for Pretrained Language Models. NeurIPS 2024.

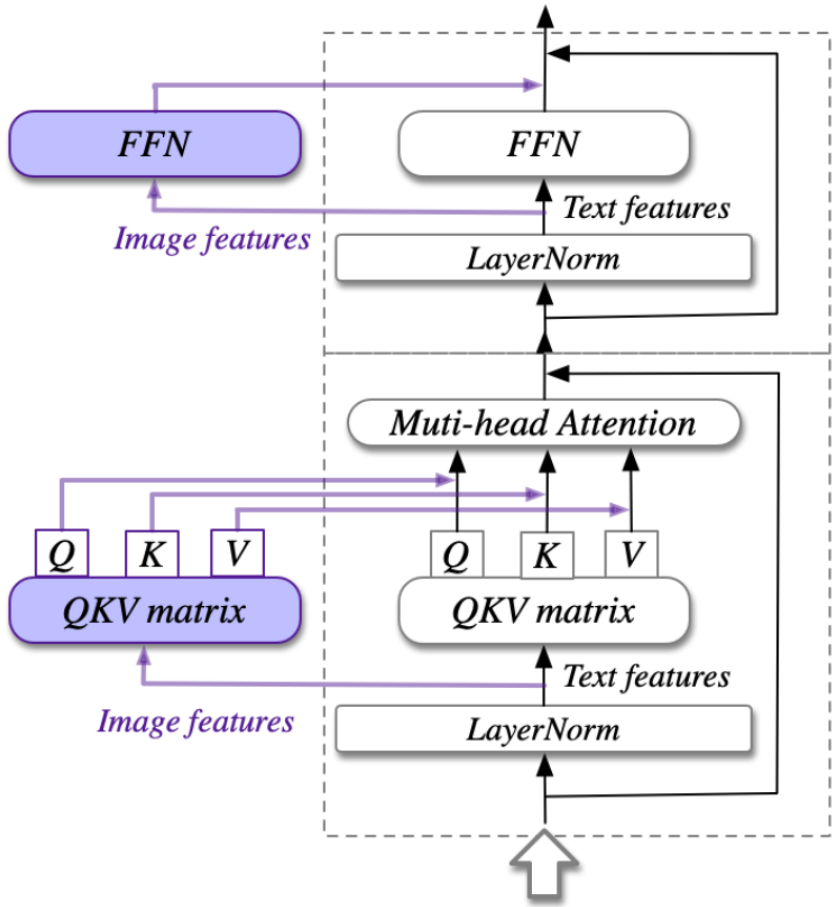
CogVLM

- **Visual expert module**

$$\text{Attention}(X, W_I, W_T) = \text{softmax}(\frac{\text{Tril}(QK^T)}{\sqrt{D}})V,$$

$$Q = \text{concat}(X_I W_I^Q, X_T W_T^Q),$$
$$K = \text{concat}(X_I W_I^K, X_T W_T^K),$$
$$V = \text{concat}(X_I W_I^V, X_T W_T^V),$$

$$\text{FFN}(X) = \text{concat}(\text{FFN}_I(X_I), \text{FFN}_T(X_T)),$$



(b) The visual expert built on the language model

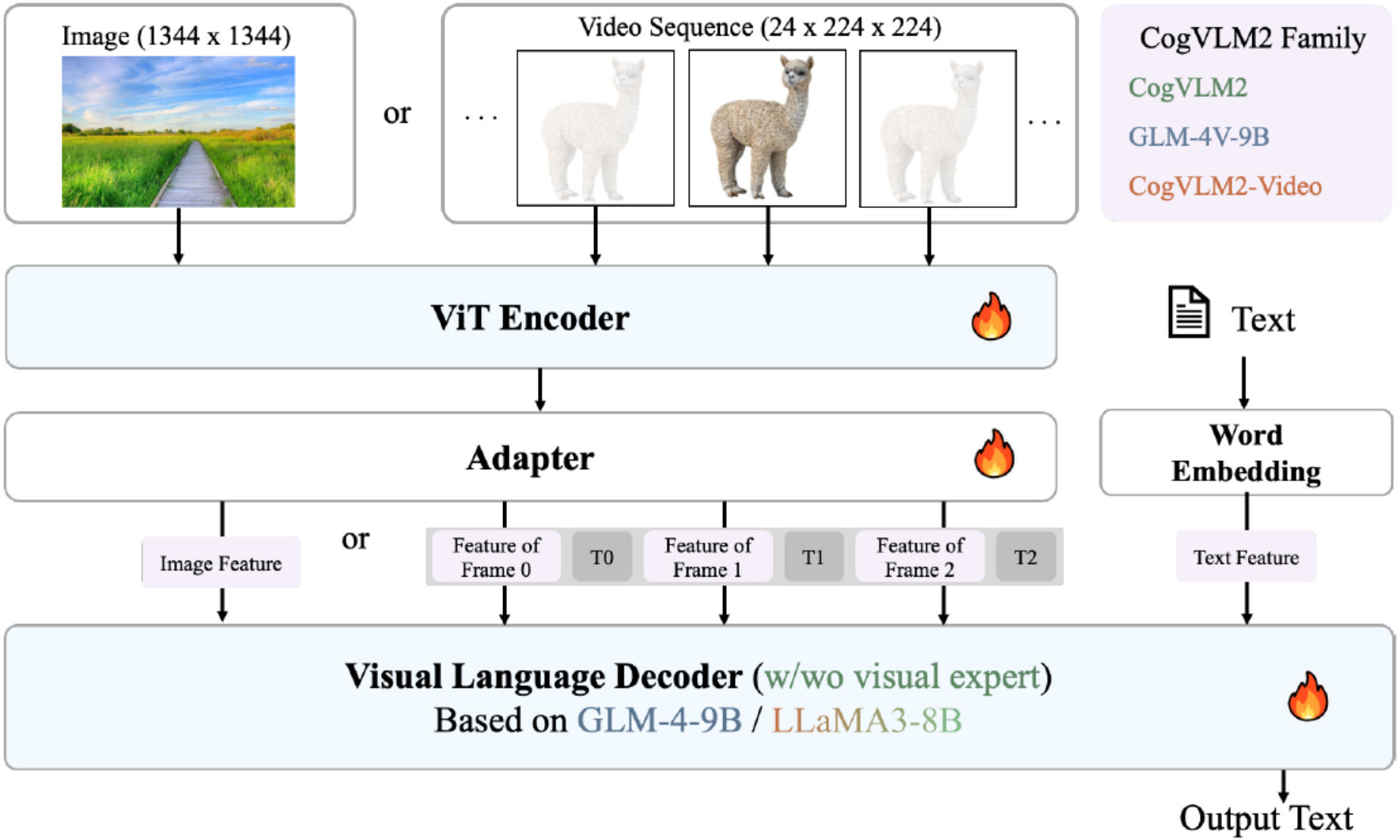
Wang et al. CogVLM: Visual Expert for Pretrained Language Models. NeurIPS 2024.

CogVLM

- Training.
  - The first stage of pretraining is for image captioning loss, i.e. next token prediction in the text part. 1.5B image-text pairs, 120,000 iterations, batch size of 8,192.
  - The second stage of pretraining is a mixture of image captioning and Referring Expression Comprehension (REC).
    - REC is a task to predict the bounding box in the image given the text description of an object, which is trained in the form of VQA, i.e., Question: Where is the object? and Answer: [[x0, y0, x1, y1]]. Both x and y coordinates range from 000 to 999, meaning the normalized position in the image.
    - Only consider the loss of the next token prediction in the "Answer" part.
    - 60,000 iterations with a batch size of 1,024 on the text-image pairs and visual grounding datasets introduced above.
    - During the final 30,000 iterations, change the input resolution from 224 × 224 to 490 × 490. The total number of trainable parameters is 6.5B.

Wang et al. CogVLM: Visual Expert for Pretrained Language Models. NeurIPS 2024.

CogVLM 2



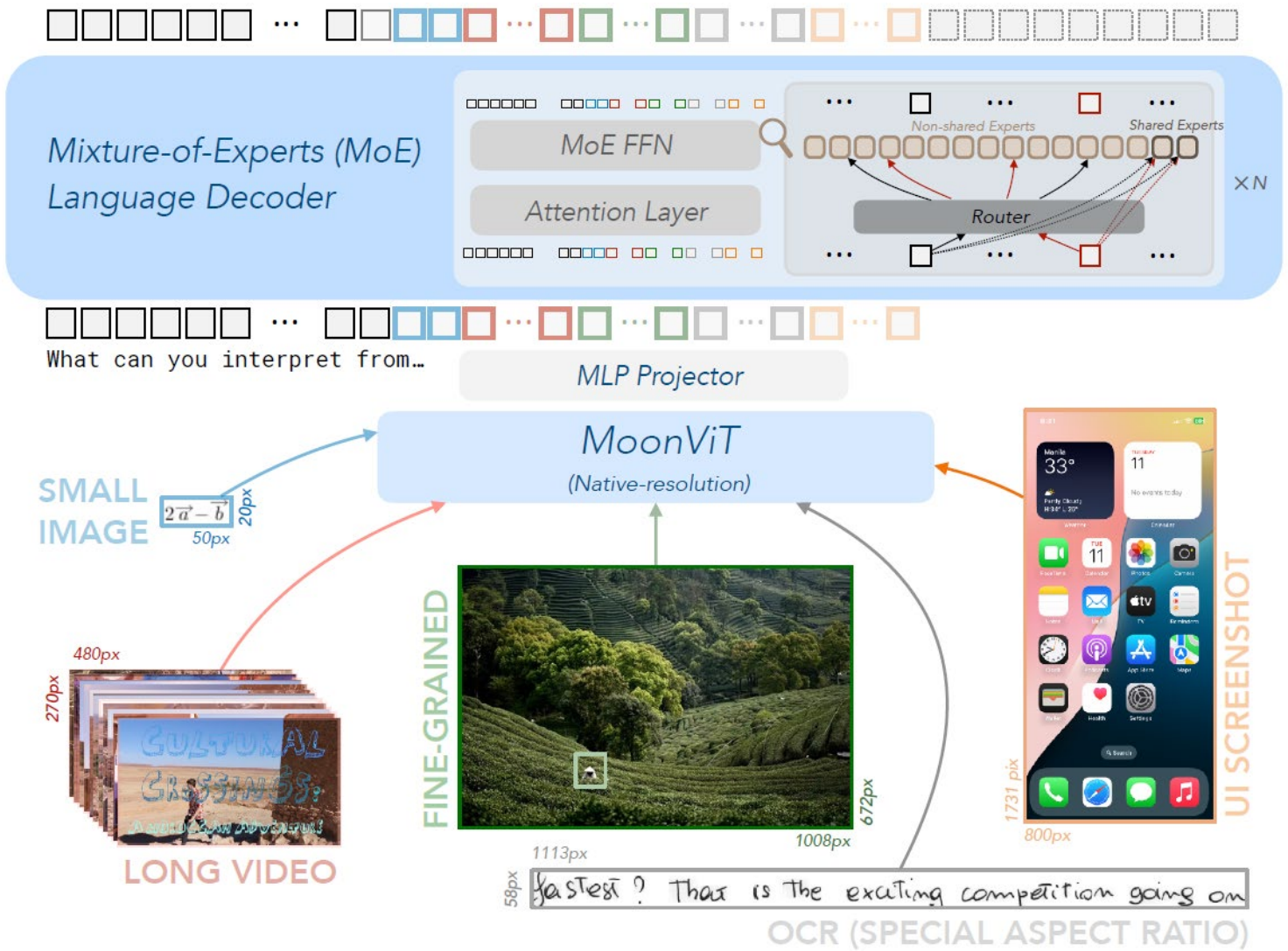
CogVLM2: Visual Language Models for Image and Video Understanding. 2024.8.

CogVLM 2

- Taking a high resolution image or the extracted frames from a given video, CogVLM models embed visual information with a pre-trained ViT Encoder and an Adapter.
- The embedded visual features are sent to a Visual Language Decoder.
- CogVLM2-Video is capable of answering image-related and video-related queries.

CogVLM2: Visual Language Models for Image and Video Understanding. 2024.8.

Kimi-VL



MoonViT: Native-resolution Vision Encoder

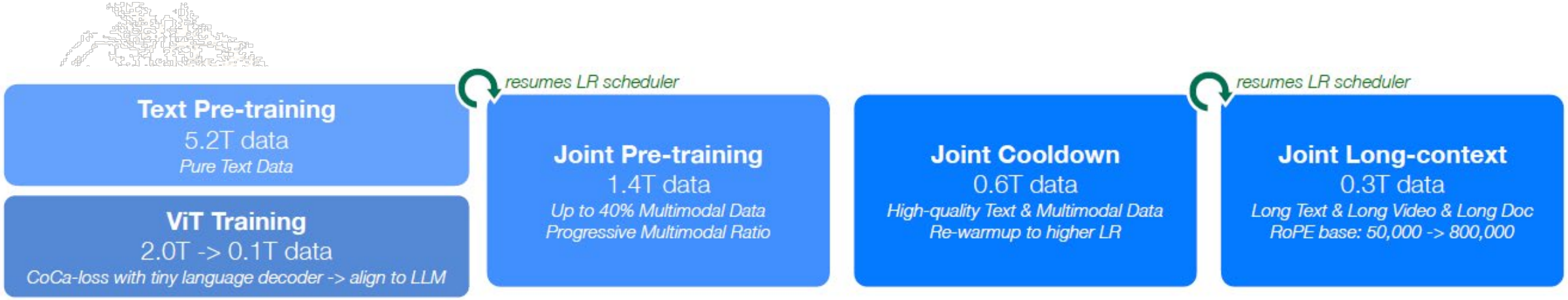
MLP Projector: a two-layer MLP to bridge the vision encoder (MoonViT) and the LLM

Mixture-of-Experts (MoE) Language Model

Team K, Du A, Yin B, et al. Kimi-vl technical report[J]. arXiv preprint arXiv:2504.07491, 2025.

Kimi-VL

- **Pretraining**
  - **ViT Training Stages. MoonViT is trained on image-text pairs, with two objectives, L_{siglip} (a variant of contrastive loss) and a cross-entropy loss $L_{caption}$ for caption generation conditioned on images.**
  - **Joint Pre-training Stage.**
  - **Joint Cooldown Stage.**
  - **Joint Long-context Activation Stage.**



Team K, Du A, Yin B, et al. Kimi-vl technical report[J]. arXiv preprint arXiv:2504.07491, 2025.

Kimi-VL

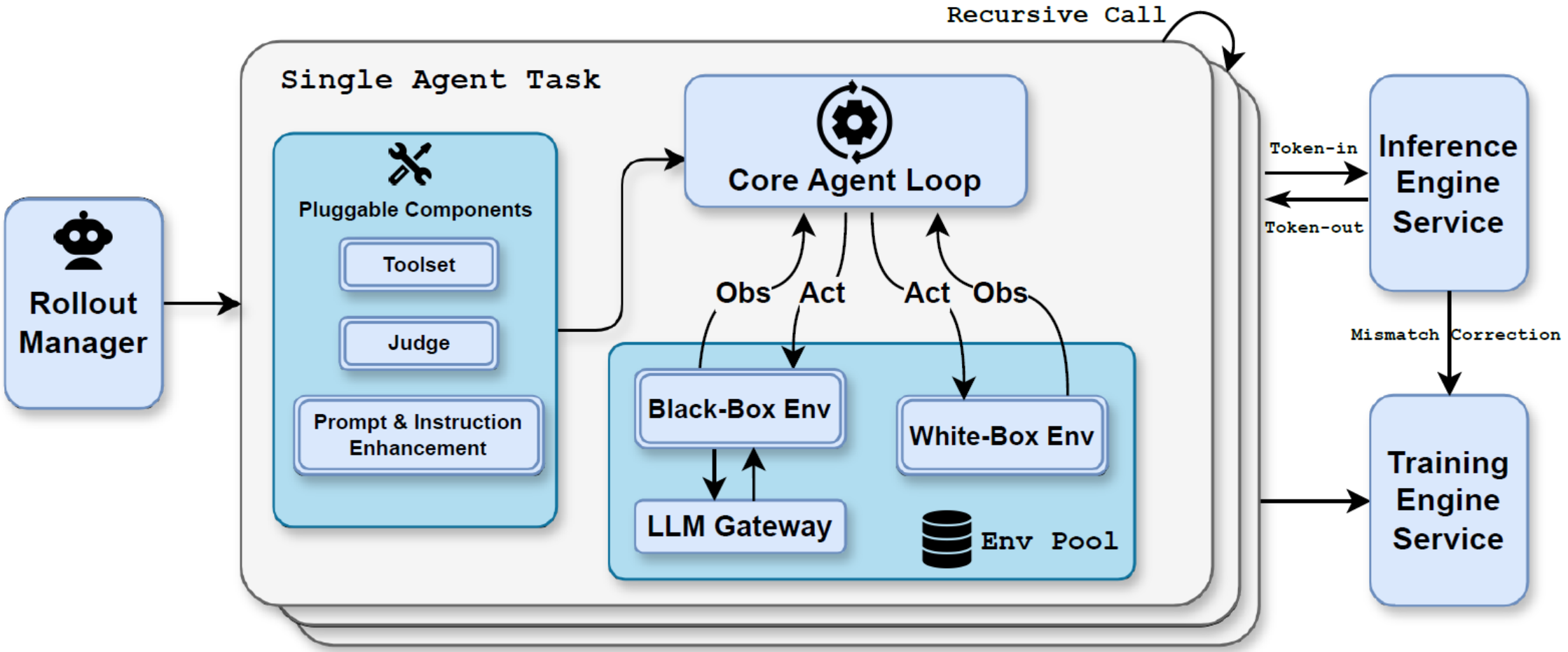
- **Post-training**
  - **Joint Supervised Fine-tuning (SFT)**
  - **Long-CoT Supervised Fine-Tuning**
  - **Reinforcement Learning**



Team K, Du A, Yin B, et al. Kimi-vl technical report[J]. arXiv preprint arXiv:2504.07491, 2025.

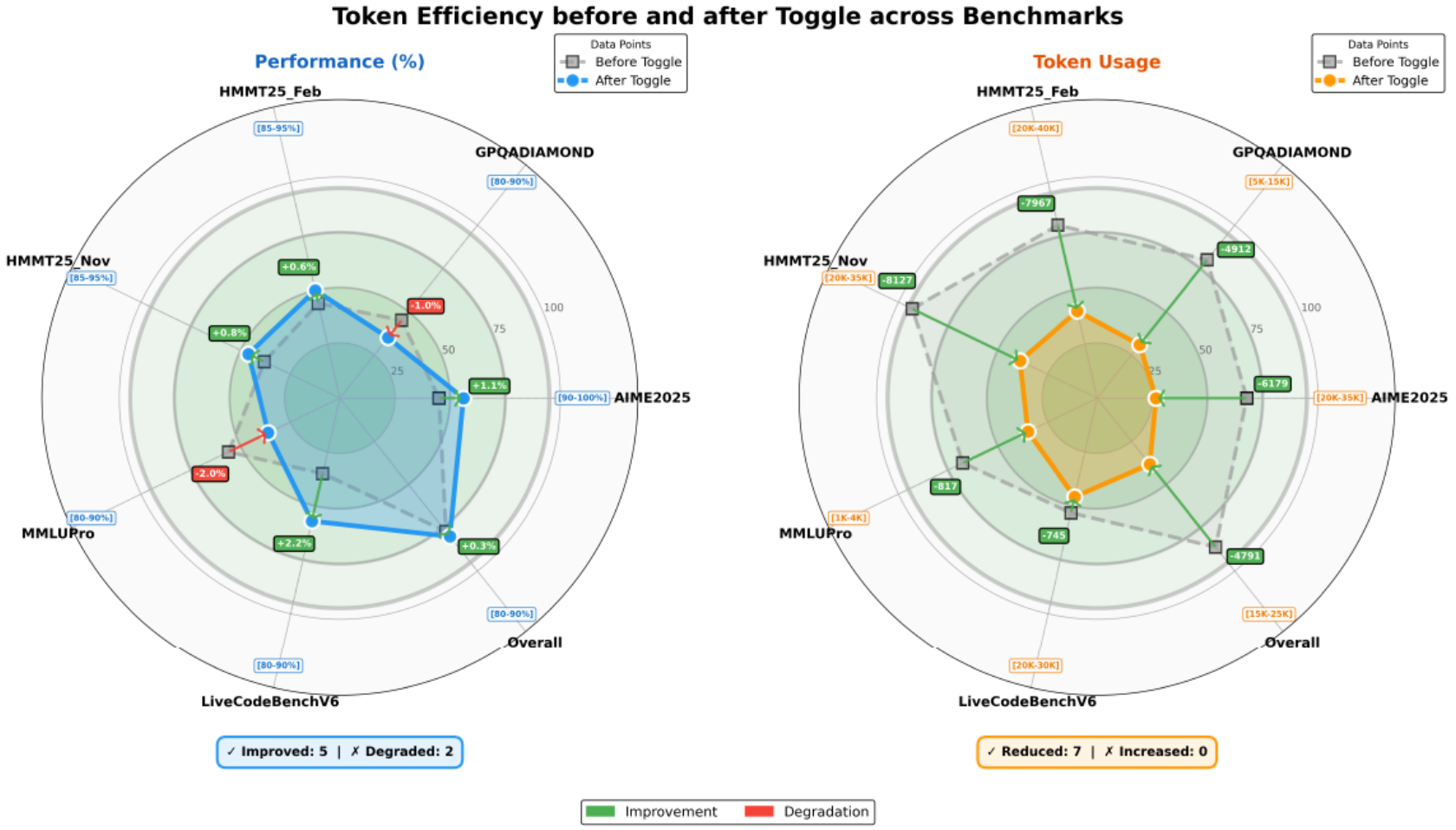
Kimi K2.5: Visual Agentic Intelligence

- Agentic RL



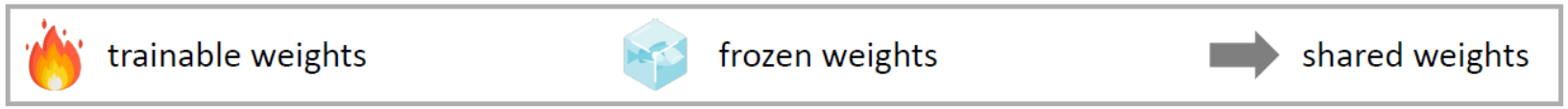
Team K, Bai T, Bai Y, et al. Kimi K2. 5: Visual Agentic Intelligence[J]. arXiv preprint arXiv:2602.02276, 2026.

Kimi K2.5: Visual Agentic Intelligence



Team K, Bai T, Bai Y, et al. Kimi K2. 5: Visual Agentic Intelligence[J]. arXiv preprint arXiv:2602.02276, 2026.

Intern-VL



stage 1: contrastive pre-training

InternViT-6B

LLaMA-7B

contrastive loss

supported tasks:

1. zero-shot image classification (new)
2. zero-shot image-text retrieval (new)

stage 2: generative pre-training

InternViT-6B → cross attention → QLLaMA

1. matching loss
2. contrastive loss
3. generative loss

supported tasks: 1. zero-shot image classification 2. zero-shot image-text retrieval

3. zero-shot image captioning (new)

stage 3: supervised fine-tuning

InternViT-6B → cross attention → QLLaMA → MLP → Vicuna-13B

MLP

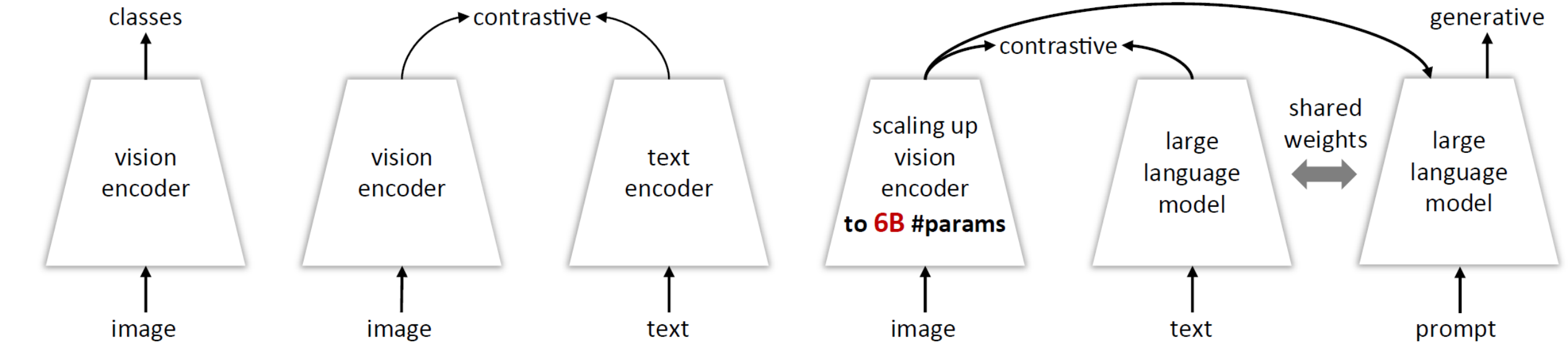
generative loss

supported tasks: 4. multi-modal dialogue (new)

5. visual question answering (new)

InternVL: Scaling up Vision Foundation Models and Aligning for Generic Visual-Linguistic Tasks. CVPR 2024.

Intern-VL



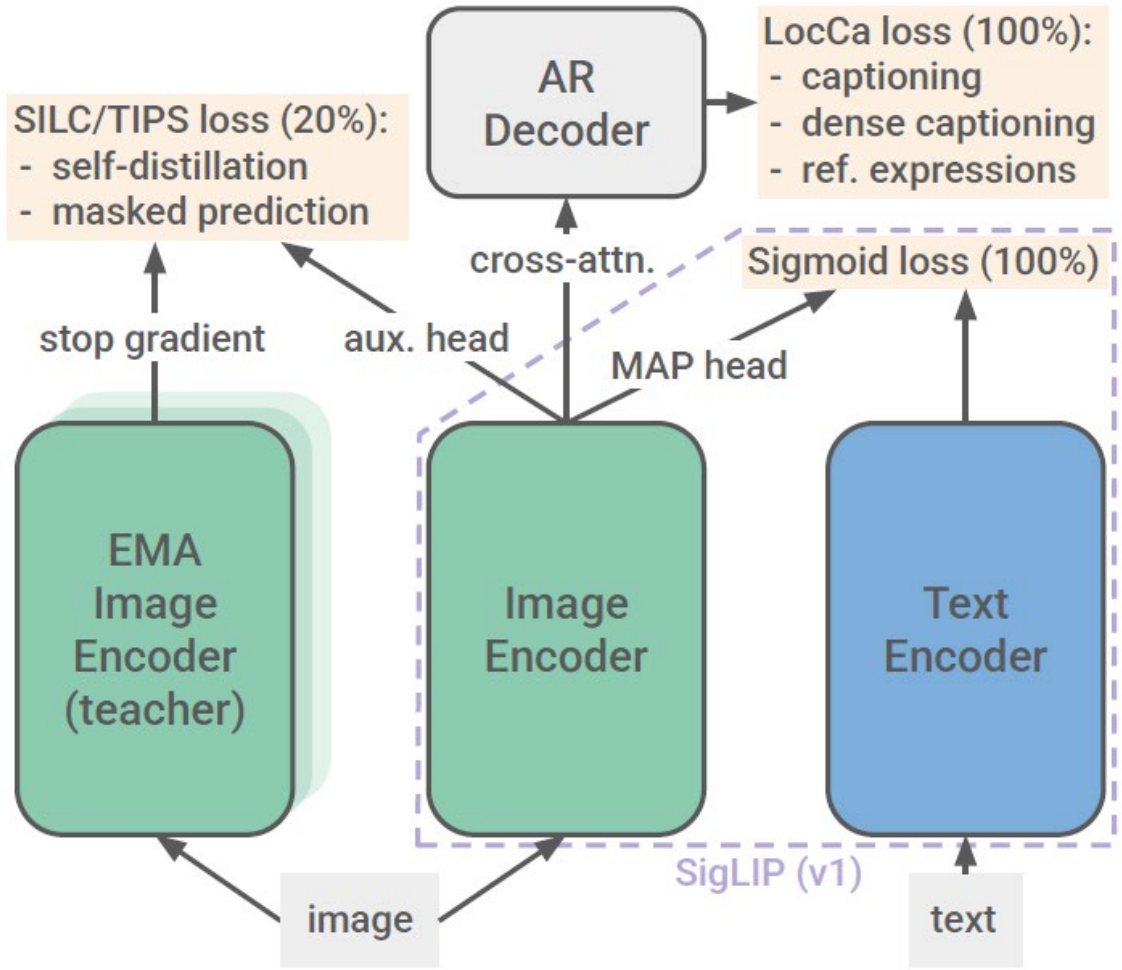
(a) Supervised pre-training (b) Contrastive pre-training (c) InternVL: Scaling up vision encoder and aligning with LLM (ours)

InternVL: Scaling up Vision Foundation Models and Aligning for Generic Visual-Linguistic Tasks. CVPR 2024.

Vision Language Models

- Instruction tuning with LLMs
  - BLIP-2, InstructBlip, LLaVA
- Vision-Language pretrained Model
  - VLMo, VL-BEIT, BEIT-3, Florence-2
- Multimodal Large Language Model
  - Qwen-VL, Qwen2-VL, Qwen3-VL
  - DeepSeek-VL
  - Kimi-VL
- Clip extended
  - Siglip2 related

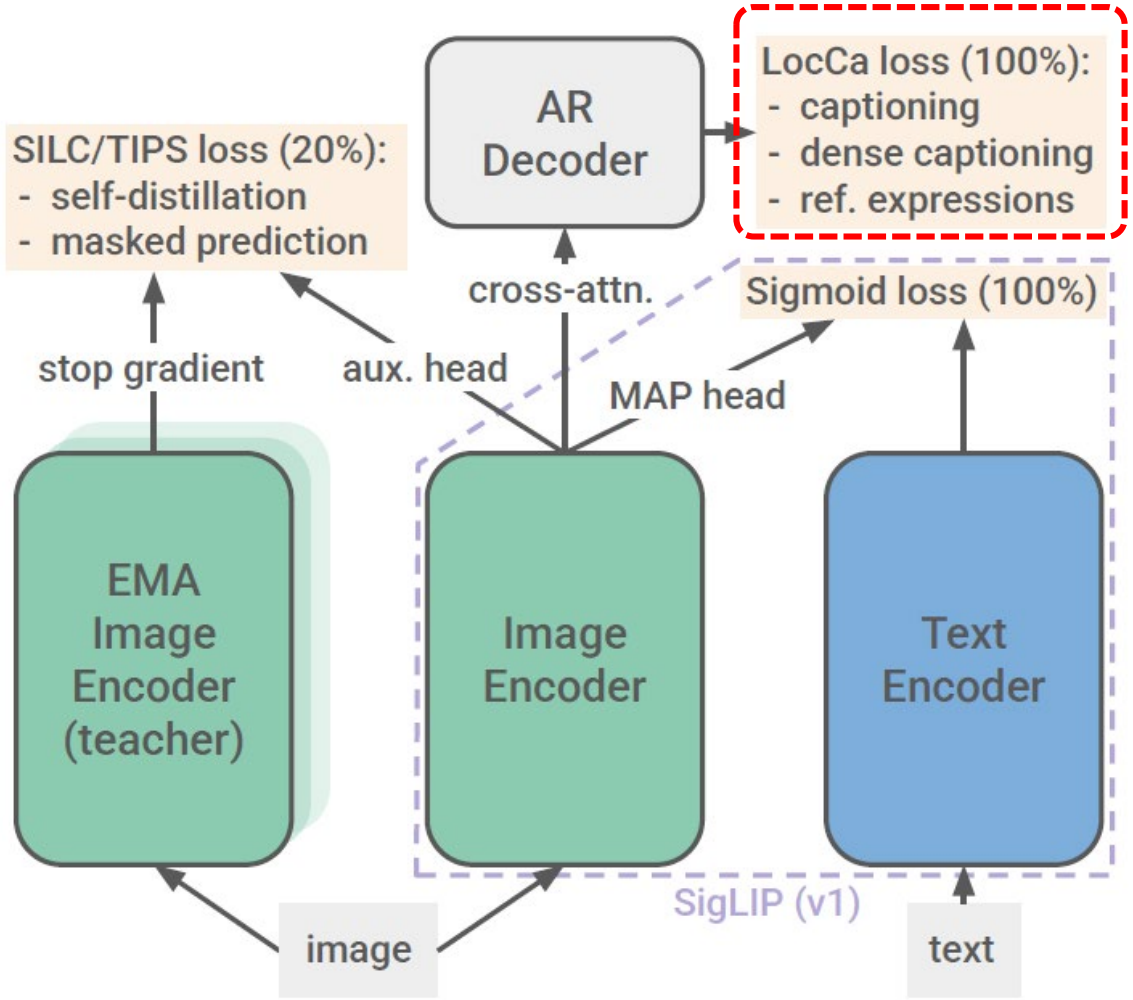
SigLIP-2: Multilingual Vision-Language Encoders



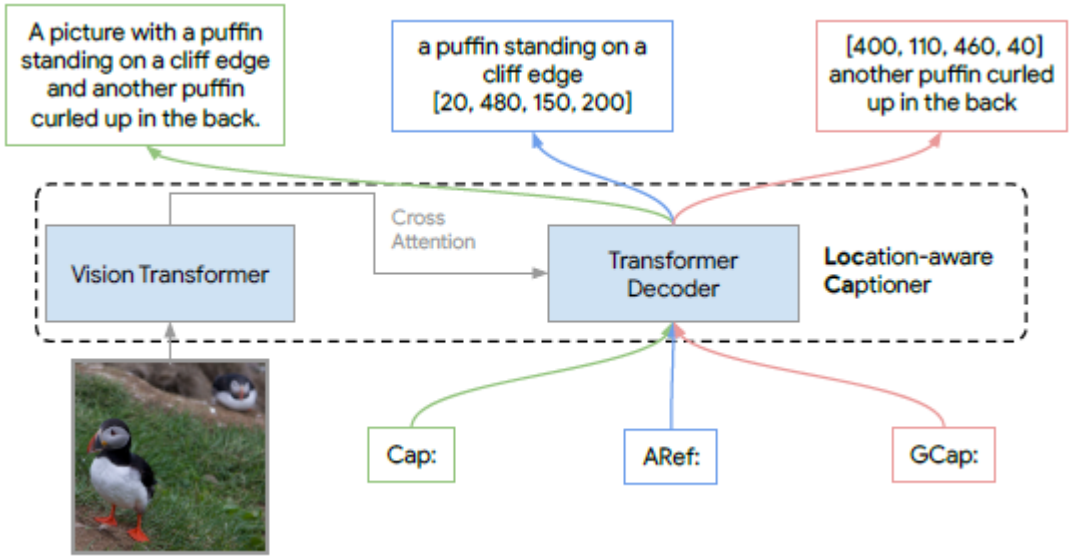
- **Training data**
  - WebLI dataset: containing 10 billion images and 12 billion alt-texts covering 109 languages.
  - 90% from English web pages, 10% from non-English ones.
  - batch size to 32k, cosine schedule with 20k warmup steps, training for a total of 40B examples 2048 TPUv5e chips

SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features. Google DeepMind, 2025.2

SigLIP-2: Multilingual Vision-Language Encoders

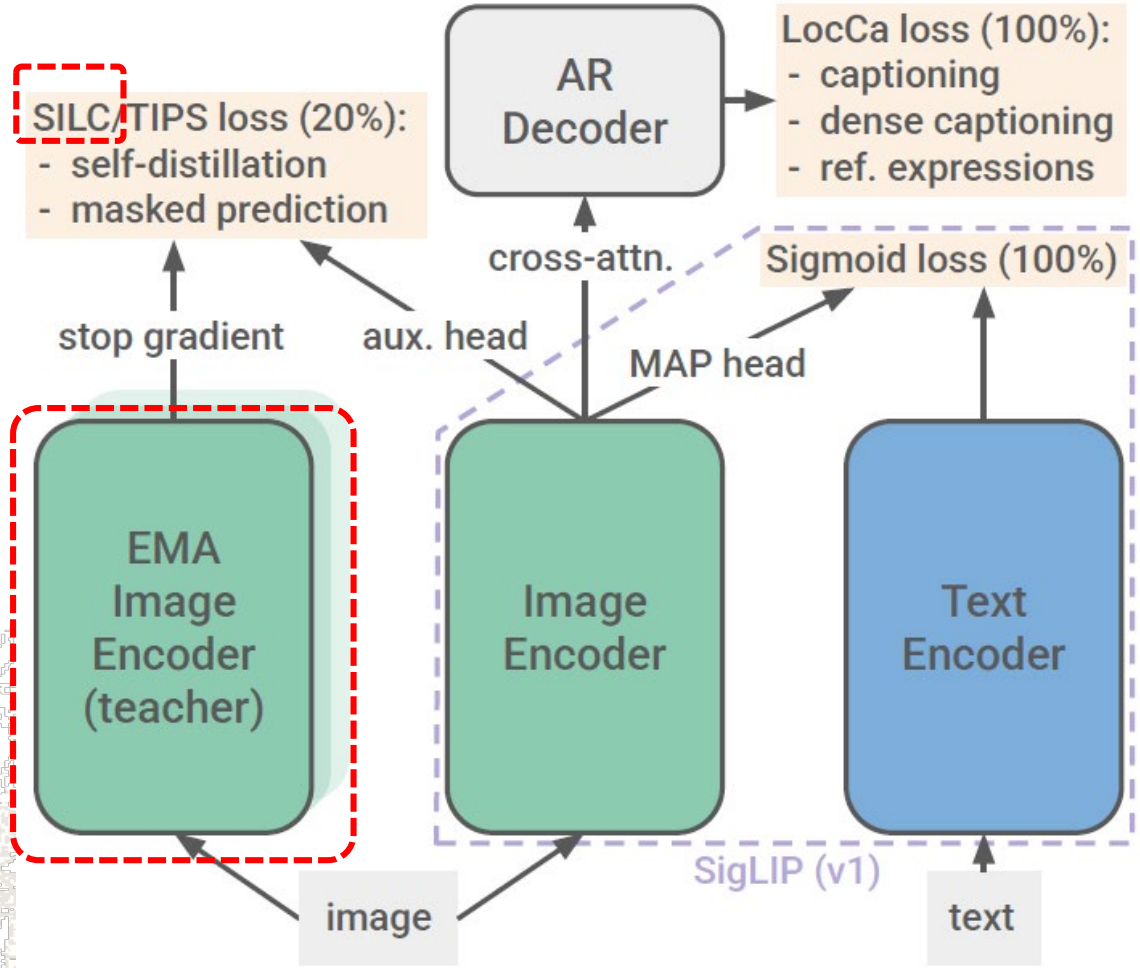


LocLa: captioning, automatic referring expression (AREF), grounded captioning (GCAP)



LocCa: Visual Pretraining with Location-aware Captioners. NeurIPS 2024

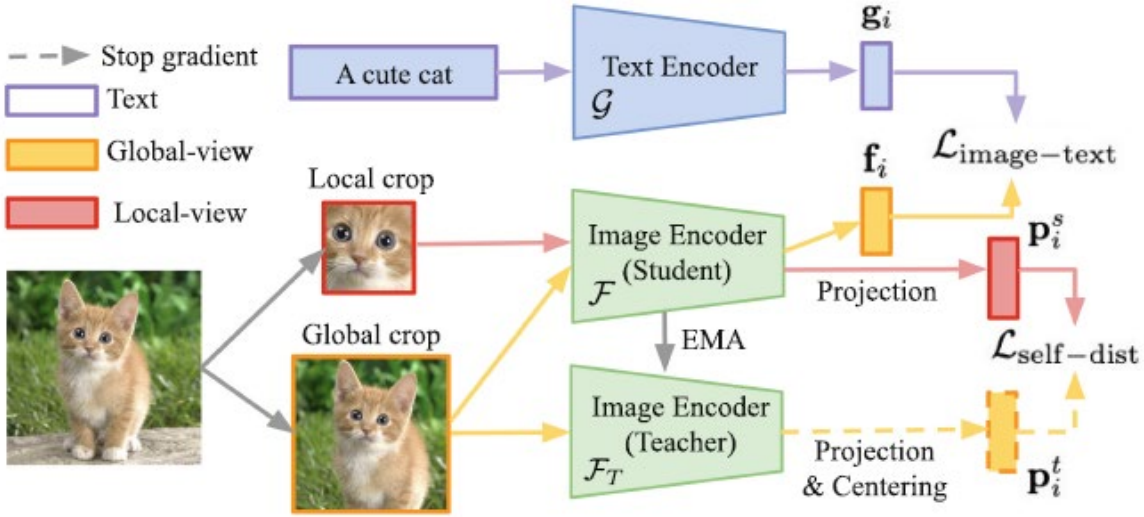
SigLIP-2: Multilingual Vision-Language Encoders



SILC:

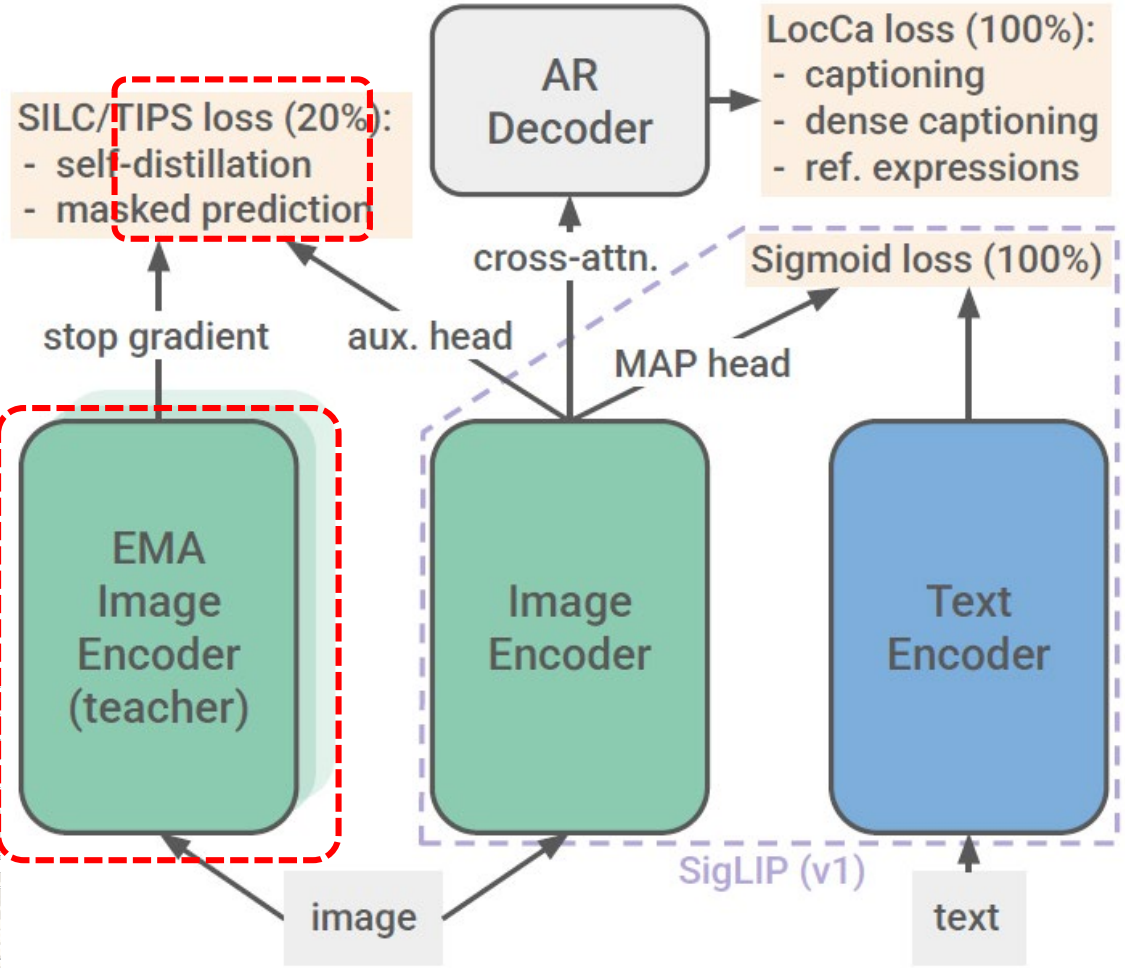
image-text contrastive learning,

local-to-global self-distillation

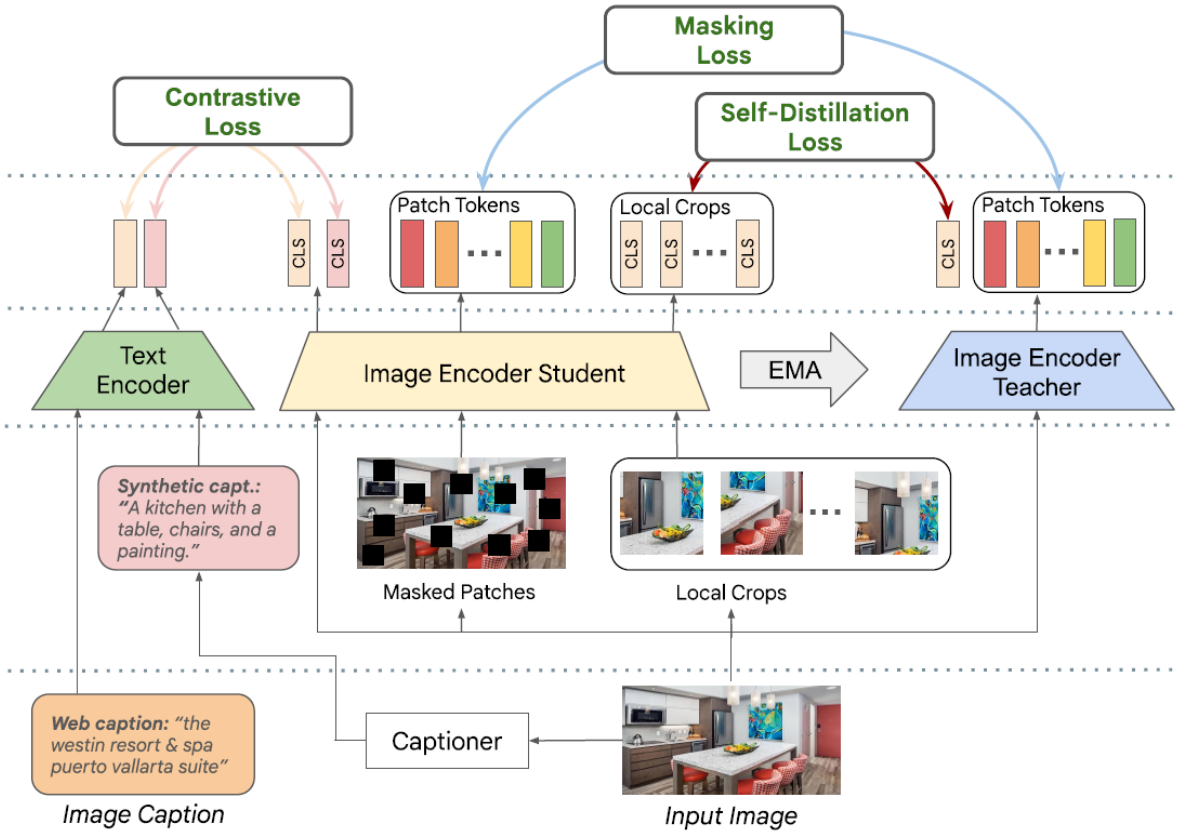


SILC: Improving Vision Language Pretraining with Self-distillation. ECCV 2024.

SigLIP-2: Multilingual Vision-Language Encoders

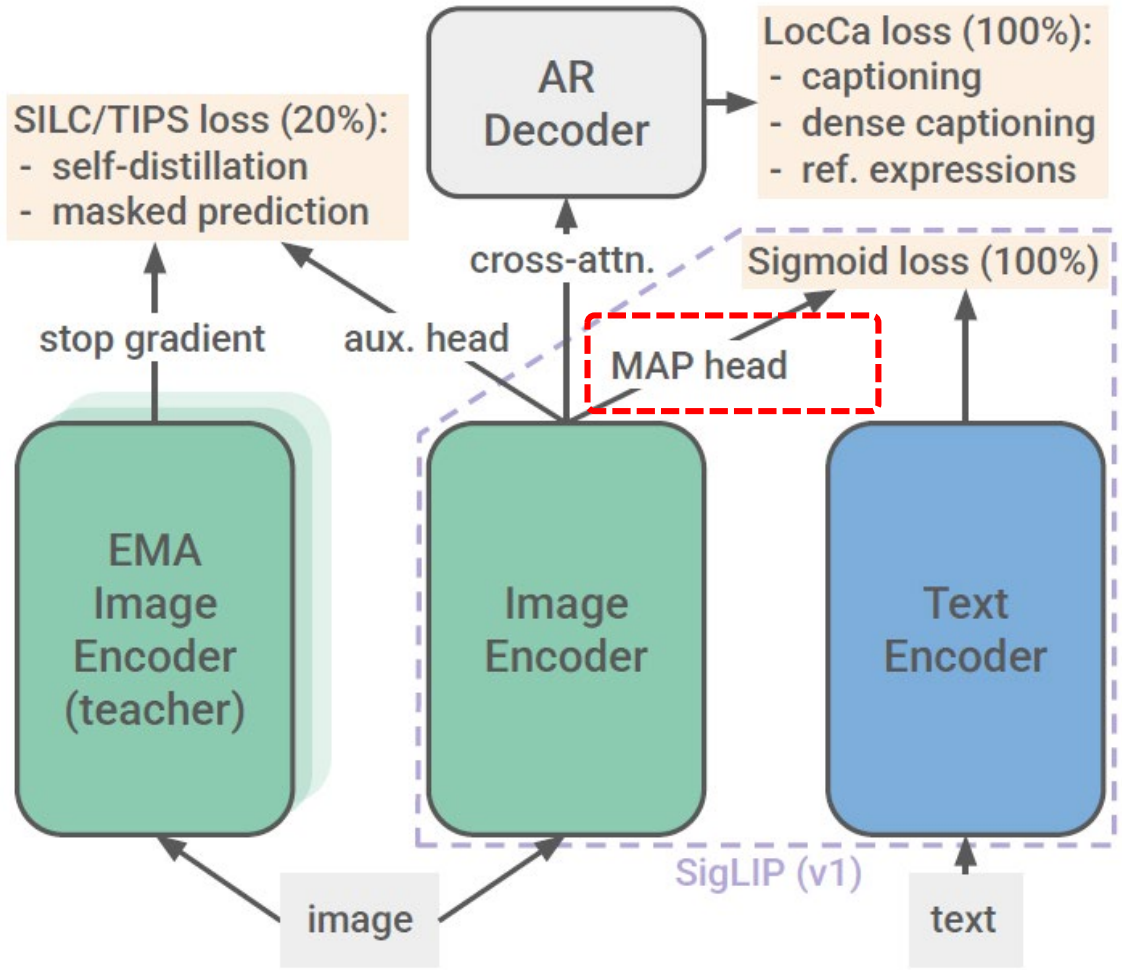


TIPS: self-distillation + mask modeling



TIPS: Text-Image Pretraining with Spatial Awareness, ICLR 2025.

SigLIP-2: Multilingual Vision-Language Encoders



MAP: multi-head pooling attention

Models: ViT-B (86M), L (303M), So400m (400M), and g (1B)

NaFlex Models: Variable aspect and resolution

SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features. Google DeepMind, 2025.2

New Research Directions

Tsinghua University

■Recent Approaches in Multimodal ML

- Expanding modalities: Incorporating new data types (e.g., 3D, physiological signals)
- Multimodal reasoning: Enhancing AI's ability to reason across multiple modalities
- Human-AI collaboration: Developing AI that can better understand and interact with humans
- Leveraging GPTs: Adapting generative pre-trained transformers (GPTs) for multimodal tasks

谢谢大家！

